Modeling and Analysis of Time Series Data: Bitcoin Price Analysis

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Introduction

Cryptocurrency is digital money that operates using blockchain technology and uses cryptography for security. Cryptocurrency is decentralized with a public ledger that records all transactions. They act without central bank supervision and without intermediaries like banks. With the passage of the recent GENIUS Act, stablecoins, one form of cryptocurrency have gained legitimacy from the US government and will soon be regulated like other products like securities and commodities.

Bitcoin is the most popular cryptocurrency and widely followed and owned. Bitcoin was the first cryptocurrency to be created. Earliest available data for Bitcoin is from September 2014 from Yahoo Finance. Over a span of almost a decade, the price of bitcoin has jumped from around $500 to above $117,000 as of day. The price growth of bitcoin has been phenomenal over the last decade with increased awareness, and institutional acceptance.

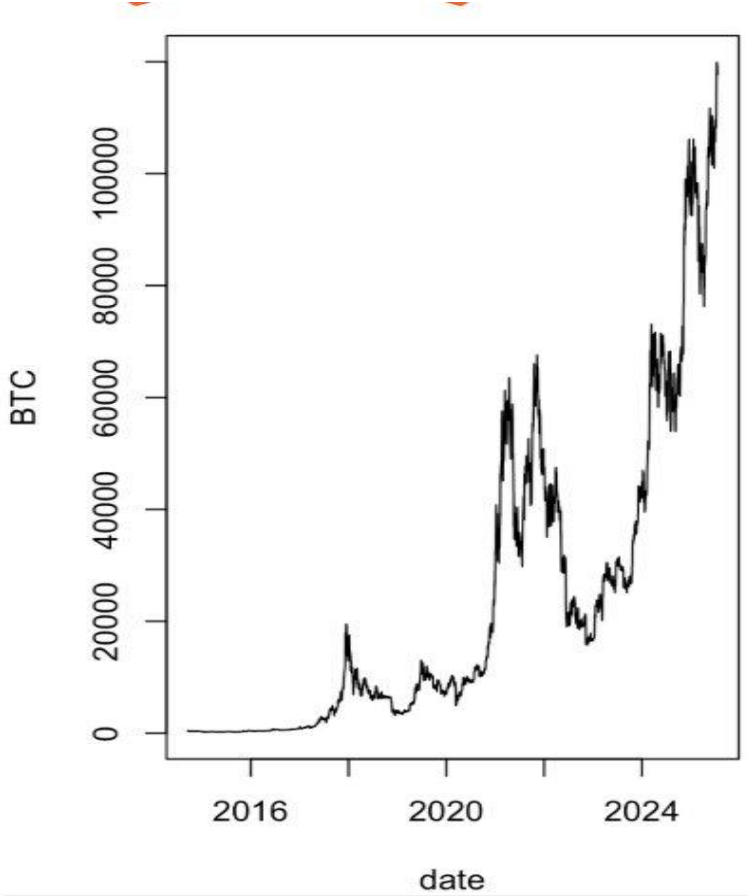
Our dataset spans from September 17, 2014 till July 19, 2025. I am using the Closing price for our time series modeling. Given that Bitcoin trades at all hours, there is no difference between the closing price and the adjusted closing price. Understanding the daily price movement of bitcoin is important for understanding future price movements and managing the risk..Our goal is to understand time series modeling and improve accuracy of predictions of future Bitcoin prices. To do that, it is important to understand volatility and variability of returns by fitting the ARCH, GARCH model and analyzing results.

Daily Bitcoin Price Trend

The graph on the left displays daily price of Bitcoin (BTC) from September 2014 till 2025.

Data set from Yahoo Finance was used for creating this graph. The time series shows that the Bitcoin spiked in 2021 to around $60,000 and dropped in 2022 to $20,000. Since 2023, it has been on an upward trajectory and has gone up substantially with recent price of $117,000. As we can see from the graph, Bitcoin is highly volatile and the pricing is highly unstable.

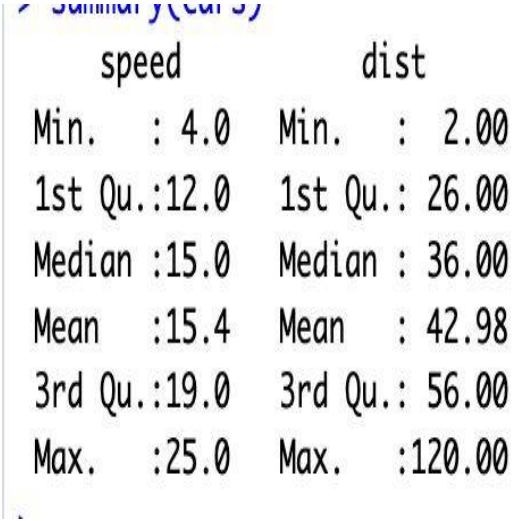
I checked the data if it is stationary or not using the adf test based on the daily closing price of Bitcoin. The immediate takeaway is that the series is non-stationary which implies that the mean and variance are not constant over time.. The stationarity test revealed that the p-value of the data was above the significance level of 0.05.



Exploratory Data Analysis

As a starting point for our time series modeling, I computed summary statistics for Bitcoin daily log returns. These data analysis provides a summary of the data and potential outliers in the data. Metrics such as the mean, quartiles, and minimum/maximum values are helpful in understanding the scale and data variability.

Understanding the scale and variability will be helpful in completing the project's data-driven exploration goal.



```
> summary(cars)
     speed           dist       
 Min.   : 4.0   Min.   :  2.00  
 1st Qu.:12.0   1st Qu.: 26.00  
 Median :15.0   Median : 36.00  
 Mean   :15.4   Mean   : 42.98  
 3rd Qu.:19.0   3rd Qu.: 56.00  
 Max.   :25.0   Max.   :120.00  
```

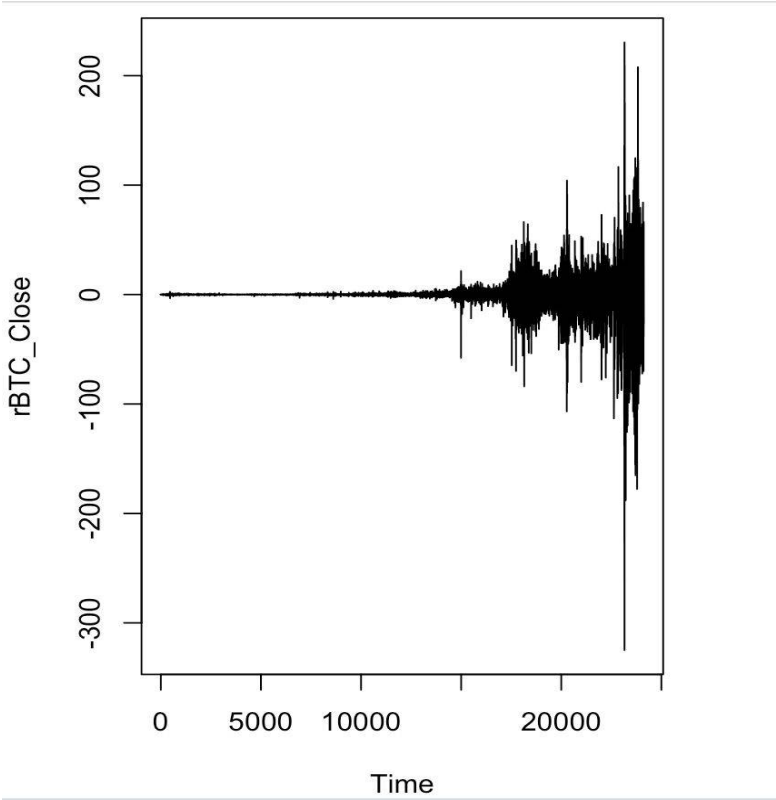
Differencing Daily Bitcoin Price for Stationarity

For making the data stationary, differencing method was used.
Code for making the data stationary:

```
> rBTC_Close=diff(Close);
> plot.ts(rBTC_Close)
```

The revised plot of the stationary data shows the daily returns. The p-value of the data was checked using adf test and the p-value is 0.01 which implies that the data is stationary but there are data outliers which implies that the returns do not converge around mean of zero.

```
> adf.test(rBTC_Close)

        Augmented Dickey-Fuller Test

data:  rBTC_Close
Dickey-Fuller = -30.613, Lag order = 28, p-value = 0.01
alternative hypothesis: stationary
```

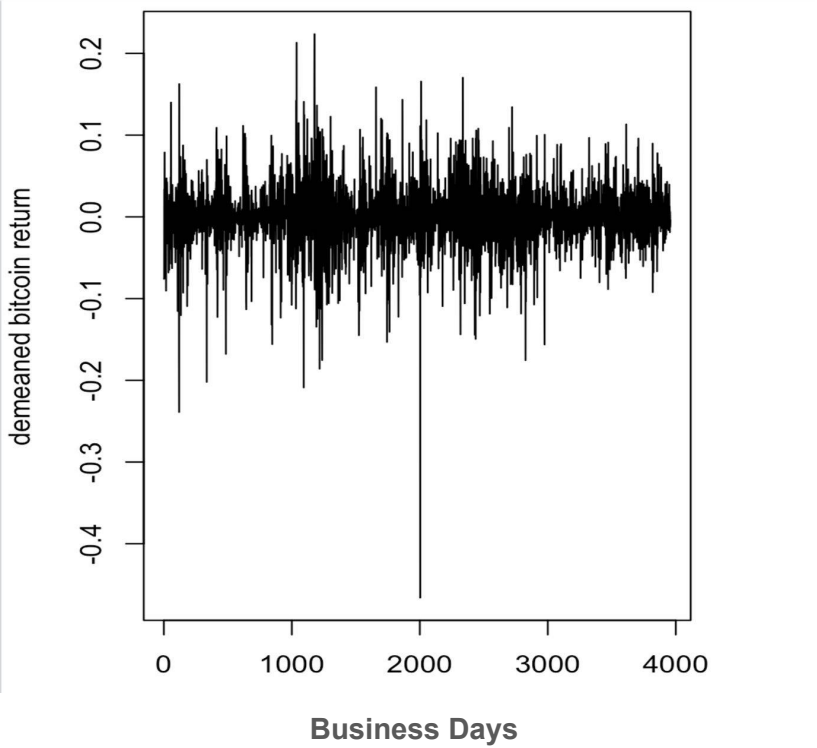


Demeaned Bitcoin Daily Returns

Demeaned return reflects deviation of an asset's return from its average return and shows periods of above-average or below-average performance.

The demeaned returns show that the time series is centered around a mean of zero. The chart reflects that high volatile periods are clustered together and followed by less volatility periods..

In this scenario,GARCH model (Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model) is useful in situations where high volatility is followed by low volatility. GARCH model helps in analyzing and forecasting the volatility of asset prices over time.

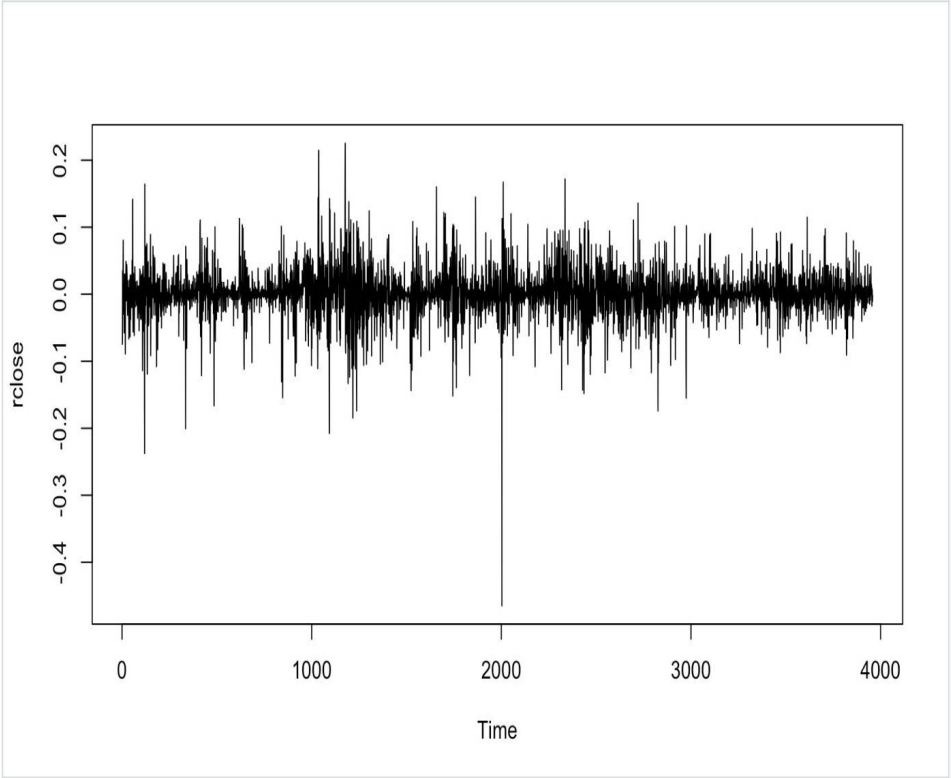


Daily Bitcoin Price Trend using Log Scale

Even though the daily bitcoin price series was stationary using differencing but the means did not converge around zero, log scale is used to check for mean convergence around 0.

Bitcoin pricing data is from September 2014 till 2025.

By plotting the price on a logarithmic y-axis, it shows the exponential price growth and stabilizes the variance. The price change is more stable but it still has outliers indicating high volatility. I tested the series and it is now stationary. See details on next sheet.



ADF Test for Checking Stationarity of Data

ADF test function is used to check stationarity of data once the data is converted to logarithmic scale.

```
> rclose=diff(log(Close))
> plot.ts(rclose)
> adf.test(rclose)

        Augmented Dickey-Fuller Test

data:  rclose
Dickey-Fuller = -15.146, Lag order = 15, p-value = 0.01
alternative hypothesis: stationary
```

Given p value is less than 0.05, the series in stationary after conversion to logarithmic scale. Next I will try to find the best fit model using ARIMA function.

ARIMA Function for Bitcoin

The auto.arima function was used for the best model fit using ARIMA. Here is the function and best fit ARIMA model from of that function.

```
auto.arima(rclose)
```

```
Series: rclose
ARIMA(2,0,0) with non-zero mean

Coefficients:
          ar1     ar2    mean
      -0.0227  0.0109  0.0014
s.e.   0.0159  0.0159  0.0006

sigma^2 = 0.00129:  log likelihood = 7551.15
AIC=-15094.31   AICc=-15094.3   BIC=-15069.17
```

It implies based on the ARIMA model, the best fitted model is (2,0,0) where p =2 and q=0.

ARIMA Model for Bitcoin

Given values of (2,0,0) for p, i and q, the best model fit was created using model function. Here is the code and result for that function:

```
modelrclose=arima(rclose,order = c(2,0,0))
modelrclose
```

```
Call:
arima(x = rclose, order = c(2, 0, 0))

Coefficients:
          ar1     ar2  intercept
      -0.0227  0.0109     0.0014
s.e.   0.0159  0.0159     0.0006

sigma^2 estimated as 0.001289:  log likelihood = 7551.15,  aic = -15094.31
```

ARIMA Model(2,0) and Limitations

The goal of the ARIMA(2,0) model is to predict the log return by using the data from the previous two log returns. The problem with the ARIMA model is the constant variance. As the Bitcoin data showed signs of volatility clustering, more stable model is needed which was able to fix this issue.

GARCH models such as standard GARCH along with both EGARCH and GJR-GARCH were used for addressing this major issue. The next slides will build on the ARIMA base to model conditional variance over time. This should help in identifying volatility clustering and provide a more accurate representation of the Bitcoin’s data.

Equations & Hypothesis

In the EGARCH(1,1) model I used the log variance equation (right side) and for the ARMA(1,1) model I used the mean equation (left side) as seen below. A hypothesis test is used for each model to find the significance.

Under the null hypothesis, any changes are white noise, are not statistically significant and have no effect on the model. The alternative hypothesis implies that the change are significant and can be explained by the model. Null hypothesis can be rejected only if the evidence from the model output is significantly different from our assumptions under null hypothesis.

$$r_t = \mu + \phi_1 r_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t$$

$$\log(\sigma_t^2) = \omega + \beta \log(\sigma_{t-1}^2) + \alpha \left(\frac{|\varepsilon_{t-1}|}{\sigma_{t-1}} - \sqrt{\frac{2}{\pi}} \right) + \gamma \cdot \frac{\varepsilon_{t-1}}{\sigma_{t-1}}$$

GARCH Model Introduction

The GARCH Model or the Generalized Autoregressive Conditional Heteroskedasticity is financial time series model - **GARCH(p,q).**

Key GARCH model properties:

1.**Volatility clustering** - The parameters of the model help in identifying volatility clustering which implies that periods of high volatility are followed by more high volatility.

2. **Normalized distribution** - The model assumes the standardized errors follow a normal distribution.

3. **Persistence** - To show volatility persistence, the sum of the alpha and beta coefficients must be close to 1..

4. **Sign Bias Test** -The GARCH model uses the **Sign Bias Test results** to state analysis performed on residuals and reveal asymmetry.

Results of GARCH Model Fit - 1

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : sGARCH(1,1)
Mean Model       : ARFIMA(1,0,1)
Distribution     : norm

Optimal Parameters
------------------------------------
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.007472    0.001713   4.3630 0.000013
ar1    -0.134144    0.086768  -1.5460 0.122103
ma1     0.203012    0.085720   2.3683 0.017870
omega   0.000254    0.000033   7.7651 0.000000
alpha1  0.077659    0.003419  22.7126 0.000000
beta1   0.921341    0.003515 262.0871 0.000000

Robust Standard Errors:
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.007472    0.002059   3.6286 0.000285
ar1    -0.134144    0.083574  -1.6051 0.108474
ma1     0.203012    0.083442   2.4330 0.014976
omega   0.000254    0.000132   1.9237 0.054394
alpha1  0.077659    0.019266   4.0309 0.000056
beta1   0.921341    0.020438  45.0798 0.000000

LogLikelihood : -40868.21

Information Criteria
------------------------------------
```

Results of GARCH Model Fit - 2

```
Information Criteria
------------------------------------

Akaike       3.3885
Bayes        3.3905
Shibata      3.3885
Hannan-Quinn 3.3892

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic  p-value
Lag[1]                      3.028 0.081822
Lag[2*(p+q)+(p+q)-1][5]     5.785 0.000184
Lag[4*(p+q)+(p+q)-1][9]    11.552 0.001879
d.o.f=2
H0 : No serial correlation
```

```
Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      25.85 3.697e-07
Lag[2*(p+q)+(p+q)-1][5]     26.59 2.868e-07
Lag[4*(p+q)+(p+q)-1][9]     30.02 4.309e-07
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]   0.05186 0.500 2.000  0.8199
ARCH Lag[5]   0.98237 1.440 1.667  0.7382
ARCH Lag[7]   4.72069 2.315 1.543  0.2543

Nyblom stability test
```

Results of GARCH Model Fit - 3

```
Nyblom stability test
------------------------------------
Joint Statistic:  27.0206
Individual Statistics:
mu     0.3620
ar1    5.9721
ma1    6.0126
omega  0.2546
alpha1 8.2881
beta1  5.2529

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          1.49 1.68 2.12
Individual Statistic:     0.35 0.47 0.75
```

```
Sign Bias Test
-----------------------------------
                   t-value      prob sig
Sign Bias            5.926 3.145e-09 ***
Negative Sign Bias   1.155 2.481e-01
Positive Sign Bias   1.020 3.075e-01
Joint Effect        50.496 6.263e-11 ***


Adjusted Pearson Goodness-of-Fit Test:
-----------------------------------
  group statistic p-value(g-1)
1    20     430.3    2.114e-79
2    30     505.8    1.884e-88
3    40     559.0    2.945e-93
4    50     615.8    4.854e-99
```

Key Findings from GARCH Model - 1

The GARCH model is designed to capture the leverage effect..

Model Specification

- GARCH Model: sGARCH(1,1): A standard GARCH model with one Autoregressive ARCH component (alpha1), one moving average GARCH component (beta1),
- Mean Model: ARFIMA(1,0,1): An ARMA(1,1) model for the daily returns. Models the current return based on the previous day's return (ar1) and the previous day's forecast error (ma1). (ar1) and (ar2) are both used to help show how yesterday's return help predict today's return.
- Distribution: Assumes the standardized residuals follow a normal distribution.

Findings

- Mean Equation Parameters (mu, ar1, ma1): All three parameters in the mean equation are statistically significant (all p-values are 0.000000). This indicates linear dependence on past values is not present. Significant ma1 implies errors in model correlation and implies presence of moving average component. The ar1 coefficient is not statistically significant (p-value = 0.122103 > 0.05). This suggests the AR(1) component may not be necessary.

Key Findings from GARCH Model - 2

- Volatility: Alpha1 and beta1 measure conditional variance and indicate dependency on both past variance values and past errors to model volatility.
- Alpha1 measures whether the magnitude of a shock (how absolute shock scales volatility), is statistically significant (0.077659) and indicates some volatility clustering. This implies that the size of the shock does have a significant impact on volatility.
- Beta1(persistence of shocks in log-variance) is statistically significant (0.921341). It implies that volatility is persistent and past shocks have a positive impact on conditional variance. It takes a very long time for volatility to return to its long-run average after a shock. .

Other Variance Equation Parameters:

- omega: The constant term in the log-variance equation and represents unconditional variance. The value is not significant and implies lack of variance in data.
- All the above parameters (omega, alpha1, beta1) are statistically significant and confirm presence of GARCH effects.

Key Findings from GARCH Model - 3

Weighted Ljung-Box Test on Standardized Residuals: This test checks for remaining autocorrelation in the model's errors. The p-values are close to zero for Lag 5 and Lag 9 which is significant. Null hypothesis of no serial correlation is rejected. The mean equation coefficients are individually significant but the model fails to capture all of the linear dependence in the returns. The mean model is misspecified. Tests randomness of a time series by examining autocorrelations at multiple lags.

The p-value at lag 1 is 0.081822 which is not statistically significant but is significant at lags 5 and 9.. This suggests the model residuals have significant autocorrelation..

Weighted Ljung-Box Test on Standardized Squared Residuals: Checks if there is any volatility clustering after modeling (ARCH effects). The null Hypothesis (H0) checks for no ARCH effects.

The p-values are small and null hypothesis must be rejected. This is a failure of the model. It means that the model does not capture all volatility clustering in the data.

Key Findings from GARCH Model - 4

Nyblom Stability Test: Tests if the model's parameters are stable. The null hypothesis assumes that the the parameters are stable.

The Joint Statistic of 27.0206 is greater than the 10% (1.69), 5% (1.9), 1% (2.12) critical value. The null hypothesis is rejected. This implies that the parameters are not stable.

Sign Bias Test: Tests for "leverage effects," where negative returns have a different impact on volatility than positive returns. The standard GARCH model assumes the effect is symmetric.

The null hypothesis assumes that there is no sign bias. The "Sign Bias" and "Joint Effect" p-values are close to zero and null hypothesis is rejected. It means that there are leverage effects, which are not captured by the GARCH model. Thus I used a asymmetric model like EGARCH or GJR-GARCH to check for best fit.
as GARCH model data analysis goals are not met.

Adjusted Pearson Goodness-of-Fit Test: Tests if the standardized residuals follow normal distribution.Null hypothesis assumes that data is normally distributed.
The p-values are small and null hypothesis is rejected. This confirms the data has "fat tails" and that the normal distribution is a poor fit..

Summary - GARCH Model

The GARCH(1,1) model is useful in identifying high volatility persistence but the test has several critical failures:

- It fails to capture all volatility clustering (according to the Ljung-Box test).
- The parameters are highly unstable.
- It fails to account for leverage effects (asymmetry).
- It incorrectly assumes a normal distribution for the errors.

Plots from the GARCH Model

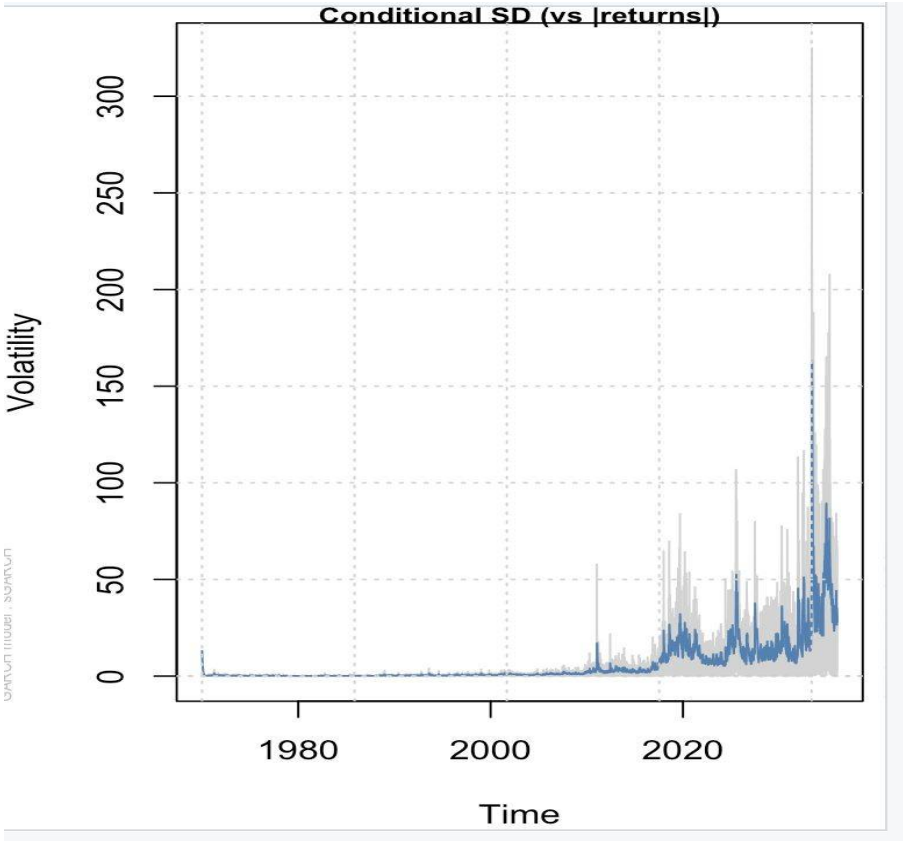
The following screens reflect the multiple plots for the GARCH(1,1) model. Details about each graph included in the following slides.

Estimated Conditional Volatility of Bitcoin from the GARCH Model

Conditional volatility measures volatility of returns based on all available historical information. GARCH model estimates the conditional volatility based on past values of itself and of model errors.

GARCH model estimates future volatility of an asset based on a weighted average of past squared returns and past conditional variances. GARCH(1,1) model was used to create this graph and estimate conditional variance. Measurement of conditional volatility helps in risk management for investors and optimizing the portfolio by asset diversification.

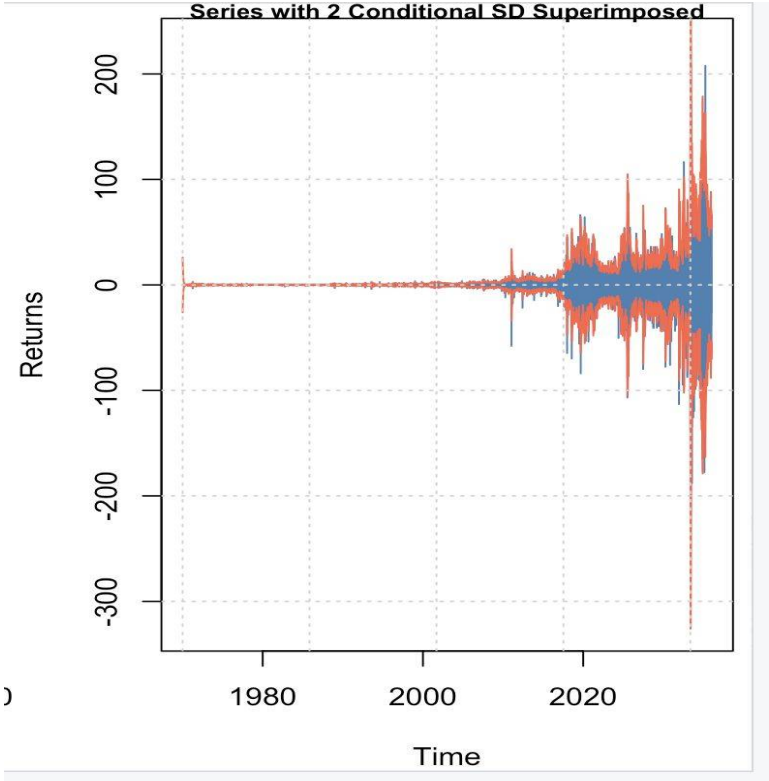
This graph shows the time-varying estimate of Bitcoin's daily volatility (standard deviation). and volatility clustering. The graph shows increased volatility in turbulent periods in the market.



Bitcoin Returns with Superimposed GARCH Conditional Volatility Bands

This graph overlays the estimated conditional volatility from the GARCH model (the red and blue bands) on top of the actual daily returns.

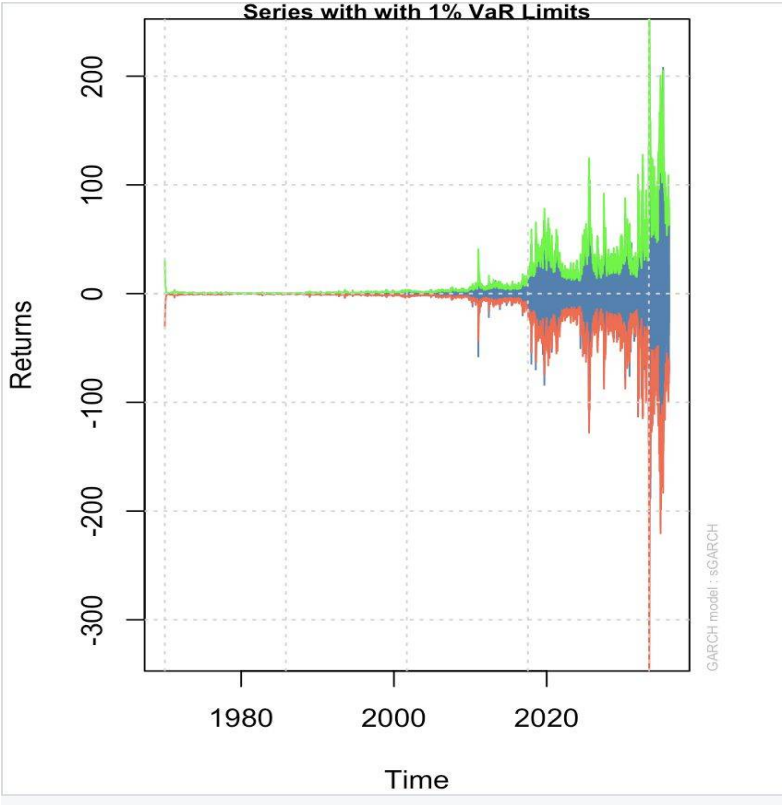
The width of the bands indicates the volatility of Bitcoin return. Narrow bands reflect lower volatility and risk and wider bands reflect increased volatility and increased risk. The chart reflects GARCH model fit as bands expand during high volatile periods and contract during low volatility periods.



GARCH-Based 1% Value-at-Risk Limits for Bitcoin Returns

GARCH-based 1% Value-at-Risk (VaR) estimates the worst potential loss a portfolio might face, with only a 1% chance that the actual loss will be higher.

The graph shows use of GARCH models in financial risk management. GARCH model volatility forecast is used to calculate the 1% VaR limit. It implies that we are 99% confident that the losses will not exceed the boundary shown by the model. VaR bands widen in periods of high volatility and shrink during periods of low volatility.

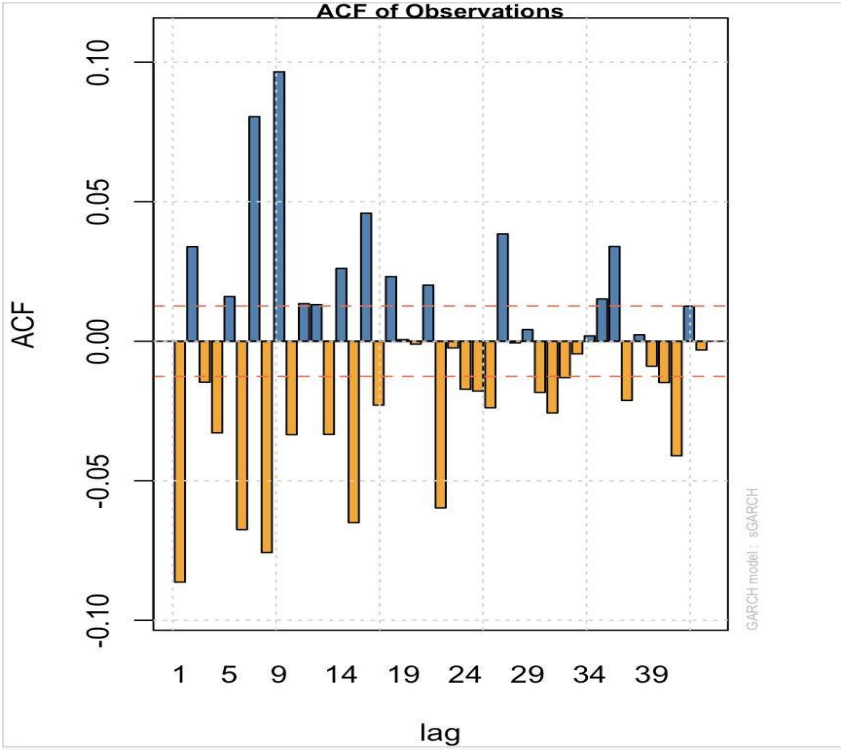


Autocorrelation Function of Daily Bitcoin Returns

The autocorrelation function (ACF) of daily Bitcoin returns shows weak autocorrelation.

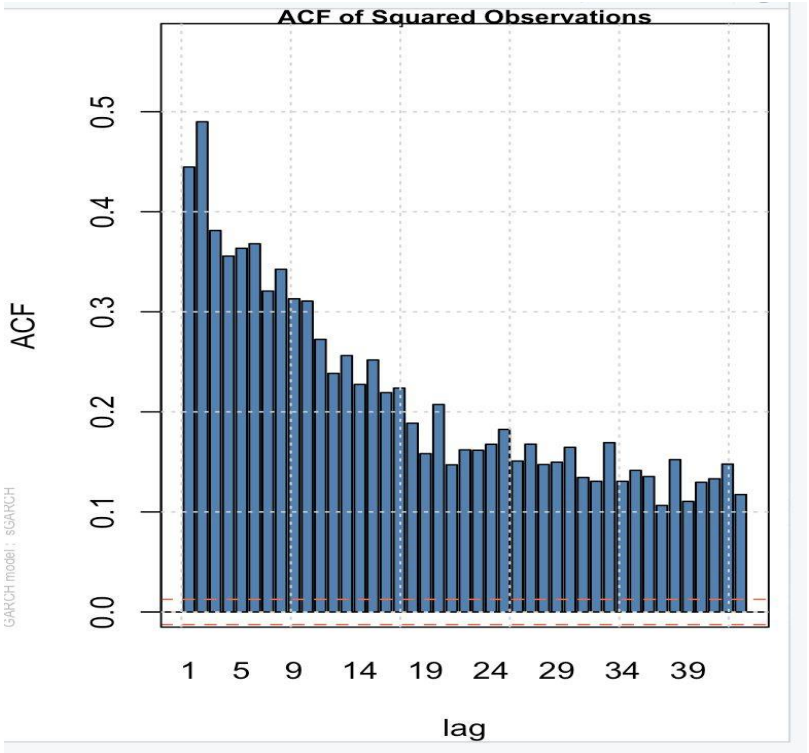
The ACF graph shows the correlation of the returns series with its own past values and graph shows weak autocorrelation.

This is justified by Efficient Market Hypothesis theory under which asset prices reflect all available information and impossible to outperform the market by using past information. Thus, a simple ARMA model for the mean would likely be insufficient in using past returns to predict future returns



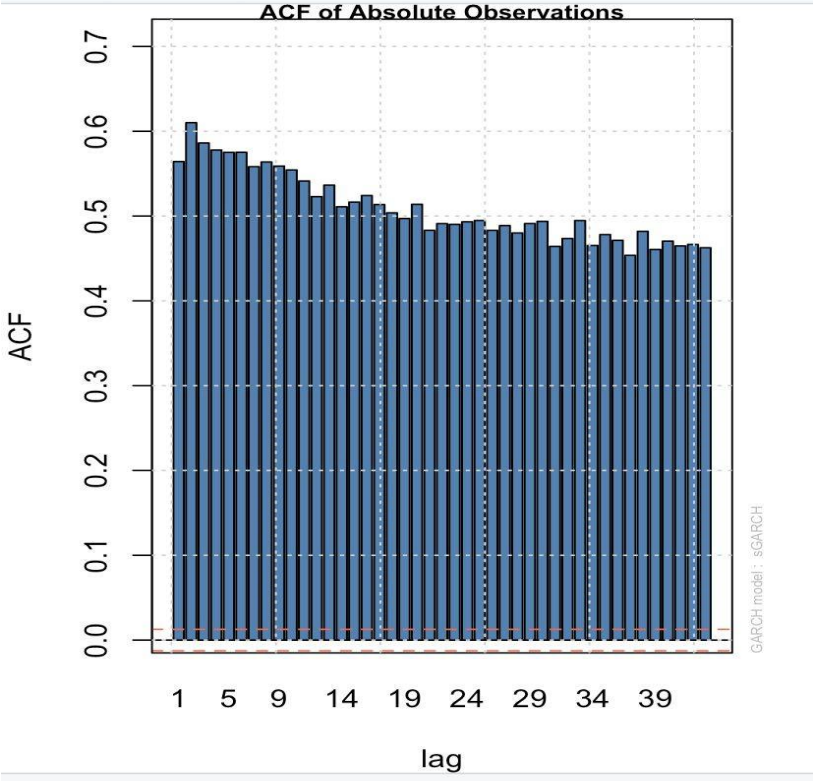
Autocorrelation Function of Squared Daily Bitcoin Returns

As outlined in the previous graph, the returns are not correlated but their squared values, which denotes variance show significant though declining autocorrelations. This graph reflects volatility clustering, which justifies the use of ARCH model (Autoregressive Conditional Heteroskedasticity) effects.



Autocorrelation Function of Absolute Daily Bitcoin Returns

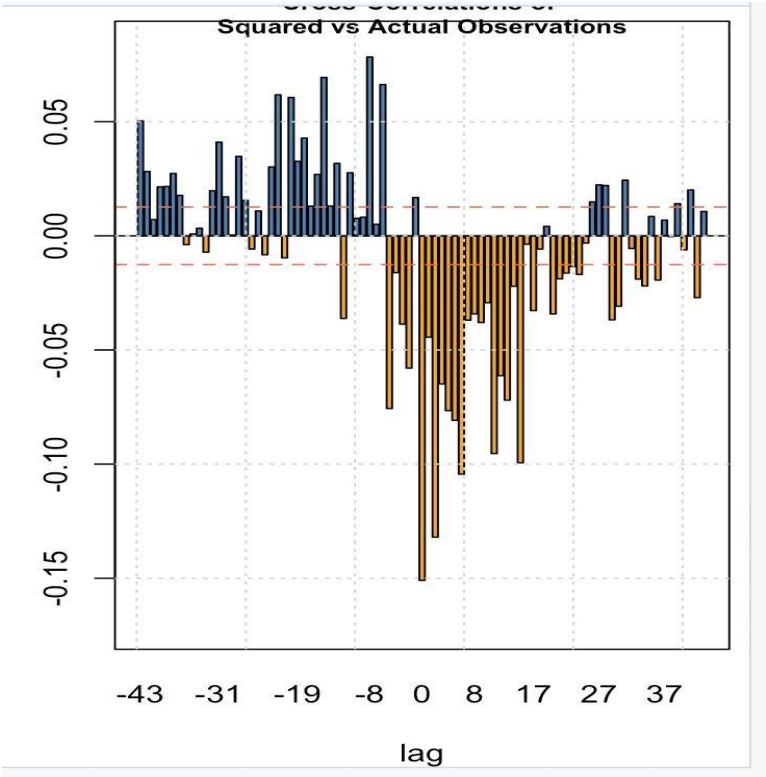
This graph uses absolute returns to measure autocorrelation. This graph also measures volatility, but the data outliers have less impact on the graph. The graph confirms the finding from the previous graph that there are significant autocorrelations which confirmed volatility clustering and thus validates the use of GARCH model.



Cross-Correlation Between Squared & Raw Returns to Test for Leverage Effects

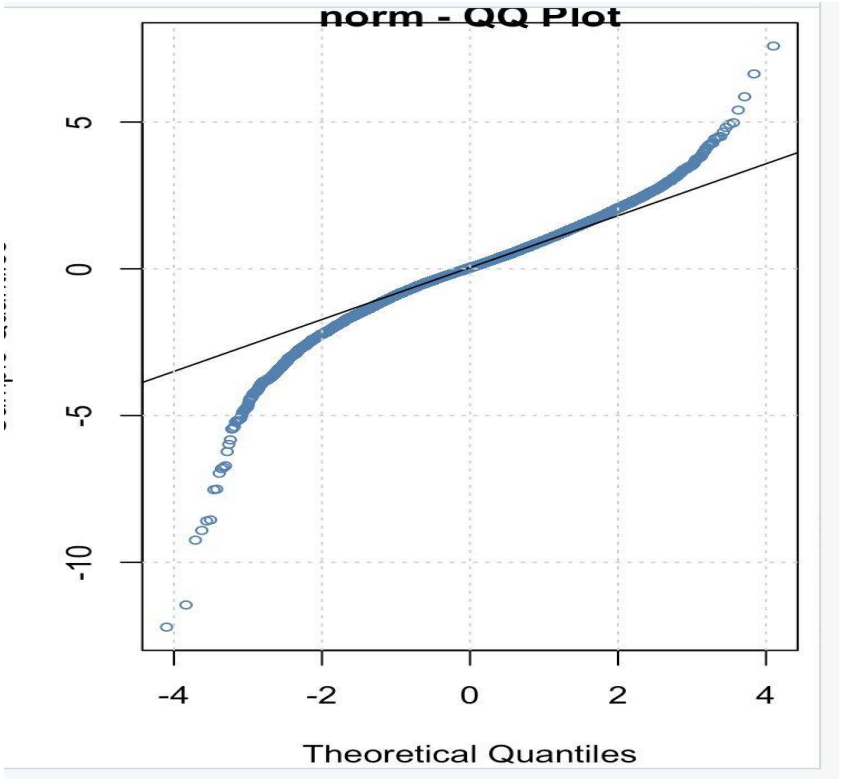
This graph tests for the leverage effect, where negative news has more impact on future volatility more than positive news.

The graph plots correlation between past returns and future squared returns (volatility). The pattern of the graph is asymmetric and given negative correlations for past returns implies a leverage effect.



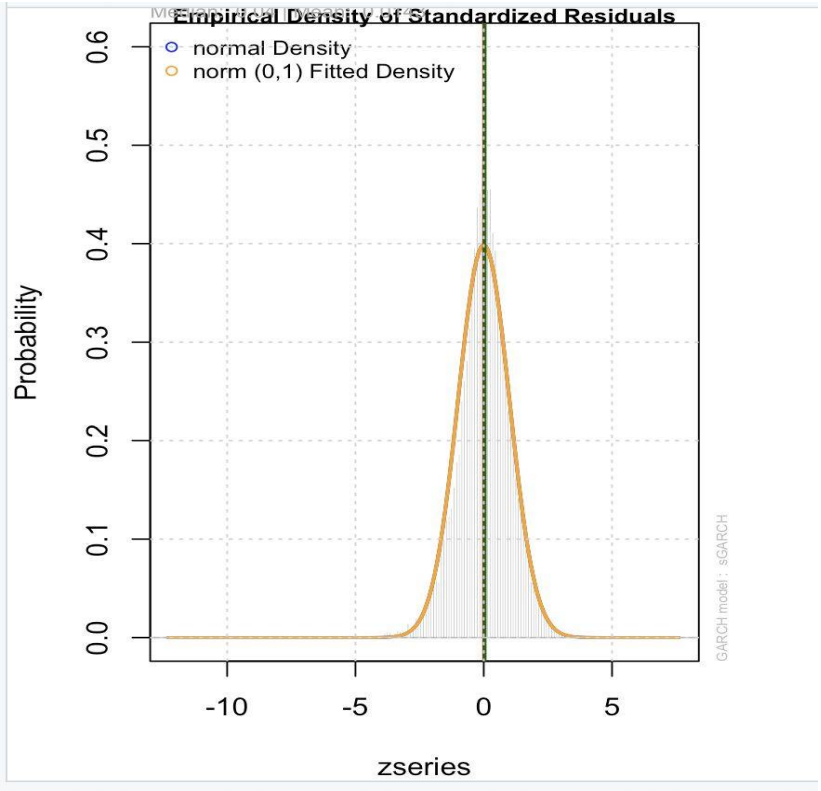
Normal Quantile-Quantile (Q-Q) Plot of Standardized GARCH Residuals

This graph analyzes the standardized GARCH residuals (the raw residuals divided by the estimated conditional volatility). This Q-Q plot compares to the Q-Q plot of a normal distribution. The "S" shape shows that the residuals are not normally distributed. The fat tails implies that extreme events are more likely than a normal distribution.



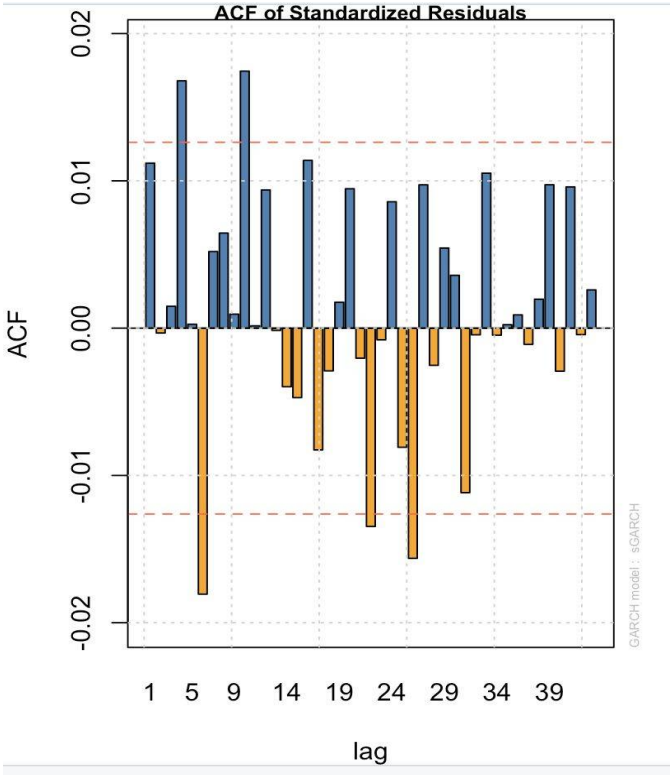
Empirical Density of Standardized GARCH Residuals

This graph confirms the finding from the previous Q-Q graph. It shows the probability distribution of the standardized residuals (in grey) compared to a normal distribution (yellow line). The graph shows that the distribution of residuals peaks in the center and has heavier tails.



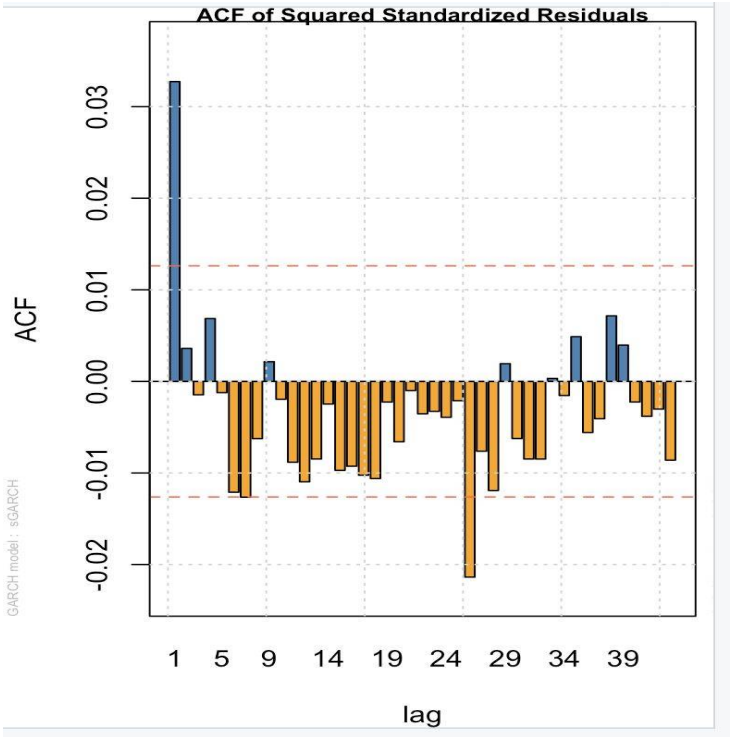
ACF of Standardized GARCH Residuals

This graph checks for the model's mean equation. As there are no major spikes, it indicates that our model has accounted for autocorrelation in the returns.



ACF of Squared Standardized GARCH Residuals

This graph checks for ARCH effects after fitting the model. The graph shows no major spikes in the ACF of the squared standardized residuals. It implies that our GARCH model has addressed the volatility clustering reflected in Bitcoin prices.



Introduction - EGARCH Model

The Exponential Generalized Autoregressive Conditional Heteroskedastic (EGARCH) model is another GARCH model variant, used for modeling and forecasting financial time series.

- EGARCH is more efficient and models volatility clustering better than GARCH model.
- The Weighted Ljung-Box Test on Standardized Squared Residuals shows very high p-values which means that any ARCH effects on the model are removed.
- The EGARCH model accounts for the leverage effect using the gamma coefficient.

Code for EGARCH Model

```
> rclose_egarch=ugarchspec(variance.model = list(model="eGARCH",garchOrder=c(1,1)),
odel = list(armaOrder=c(1,1)))
> rclose_egarch_fit=ugarchfit(rclose_egarch,data = rclose)
> rclose_egarch_fit
```

```
> install.packages("tseries")
trying URL 'https://cran.rstudio.com/bin/
8.tgz'
Content type 'application/x-gzip' length
==========================================
downloaded 414 KB


The downloaded binary packages are in
        /var/folders/06/sg_59qv509l9jnc8v
Loading required namespace: pomp
Loading required namespace: rugarch
> library(rugarch)
```

```
Loading required package: parallel
```

```
> library(FinTS)
```

```
> library(tseries)
```

```
Registered S3 method overwritten by '
  method            from
  as.zoo.data.frame zoo

    'tseries' version: 0.10-58

    'tseries' is a package for time s
    finance.

    See 'library(help="tseries")' for
```

```
> install.packages("tseries")
```

```
Error in install.packages : Updating
```

```
attach(BTC6)
```

```
),mean.model = list(armaOrder=c(1,1)))
```

EGARCH Model Fit - 1

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(1,1)
Mean Model       : ARFIMA(1,0,1)
Distribution     : norm
```

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001283	0.000186	6.8785	0.000000
ar1	0.341614	0.010827	31.5535	0.000000
ma1	-0.376709	0.023559	-15.9899	0.000000
omega	-0.423511	0.120848	-3.5045	0.000457
alpha1	-0.039468	0.025604	-1.5414	0.123210
beta1	0.933350	0.018181	51.3374	0.000000
gamma1	0.253200	0.040390	6.2688	0.000000

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001283	0.000276	4.6527	3.0e-06
ar1	0.341614	0.052932	6.4538	0.0e+00
ma1	-0.376709	0.052728	-7.1444	0.0e+00
omega	-0.423511	0.055805	-7.5891	0.0e+00
alpha1	-0.039468	0.009661	-4.0851	4.4e-05
beta1	0.933350	0.008361	111.6281	0.0e+00
gamma1	0.253200	0.020644	12.2653	0.0e+00

LogLikelihood : 7913.403

Information Criteria

Akaike	-3.9952
Bayes	-3.9840
Shibata	-3.9952
Hannan-Quinn	-3.9912

EGARCH Model Fit - 2

```
Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      17.28 3.228e-05
Lag[2*(p+q)+(p+q)-1][5]     24.28 0.000e+00
Lag[4*(p+q)+(p+q)-1][9]     29.36 2.248e-13
d.o.f=2
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.2537  0.6145
Lag[2*(p+q)+(p+q)-1][5]    2.1315  0.5877
Lag[4*(p+q)+(p+q)-1][9]    2.8389  0.7852
d.o.f=2
```

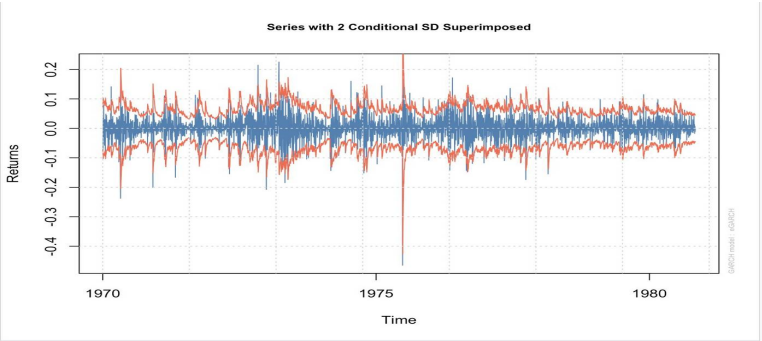
```
Nyblom stability test
------------------------------------
Joint Statistic:  1.7572
Individual Statistics:
mu      0.13980
ar1     0.07533
ma1     0.07358
omega   0.46587
alpha1  0.07906
beta1   0.45343
gamma1  0.24585
```

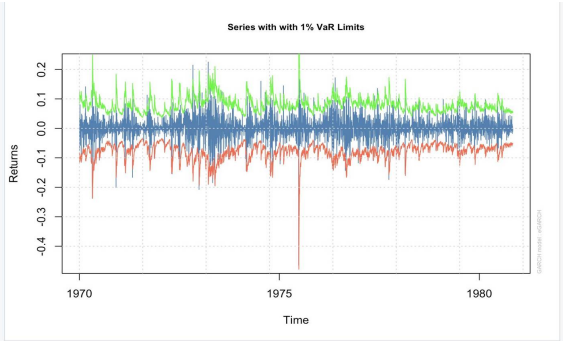
```
Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     578.1   2.058e-110
2    30     611.1   3.237e-110
3    40     619.1   1.669e-105
4    50     660.0   6.335e-108


Elapsed time : 0.2809451
```

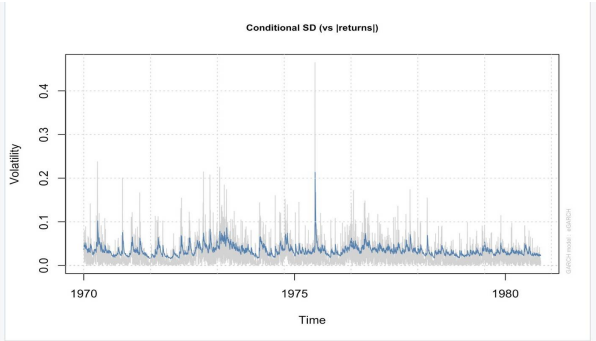
```
Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]     1.482 0.500 2.000  0.2234
ARCH Lag[5]     2.181 1.440 1.667  0.4327
ARCH Lag[7]     2.290 2.315 1.543  0.6560
```

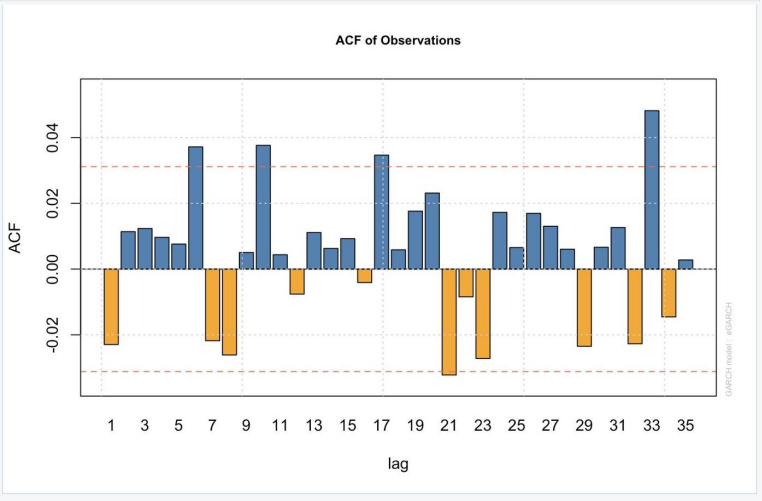
```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          1.69 1.9 2.35
Individual Statistic:     0.35 0.47 0.75

Sign Bias Test
------------------------------------
                    t-value   prob sig
Sign Bias           0.05410 0.9569
Negative Sign Bias  0.07256 0.9422
Positive Sign Bias  0.13952 0.8890
Joint Effect        0.06629 0.9955
```

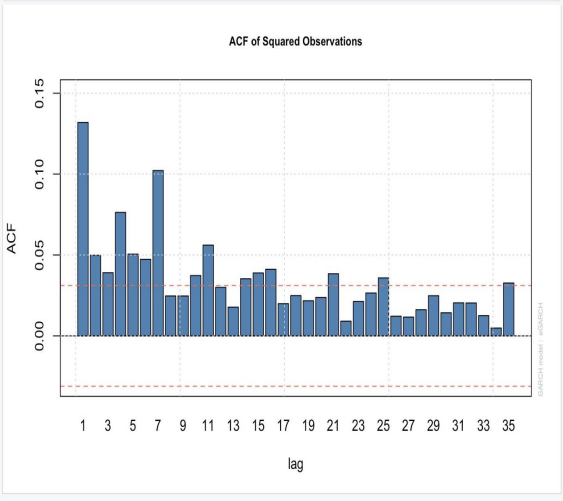
EGARCH Model Graphs - 1

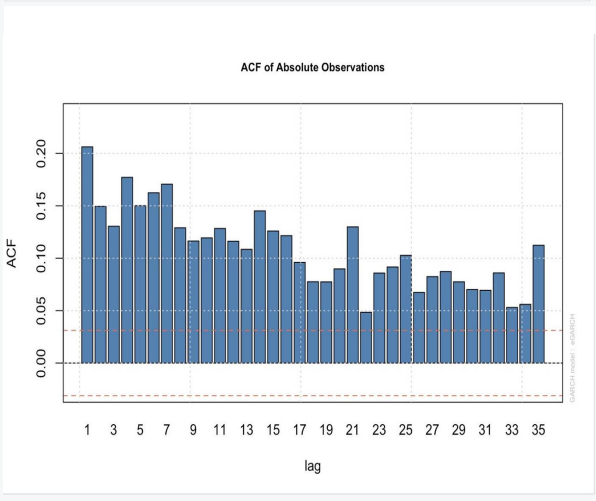




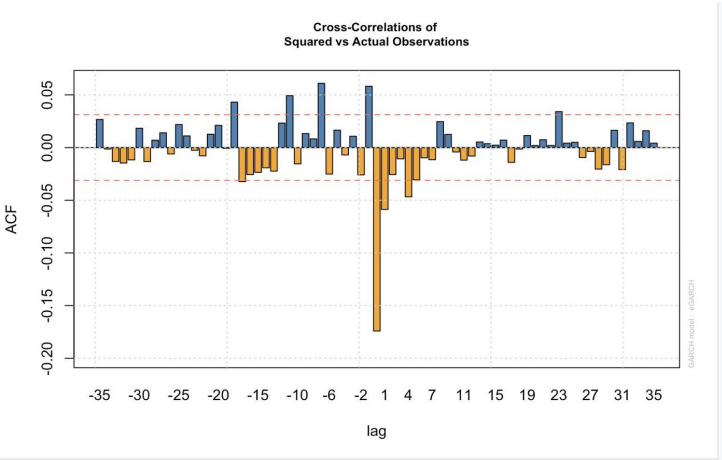


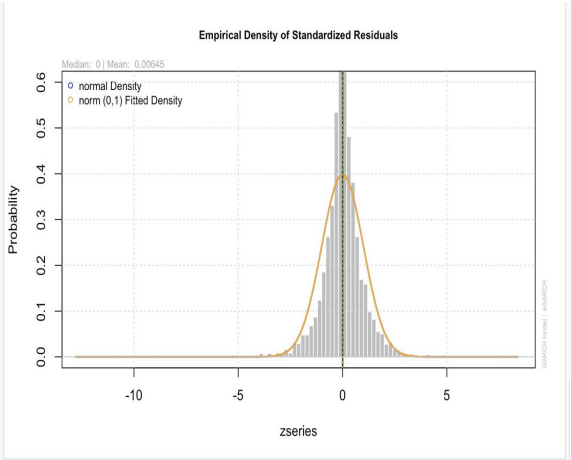


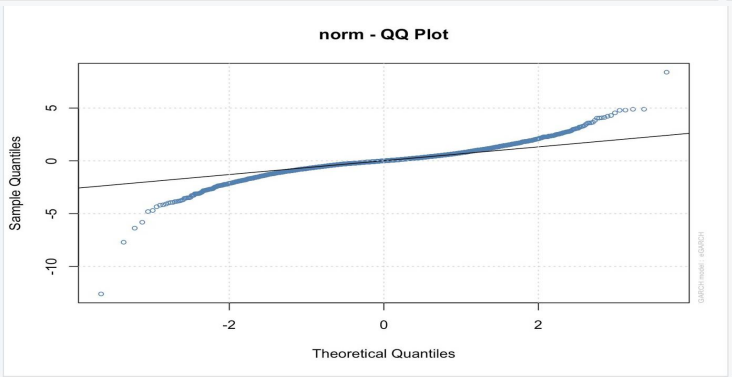


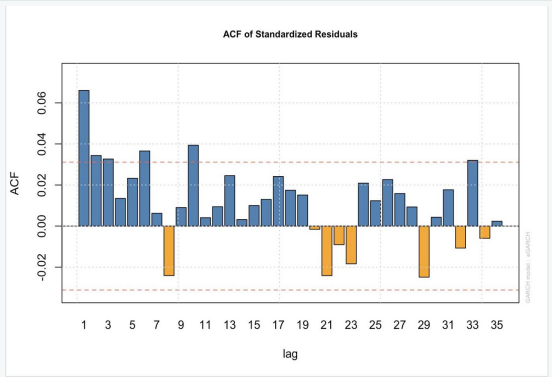


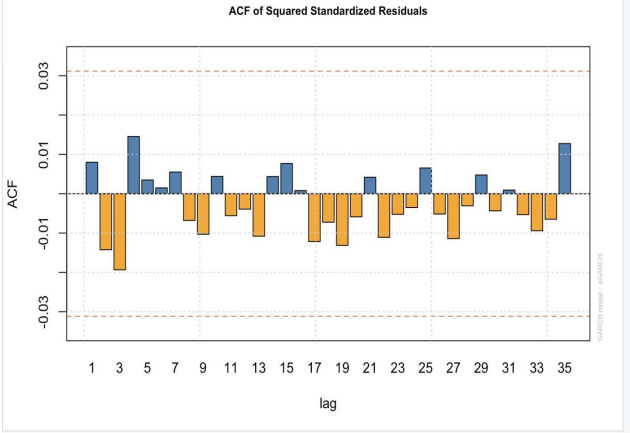
EGARCH Model Graphs - 2











Key Findings from EGARCH Model - 1

The EGARCH model is designed to capture the leverage effect..

Model Specification

- GARCH Model: `eGARCH(1,1)`: An Exponential GARCH model with one Autoregressive ARCH component (`alpha1`), one moving average GARCH component (`beta1`), and one leverage component (`gamma1`).
- Mean Model: `ARFIMA(1,0,1)`: An ARMA(1,1) model for the daily returns.
- Distribution: Assumes the standardized residuals follow a normal distribution.

Findings

- Mean Equation Parameters (`mu`, `ar1`, `ma1`): All three parameters in the mean equation are statistically significant (all p-values are 0.000000). This indicates linear dependence on past values is not present. Significant ma1 implies errors in model correlation and implies presence of moving average component.

Key Findings from EGARCH Model - 2

- Volatility: Alpha1 and beta1 measure conditional variance and indicate dependency on both past variance values and past errors to model volatility.
- Alpha1 measures the magnitude of a shock , is not statistically significant (-0.039468) which indicates lower volatility clustering. This implies that the size of a shock does not have a significant impact on volatility.
- Beta1 (persistence of shocks in log-variance) is statistically significant (0.933350) imply that volatility is persistent and past shocks have a positive impact on conditional variance.

Other Variance Equation Parameters:

- omega: The constant term in the log-variance equation, which is significant. It implies inherent variance in data even without past values.
- gamma1: This is a asymmetry or leverage term. It is statistically significant (p-value = 0.000000). The positive value (0.253200) suggests positive news has a bigger impact on volatility than negative news. This is the inverse of the standard leverage effect seen in equity markets.

Key Findings from EGARCH Model - 3

Variance Equation Parameters:

- Log Likelihood: The log-likelihood value is 7913.403, indicating that the GARCH model has a good overall fit to the observed data. However, it is important to note that log Likelihood only provides a relative measure of fit and cannot be used in isolation to assess model quality.

Tests for Remaining Patterns in Residuals

- Weighted Ljung-Box Test on Standardized Residuals: This test checks for remaining autocorrelation in the model's errors. The p-values are close to zero. Null hypothesis of no serial correlation is rejected. The mean equation coefficients are individually significant but the model fails to capture all of the linear dependence in the returns. The mean model is misspecified and there is significant autocorrelation.

Key Findings from EGARCH Model - 4

Weighted Ljung-Box Test and ARCH LM Tests on Squared Residuals: Both of these tests check for remaining ARCH effects. In both cases, the p-values are large (e.g., Ljung-Box p-value at lag 1 is 0.6145; ARCH LM p-value at lag 3 is 0.2234). Null hypothesis is not rejected. The EGARCH model has successfully accounted for the volatility clustering in the data. This implies that there is no evidence of remaining autocorrelation in the residuals and model **captures all autocorrelation in the residuals** .

Nyblom Stability Test: This tests for parameter stability over time. The Joint Statistic (1.7572) is between the 10% (1.69) and 5% (1.9) critical values. This borderline result suggests potential parameter instability, which could be a weakness of the model.

Sign Bias Test: Tests for "leverage effects," where negative returns have a different impact on volatility than positive returns. The EGARCH model assumes the effect is symmetric.
The null hypothesis assumes that there is no sign bias. This test checks if any uncaptured asymmetric effects remain. The p-values for all terms are large (e.g., Joint Effect p-value = 0.9955) and null hypothesis is not rejected. This confirms that the EGARCH model has captured the asymmetry in the data.

Adjusted Pearson Goodness-of-Fit Test: This tests if the residuals conform to the assumed normal distribution. The p-values are close to zero. Null hypothesis is not rejected. This is a clear indication that the normality assumption is incorrect.

Summary - EGARCH Model

The EGARCH model is successful in capturing the volatility clustering and the asymmetric volatility to shocks.

The model identifies a inverse leverage effect. The gamma coefficient produced was highly effective. The positive value of gamma suggests positive news has a bigger impact on volatility than negative news. This is the inverse of the standard leverage effect seen in equity markets, a key feature that makes Bitcoin from traditional assets

The results from the Sign Bias Test show that the model was able to account for any asymmetry within the model. This confirms that the EGARCH model has captured the asymmetry in the data.

Weaknesses:

The ARMA(1,1) mean model is misspecified, leaving behind predictable patterns in the residuals.

The assumption that the model is normal is incorrect. It mean that the model will underestimate the probability of extreme events.

There is evidence of potential parameter instability

Introduction - GJR-GARCH Model

The Glosten-Jagannathan-Runkle Generalized Autoregressive Conditional Heteroskedasticity (GJR-GARCH) model is for financial time series analysis. A key feature of GJR-GARCH is modeling for "leverage effect," where negative shocks tend to increase volatility more than positive shocks.

- The GJR-GARCH model was effective in capturing leverage effect like EGARCH model.
- The Sign Bias Test shows there was no sign bias and that GJR-GARCH model also provided the good fit.
- Gamma1 is significant; calculated value > table t-value. This implies that null hypothesis is wrong and there is a leverage effect. Negative shocks have bigger effect on volatility than positive ones.
- Model showed that asymmetry exists as gamma1 term was positive and more significant. The negative returns have a much larger impact on the Bitcoin’s future volatility in comparison to more positive returns.
- GJR-GARCH was more effective in volatility clustering based on results from Weighted Ljung-Box Test on Standardized Squared Residuals and Weighted ARCH LM Tests. Both produced high p-values which shows us that no GARCH effects are left over after fitting the model.

Code for GJR-GARCH(1,1) Model Fit

```
),mean.model = list(armaOrder=c(1,1)))
```

```
> rclose_gjrgarch=ugarchspec(variance.model = list(model="gjrGARCH",garchOrder=c(1,1)),
an.model = list(armaOrder=c(1,1)))
> rclose_gjrgarch_fit=ugarchfit(rclose_gjrgarch,data = rclose)
> rclose_gjrgarch_fit

Loading required namespace: pomp
> BTC6 <- read.csv("~/POMP Model for AMZN Stock/AMZN_Stock/BTC6.csv")
>   View(BTC6)
> attach(BTC6)
> rclose=diff(log(Close))
> library(rugarch)
```

GJR-GARCH(1,1) Model Fit - 1

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(1,1)
Mean Model       : ARFIMA(1,0,1)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001416	0.000474	2.98666	0.002820
ar1	-0.401980	0.513011	-0.78357	0.433293
ma1	0.389821	0.515973	0.75551	0.449945
omega	0.000062	0.000008	7.41914	0.000000
alpha1	0.099814	0.012387	8.05782	0.000000
beta1	0.834956	0.014283	58.45599	0.000000
gamma1	0.054120	0.015786	3.42833	0.000607

Robust Standard Errors:

	Estimate	Std. Error
mu	0.001416	0.000486
ar1	-0.401980	0.219594
ma1	0.389821	0.224147
omega	0.000062	0.000020
alpha1	0.099814	0.022741
beta1	0.834956	0.029203
gamma1	0.054120	0.045542

LogLikelihood : 7903.851

Information Criteria

Akaike	-3.9903
Bayes	-3.9792
Shibata	-3.9903
Hannan-Quinn	-3.9864

Weighted Ljung-Box Test on Standardized Squared Resi

	statistic	p-value
Lag[1]	0.1415	0.7068
Lag[2*(p+q)+(p+q)-1][5]	2.5328	0.4990
Lag[4*(p+q)+(p+q)-1][9]	3.4075	0.6902

d.o.f=2

Weighted Ljung-Box Test on Standardized Residual

	statistic	p-value
Lag[1]	9.056	2.618e-03
Lag[2*(p+q)+(p+q)-1][5]	12.841	0.000e+00
Lag[4*(p+q)+(p+q)-1][9]	16.922	3.696e-06

d.o.f=2

H0 : No serial correlation

GJR-GARCH(1,1) Model Fit - 2

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[3]	1.789	0.500	2.000	0.1811
ARCH Lag[5]	2.817	1.440	1.667	0.3174
ARCH Lag[7]	2.909	2.315	1.543	0.5310

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic: 1.69 1.9 2.35
Individual Statistic: 0.35 0.47 0.75

Sign Bias Test

	t-value	prob	sig
Sign Bias	0.14933	0.8813	
Negative Sign Bias	0.09391	0.9252	
Positive Sign Bias	0.32935	0.7419	
Joint Effect	0.24517	0.9700	

Nyblom stability test

Joint Statistic: 1.7539
Individual Statistics:

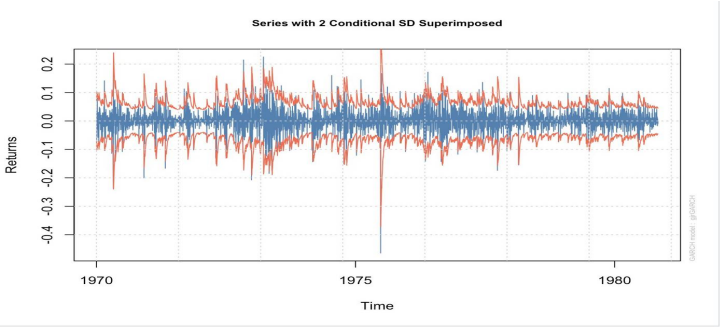
mu	0.08391
ar1	0.28973
ma1	0.28767
omega	0.36476
alpha1	0.35173
beta1	0.36049
gamma1	0.11536

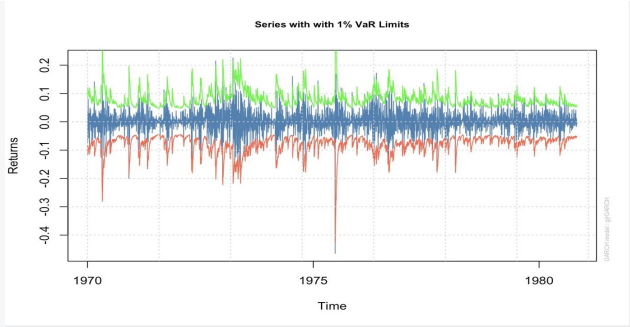
Adjusted Pearson Goodness-of-Fit Test:

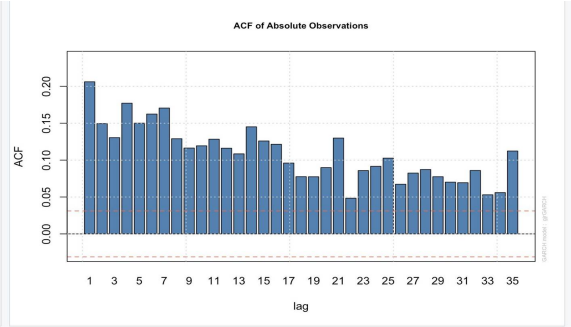
	group	statistic	p-value(g-1)
1	20	582.6	2.389e-111
2	30	610.4	4.452e-110
3	40	626.6	5.045e-107
4	50	645.4	5.628e-105

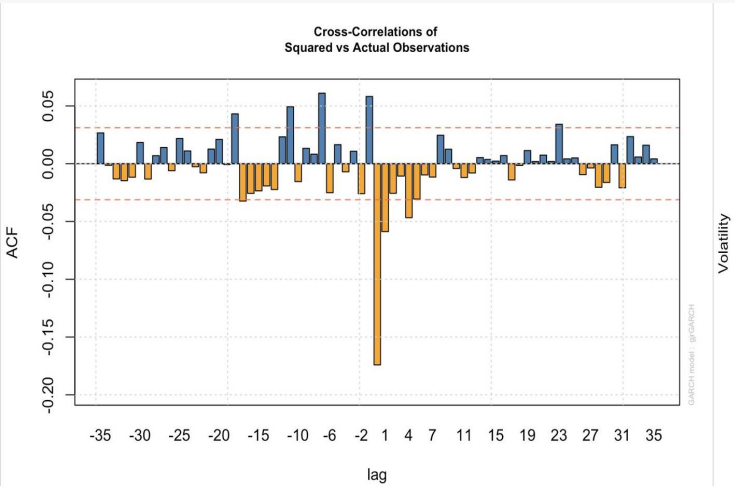
Elapsed time : 0.8642571

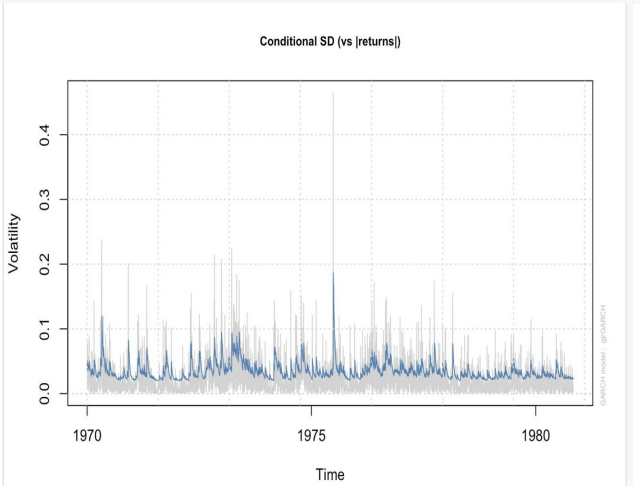
GJR-GARCH(1,1) Model Graphs - 1

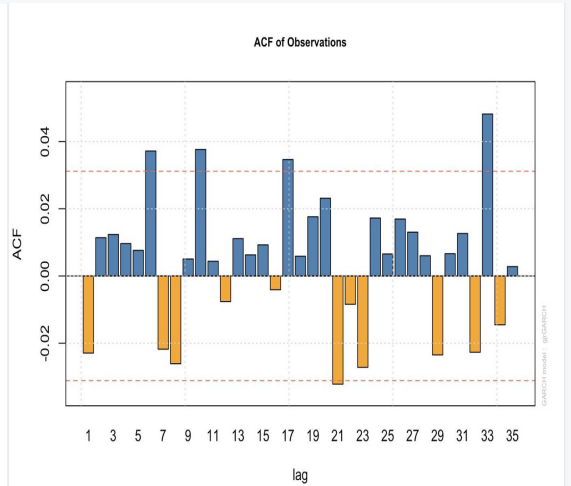




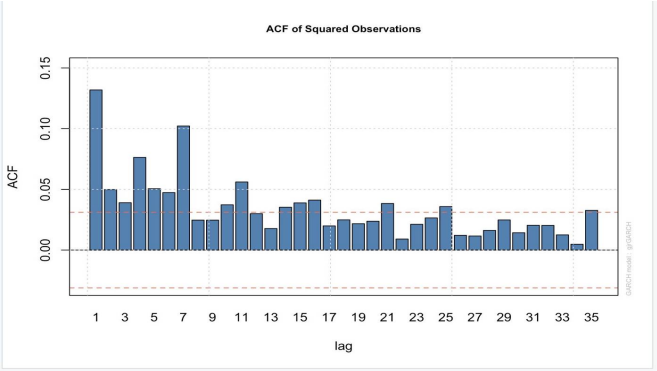


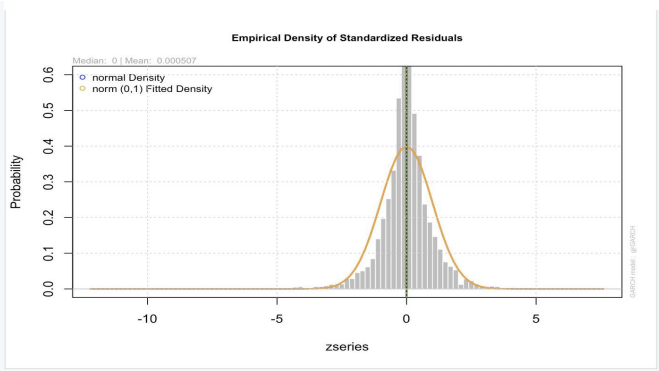


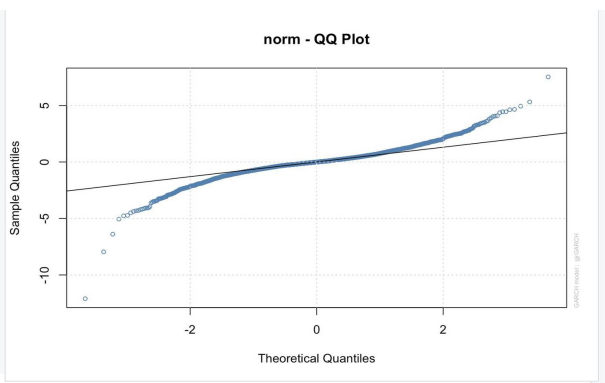


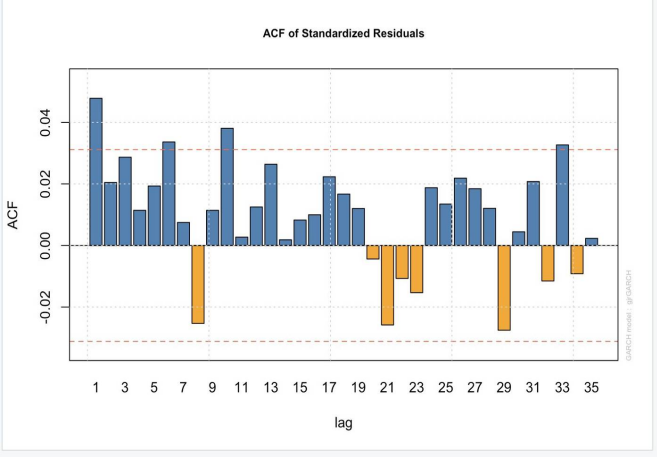


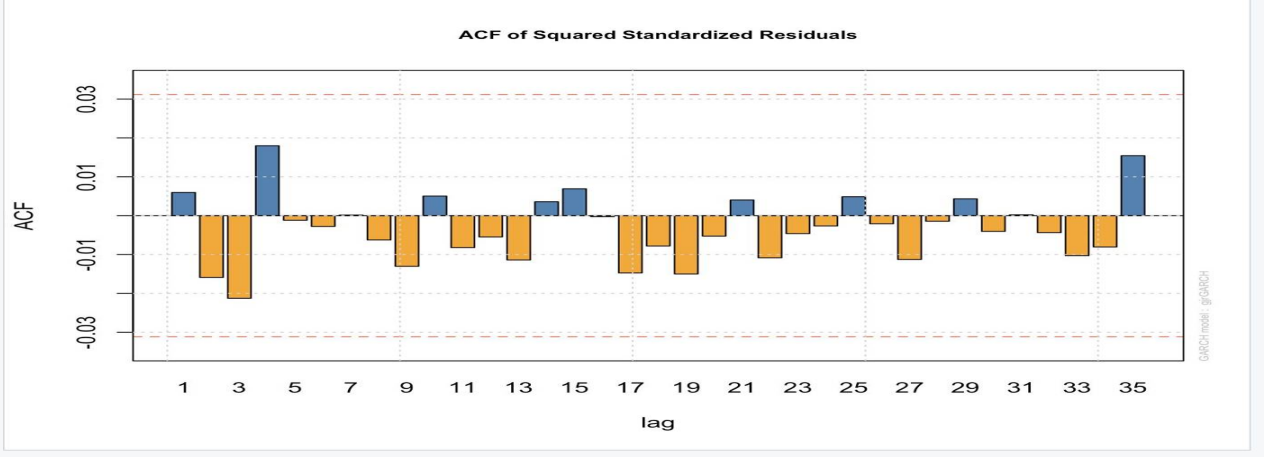
GJR-GARCH(1,1) Model Graphs - 2











Key Findings from GJR-GARCH(1,1) Model - 1

GJR-GARCH model, extension of the standard GARCH model for capturing leverage effect.

Model Specification

- GARCH Model: `gjr GARCH(1,1)`: GJR-GARCH model with one Autoregressive ARCH component (`alpha1`), one moving average GARCH component (`beta1`), and one leverage component (`gamma1`).
- Mean Model: `ARFIMA(1,0,1)`: An ARMA(1,1) model for the daily returns. Models the current return based on the previous day's return (ar1) and the previous day's forecast error (ma1). (ar1) and (ar2) are both used to help show how yesterday's return help predict today's return.
- Distribution: Assumes the standardized residuals follow a normal distribution.

Findings

Mean Equation Parameters (`mu`, `ar1`, `ma1`): Mean of time series (mu) is statistically significant (p-value = 0.000000). Autoregressive coefficient (ar1) is statistically significant (p-value =0.433293).This indicates linear dependence on past values is present. Significant moving average coefficient (ma1) implies errors in model correlation and implies presence of moving average component. .

Key Findings from GJR-GARCH(1,1) Model - 2

- Volatility: Alpha1 and beta1 measure conditional variance and indicate dependency on both past variance values and past errors to model volatility.
- Alpha1 measures the magnitude of a shock, is statistically significant (0.99814) which indicates high volatility clustering. This implies that the size of shock does have a significant impact on volatility.
- Beta1 (persistence of shocks in log-variance) is statistically significant (0.834956) and implies that volatility is persistent and past shocks have a positive impact on conditional variance.

Other Variance Equation Parameters:

- omega:Constant term in the log-variance equation is significant. It implies inherent variance in data even without past values.
- Gamma1 (captures the asymmetric effect of past shocks on current volatility): This is a asymmetry or leverage term. It is 0.054120 which indicates that evidence of leverage effect is weak. The implies that the impact is almost neutral for positive and negative shock.

Key Findings from GJR-GARCH(1,1) Model - 3

Weighted Ljung-Box Test on Standardized Residuals: This test checks for any remaining serial correlation in the model's residuals. This test checks for remaining autocorrelation in the model's errors. The p-values are close to zero. We reject the null hypothesis of no serial correlation. The mean equation coefficients are individually significant but the model fails to capture all of the linear dependence in the returns. The mean model is misspecified and there is significant autocorrelation

Weighted Ljung-Box Test on Standardized *Squared* Residuals: This is a crucial test to see if the GARCH model successfully captures the volatility clustering. The p-values are all large. Null hypothesis is not rejected and indicates the GJR-GARCH model has accounted the volatility clustering.

Weighted ARCH LM Tests: This is an alternative test for remaining ARCH effects. The p-values are all large and confirms that: the GARCH component of the model is well-specified.

Sign Bias Test: Tests for "leverage effects," where negative returns have a different impact on volatility than positive returns. GJR-GARCH model captures these leverage effects given it is asymmetric. This means that negative shocks have a greater impact on volatility then positive shocks.

The null hypothesis of the Sign Bias Test assumes that there are no asymmetric effects. P-values for negative and positive sign bias and joint effect are all greater than 0.05. This means the model fails to reject the null hypothesis which means that the model was captured the asymmetry in the data.

Key Findings from GJR-GARCH(1,1) Model - 4

- **Nyblom Stability Test:** This tests for parameter stability over time. The Joint Statistic (1.7539) is between the 10% (1.69) and 5% (1.9) critical values. This borderline result suggests potential parameter instability, which could be a weakness of the model.
-
- **Adjusted Pearson Goodness-of-Fit Test:**
- This tests if the residuals conform to the assumed normal distribution. The p-values are close to zero. Null hypothesis is not rejected. This is a clear indication that the normality assumption is incorrect

Summary - GJR-GARCH(1,1) Model

The GJR-GARCH model shows mixed but generally positive results.

- Strengths:
  - It successfully captures the core volatility clustering (GARCH effects).
  - The Sign Bias Test confirms it has effectively modeled the leverage effect, even though the significance of gamma coefficient is marginal.
  - Parameters are borderline stable in the data which suggest model's relationships may change over time..
- Weaknesses:
  - The ARMA(1,1) mean model is misspecified and fails to capture the linear dynamics of the returns.
  - The assumption that the model is normal is incorrect. It mean that the model will underestimate the probability of extreme events.

Comparison of GARCH Models

Based on the screenshot below, the EGARCH model was the most statistically significant mean model as it had the lowest AIC value and it was able to capture GARCH and leverage effects. The standard GARCH model was good baseline model but it was unable to account for the leverage effects seen in the bitcoin data. GJR-GARCH model was good as it was able to model both GARCH and leverage effects but the AIC value was higher than EGARCH model.

Metric	sGARCH(1,1) Model	GJR-GARCH(1,1) Model	EGARCH(1,1) Model
Log-Likelihood	-40868.21	7903.851	7913.403
Akaike (AIC)	3.3885	-3.9903	-3.9952
Mean Model Significant?	Partially (MA1 only)	No	Yes
Captures GARCH Effects?	No (Failed Ljung-Box)	Yes	Yes
Captures Leverage Effects?	No (Failed Sign Bias)	Yes	Yes
Residuals Normal?	No	No	No
Parameters Stable?	No (Failed Nyblom)	Borderline	Borderline

Information Criteria

Information criteria, such as the Akaike Criterion (AIC), Bayes Criterion (BIC), Shibata Criterion, and Hannan-Quinn Criterion, provide measures of the quality of model fit.

Lower values for these criteria indicate a better fit.

I tested for AIC values and summarized findings on the next screen.

Based on the lower value as a criteria, the fitted GARCH model appears to have a reasonably good fit, although the values are not significantly different.

Summary - AIC Table & Log Likelihood from EGARCH models

	q = 1	q = 2	q = 3	q = 4
p = 0	-3.8135	-3.8126	-3.8118	-3.8109
p = 1	-3.9952	-3.9944	-3.9951	-4.0037
p = 2	-3.9946	-3.9936	-4.0023	-4.0028
p = 3	-3.9968	-3.9995	-4.0020	-4.0028
p = 4	-4.0037	-4.0041	-4.0052	-4.0064

	q = 1	q = 2	q = 3	q = 4
p = 0	7550.899	7551.178	7551.469	7551.673
p = 1	7913.403	7914.889	7919.398	7939.421
p = 2	7914.378	7915.361	7934.476	7937.614
p = 3	7920.650	7930.06	7937.014	7940.589
p = 4	7936.225	7942.212	7946.306	7950.592

Summary

- We can see the AIC values and maximized log likelihood of EGARCH(1,1).
- Lowest AIC value is for EGARCH(4,4), which shows that EGARCH(4,4) performs well as a popular model.
- EGARCH(1,1) has less fitted parameters than EGARCH(4,4).
- EGARCH(4,3) and EGARCH(3,4) have largest log likelihood, given that they have more parameters. Given lower AIC values are a better fit, EGARCH(4,4) has the lowest AIC value.
- EGARCH(4,4) has the largest log likelihood, possibly due more parameters and shows the best model fit under that also.

Summary - AIC Table & Log Likelihood from GJR-GARCH models

AIC Value Table

	q = 1	q = 2	q = 3	q = 4
p = 0	-3.8135		-3.8118	-3.8146
p = 1	-3.9903	-3.9912	-3.9992	-3.9987
p = 2	-3.9898	-3.9927	-4.0001	-3.9924
p = 3	-3.9789	-3.9921	-4.0002	-3.9984
p = 4	-3.9975	-4.0059	-4.0022	-4.0077

Log likelihood

	q = 1	q = 2	q = 3	q = 4
p = 0	7550.942	7551.15	7551.499	7559.132
p = 1	7903.851	7907.645	7925.436	7926.488
p = 2	7905.729	7913.511	7930.166	7917.026
p = 3	7887.194	7915.443	7933.372	7931.843
p = 4	7927.026	7945.641	7940.344	7953.188

Summary

- The maximized log likelihood of ARCH(4) and GARCH(1,1) are close to each other.
- AIC favors GARCH(1,1), which shows that GARCH(1,1) performs well as a popular model.
- Given lower AIC values are better fit, GARCH(4,4) has the lowest AIC value.
- GARCH(4,4) has the largest log likelihood, possibly due more parameters.
- GARCH(2,3) has the largest log likelihood for p+q<=5.
- The red highlighted number have the lowest values in the Log likelihood table under different GARCH model.

Conclusion

I applied GARCH, EGARCH, GJR-GARCH to analyze the price movement of Bitcoin. Given all the models, the EGARCH model was the best fit for modeling Bitcoin price movement. Despite various weaknesses, the EGARCH model improves upon the shortcomings of other GARCH models such as the GJR-GARCH and the standard GARCH model. EGARCH(1,1) has lowest AIC and is well-specified among the various models for values of p=1 and q=1.

Given lower AIC values are better fit, among EGARCH(4,4) has the lowest AIC value and EGARCH(4,4) also has the highest log-likelihood values as it has more parameters and implies a good model fit.

For p+q values less than 5, EGARCH(4,1) has the lowest AIC among the GARCH models and also has the highest log likelihood values.

Appendix

EGARCH(0,1) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(0,1)
Mean Model       : ARFIMA(0,0,1)
Distribution     : norm

Information Criteria
------------------------------------

Akaike       -3.8135
Bayes        -3.8071
Shibata      -3.8135
Hannan-Quinn -3.8112

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                  0.0002859  0.9865
Lag[2*(p+q)+(p+q)-1][2] 0.2705242  0.9961
Lag[4*(p+q)+(p+q)-1][5] 1.0468638  0.9396
d.o.f=1
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      60.53 7.216e-15
Lag[2*(p+q)+(p+q)-1][2]     65.19 0.000e+00
Lag[4*(p+q)+(p+q)-1][5]     82.71 0.000e+00
d.o.f=1
```

```
Optimal Parameters
------------------------------------
       Estimate  Std. Error   t value Pr(>|t|)
mu     0.001402    0.000558    2.5134 0.011956
ma1   -0.022424    0.015691   -1.4291 0.152983
omega -0.658671    0.005092 -129.3418 0.000000
beta1  0.901030    0.000697 1292.8938 0.000000

Robust Standard Errors:
       Estimate  Std. Error    t value Pr(>|t|)
mu     0.001402    0.000597     2.3496 0.018793
ma1   -0.022424    0.020757    -1.0803 0.279989
omega -0.658671    0.000369 -1783.5190 0.000000
beta1  0.901030    0.001068   843.4928 0.000000

LogLikelihood : 7550.899
```

```
Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale   P-Value
ARCH Lag[2]     9.312 0.500 2.000 2.277e-03
ARCH Lag[4]    25.293 1.397 1.611 5.230e-07
ARCH Lag[6]    37.382 2.222 1.500 7.198e-10

Nyblom stability test
------------------------------------
Joint Statistic:  2.1617
Individual Statistics:
mu     0.09481
ma1    0.09901
omega 1.83431
beta1 1.83431

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         1.07 1.24 1.6
Individual Statistic:    0.35 0.47 0.75
```

```
Sign Bias Test
------------------------------------
                   t-value      prob sig
Sign Bias            0.828 4.077e-01
Negative Sign Bias   7.603 3.585e-14 ***
Positive Sign Bias   3.119 1.829e-03 ***
Joint Effect        69.038 6.859e-15 ***


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     881.6   9.558e-175
2    30     915.1   7.313e-174
3    40     945.1   6.754e-173
4    50     954.1   4.925e-168


Elapsed time : 0.201993
```

EGARCH(0,2) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(0,2)
Mean Model       : ARFIMA(0,0,2)
Distribution     : norm

Optimal Parameters
------------------------------------
       Estimate  Std. Error     t value Pr(>|t|)
mu     0.001402    0.000564     2.48481 0.012962
ma1   -0.023001    0.015906    -1.44607 0.148157
ma2    0.011767    0.015768     0.74624 0.455522
omega -0.657185    0.005392  -121.89202 0.000000
beta1  0.452064    0.000024 18900.52476 0.000000
beta2  0.449190    0.000026 17357.66317 0.000000

Robust Standard Errors:
       Estimate  Std. Error     t value Pr(>|t|)
mu     0.001402    0.000594     2.35946 0.018302
ma1   -0.023001    0.021601    -1.06482 0.286957
ma2    0.011767    0.019680     0.59789 0.549910
omega -0.657185    0.008220   -79.95283 0.000000
beta1  0.452064    0.000013 36095.22570 0.000000
beta2  0.449190    0.000017 27158.45050 0.000000

LogLikelihood : 7551.178
```

```
Information Criteria
------------------------------------

Akaike       -3.8126
Bayes        -3.8031
Shibata      -3.8126
Hannan-Quinn -3.8092

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                  0.0001014  0.9920
Lag[2*(p+q)+(p+q)-1][5] 0.5807087  1.0000
Lag[4*(p+q)+(p+q)-1][9] 4.6231307  0.5417
d.o.f=2
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      61.00 5.662e-15
Lag[2*(p+q)+(p+q)-1][5]     82.76 0.000e+00
Lag[4*(p+q)+(p+q)-1][9]    111.85 0.000e+00
d.o.f=2
```

```
Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale   P-Value
ARCH Lag[3]     5.736 0.500 2.000 1.662e-02
ARCH Lag[5]    30.032 1.440 1.667 6.160e-08
ARCH Lag[7]    54.129 2.315 1.543 5.496e-14

Nyblom stability test
------------------------------------
Joint Statistic:  4.9936
Individual Statistics:
mu     0.09263
ma1    0.
ma2    0.
omega  1.
beta1  1.
beta2  1.

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```

```
Sign Bias Test
------------------------------------
                     t-value      prob sig
Sign Bias              0.762 4.461e-01
Negative Sign Bias     7.614 3.313e-14 ***
Positive Sign Bias     3.152 1.634e-03 ***
Joint Effect          69.595 5.213e-15 ***

Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     874.7   2.806e-173
2    30     914.1   1.162e-173
3    40     937.5   2.526e-171
4    50     961.6   1.373e-169


Elapsed time : 0.1190941
```

EGARCH(0,3) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(0,3)
Mean Model       : ARFIMA(0,0,3)
Distribution     : norm


Sign Bias Test
------------------------------------
                   t-value      prob sig
Sign Bias            2.640 8.313e-03 ***
Negative Sign Bias   8.419 5.237e-17 ***
Positive Sign Bias   2.246 2.475e-02  **
Joint Effect        75.944 2.274e-16 ***


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     879.2   3.063e-174
2    30     910.4   6.843e-173
3    40     944.4   9.215e-173
4    50     966.5   1.350e-170


Elapsed time : 0.202441
```

```
Optimal Parameters
------------------------------------
       Estimate  Std. Error   t value Pr(>|t|)
mu     0.001402    0.000571   2.45679 0.014018
ma1   -0.023065    0.015897  -1.45087 0.146818
ma2    0.011235    0.015752   0.71322 0.475712
ma3    0.011725    0.015357   0.76348 0.445179
omega -0.656963    0.017086 -38.45045 0.000000
beta1  0.303319    0.001385 218.94690 0.000000
beta2  0.300481    0.001473 203.92545 0.000000
beta3  0.297490    0.001558 190.91356 0.000000

Robust Standard Errors:
       Estimate  Std. Error     t value Pr(>|t|)
mu     0.001402    0.000595  2.3549e+00 0.018528
ma1   -0.023065    0.021568 -1.0694e+00 0.284886
ma2    0.011235    0.019798  5.6747e-01 0.570393
ma3    0.011725    0.017703  6.6231e-01 0.507771
omega -0.656963    0.008242 -7.9710e+01 0.000000
beta1  0.303319    0.000001  5.0758e+05 0.000000
beta2  0.300481    0.000000  1.0754e+06 0.000000
beta3  0.297490    0.000000  2.4195e+06 0.000000

LogLikelihood : 7551.469
```

```
Information Criteria
------------------------------------

Akaike       -3.8118
Bayes        -3.7991
Shibata      -3.8118
Hannan-Quinn -3.8073

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                  5.097e-05  0.9943
Lag[2*(p+q)+(p+q)-1][8] 3.310e+00  0.9821
Lag[4*(p+q)+(p+q)-1][14] 8.953e+00  0.2275
d.o.f=3
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residu
------------------------------------
                        statistic  p-value
Lag[1]                       61.1 5.44e-15
Lag[2*(p+q)+(p+q)-1][8]     106.6 0.00e+00
Lag[4*(p+q)+(p+q)-1][14]    137.4 0.00e+00
d.o.f=3
```

```
Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale   P-Value
ARCH Lag[4]     23.53 0.500 2.000 1.231e-06
ARCH Lag[6]     37.50 1.461 1.711 1.454e-09
ARCH Lag[8]     64.69 2.368 1.583 2.220e-16

Nyblom stability test
------------------------------------
Joint Statistic:  13.7409
Individual Statistics:
mu    0.09062
ma1   0.10219
ma2   0.36958
ma3   0.08559
omega 1.84295
beta1 1.84295
beta2 1.84295
beta3 1.84295

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         1.89 2.11 2.59
Individual Statistic:    0.35 0.47 0.75
```

EGARCH(0,4) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
                                                  -----
Information Criteria
-------------------------------------

Akaike       -3.8109                                   )
Bayes        -3.7950
Shibata      -3.8109
Hannan-Quinn -3.8052

Weighted Ljung-Box Test on Standardized Residuals
-------------------------------------
                        statistic p-value
Lag[1]                  4.328e-05  0.9948
Lag[2*(p+q)+(p+q)-1][11] 6.165e+00  0.3804
Lag[4*(p+q)+(p+q)-1][19] 1.174e+01  0.2210
d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
-------------------------------------
                        statistic   p-value
Lag[1]                      61.73 3.997e-15
Lag[2*(p+q)+(p+q)-1][11]    124.19 0.000e+00
Lag[4*(p+q)+(p+q)-1][19]    155.07 0.000e+00
d.o.f=4
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001402	0.000576	2.43196	0.015017
ma1	-0.023168	0.015901	-1.45703	0.145107
ma2	0.010594	0.015932	0.66497	0.506071
ma3	0.012008	0.015379	0.78085	0.434889
				0.522579
				0.000000
				0.000000
				0.000000
				0.000000

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[5]	9.566	0.500	2.000	1.982e-03
ARCH Lag[7]	45.405	1.473	1.746	2.726e-11
ARCH Lag[9]	51.153	2.402	1.619	2.020e-12

Nyblom stability test

```
Joint Statistic:  14.2103
Individual Statistics:
mu    0.08886
ma1   0.10186
ma2   0.37603
ma3   0.08259
ma4   0.19340
omega 1.85944
beta1 1.85943
beta2 1.85943
beta3 1.85943
beta4 1.85943
```

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001402	0.000597	2.34901	0.018823
ma1	-0.023168	0.021519	-1.07664	0.281642
ma2	0.010594	0.019821	0.53448	0.593011
ma3	0.012008	0.017480	0.68697	0.492102
ma4	0.010478	0.026113	0.40126	0.688227
omega	-0.654079	0.008871	-73.73176	0.000000
beta1	0.229988	0.000029	7994.47032	0.000000
beta2	0.227018	0.000026	8678.37301	0.000000
beta3	0.223947	0.000024	9372.91560	0.000000
beta4	0.220775	0.000023	9406.15949	0.000000

LogLikelihood : 75

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          2.29 2.54 3.05
Individual Statistic:     0.35 0.47 0.75
```

Sign Bias Test

	t-value	prob	sig
Sign Bias	2.573	1.013e-02	**
Negative Sign Bias	8.409	5.694e-17	***
Positive Sign Bias	2.314	2.073e-02	**
Joint Effect	76.076	2.131e-16	***

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	879.0	3.316e-174
2	30	913.0	2.018e-173
3	40	938.9	1.293e-171
4	50	962.7	8.089e-170

```
Elapsed time : 0.3414421
```

EGARCH(2,1) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model     : eGARCH(2,1)
Mean Model      : ARFIMA(2,0,1)
Distribution    : norm

Robust Standard Errors:
        Estimate  Std. Error    t value Pr(>|t|)
mu      0.001286    0.000719   1.788211 0.073742
ar1     0.282439    0.029172   9.681820 0.000000
ar2    -0.002510    0.045037  -0.055733 0.955555
ma1    -0.320071    0.021343 -14.996820 0.000000
omega  -0.383427    0.208083  -1.842660 0.065379
alpha1 -0.021121    0.041735  -0.506072 0.612806
alpha2 -0.020171    0.060862  -0.331426 0.740322
beta1   0.939492    0.032173  29.201259 0.000000
gamma1  0.290892    0.060629   4.797934 0.000002
gamma2 -0.052658    0.110220  -0.477749 0.632829
```

```
Optimal Parameters
------------------------------------
        Estimate  Std. Error  t value Pr
mu      0.001286    0.000438  2.93377 0.
ar1     0.282439    0.042277  6.68069 0.
ar2    -0.002510    0.024592 -0.10207 0.
ma1    -0.320071    0.044544 -7.18551 0
omega  -0.383427    0.069140 -5.54566
alpha1 -0.021121    0.022358 -0.94467
alpha2 -0.020171    0.023503 -0.85823
beta1   0.939492    0.010432 90.06080
LogLikelihood : 7914.889                    4  9.78978
                                            0 -1.47378

Information Criteria
------------------------------------

Akaike       -3.9944
Bayes        -3.9785
Shibata      -3.9944
Hannan-Quinn -3.9888

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                             statistic   p-value
Lag[1]                           17.61 2.709e-05
Lag[2*(p+q)+(p+q)-1][8]          28.77 0.000e+00
Lag[4*(p+q)+(p+q)-1][14]         35.13 2.953e-14
d.o.f=3
H0 : No serial correlation
```

```
Nyblom stability test
------------------------------------
Joint Statistic:  2.4953
Individual Statistics:
mu     0.14650
ar1    0.07306
ar2    0.10857
ma1    0.07113
omega  0.45667

Sign Bias Test
------------------------------------
                   t-value   prob sig
Sign Bias          0.01034 0.9918
Negative Sign Bias 0.06010 0.9521
Positive Sign Bias 0.58035 0.5617
Joint Effect       0.46928 0.9256

Weigl
-----
Lag[
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-----
ARCH
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Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     578.8   1.496e-110
2    30     609.9   5.779e-110
3    40     614.6   1.389e-104
4    50     667.1   2.339e-109

Elapsed time : 0.620002
```

EGARCH(3,1) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(3,1)
Mean Model       : ARFIMA(3,0,1)
Distribution     : norm

Information Criteria
------------------------------------

Akaike       -3.9951
Bayes        -3.9745
Shibata      -3.9952
Hannan-Quinn -3.9878

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      15.56 7.979e-05
Lag[2*(p+q)+(p+q)-1][11]    23.52 0.000e+00
Lag[4*(p+q)+(p+q)-1][19]    29.48 1.109e-08
d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                    0.05698  0.8113
Lag[2*(p+q)+(p+q)-1][11]  2.38541  0.9317
Lag[4*(p+q)+(p+q)-1][19]  3.38659  0.9906
d.o.f=4
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000929	0.000349	2.665402	0.007
ar1	-0.157047	0.014331	-10.958489	0.000
ar2	-0.000316	0.029107	-0.010871	0.991
ar3	0.035001	0.013738	2.547745	0.010
ma1	0.120665	0.015535	7.767174	0.000
omega	-0.450916	0.075694	-5.957122	0.000
alpha1			-1.052567	0.292
alpha2			-0.903663	0.366
alpha3			0.295200	0.767
beta1			81.399103	0.000
gamma1			9.927343	0.000
gamma2			-3.488812	0.000
gamma3			3.355678	0.000

```
Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[5]   0.02549 0.500 2.000  0.8731
ARCH Lag[7]   0.07640 1.473 1.746  0.9926
ARCH Lag[9]   0.50498 2.402 1.619  0.9836

Nyblom stability test
------------------------------------
Joint Statistic:  2.893
Individual Statistics:
mu     0.09766
ar1    0.18324
ar2    0.45421
ar3    0.09415
ma1    0.16671
omega  0.48755
alpha1 0.08751
alpha2 0.06650
alpha3 0.07694
beta1  0.47163
gamma1 0.24004
gamma2 0.14502
gamma3 0.10399
```

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000929	0.000428	2.171898	0.029863
ar1	-0.157047	0.071448	-2.198059	0.027945
ar2	-0.000316	0.065934	-0.004799	0.996171
ar3	0.035001	0.043874	0.797754	0.425013
ma1	0.120665	0.049469	2.439231	0.014719
omega	-0.450916	0.252369	-1.786733	0.073981
alpha1	-0.023271	0.040813	-0.570185	0.568552
alpha2	-0.028265	0.050418	-0.560601	0.575070
alpha3	0.006868	0.047874	0.143456	0.885930
beta1	0.929211	0.039154	23.732124	0.000000
gamma1	0.288011			
gamma2	-0.134138			
gamma3	0.106433			

```
LogLikelihood : 7919.
```

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.89 3.15 3.69
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                    t-value   prob sig
Sign Bias           1.3297 0.1837
Negative Sign Bias  0.7125 0.4762
Positive Sign Bias  1.0779 0.2812
Joint Effect        2.0965 0.5526


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     571.5   5.186e-109
2    30     599.0   1.063e-107
3    40     626.0   6.711e-107
4    50     643.8   1.150e-104


Elapsed time : 0.6956658
```

EGARCH(4,1) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(4,1)
Mean Model       : ARFIMA(4,0,1)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000709	0.000491	1.44542	0.148341
ar1	-0.121016	0.020307	-5.95942	0.000000
ar2	0.014107	0.016581	0.85079	0.394886
ar3	0.039614	0.016024	2.47223	0.013427
ar4	-0.006977	0.018114	-0.38518	0.700107
ma1	0.095360	0.020430	4.66760	0.000003
omega	-0.634276	0.091008	-6.96949	0.000000
alpha1	-0.038066	0.022412	-1.69842	0.089429
alpha2	-0.017477	0.031046	-0.56295	0.573466
alpha3	-0.001025	0.031079	-0.03297	0.973698
alpha4	0.009081	0.023094	0.39321	0.694164
beta1	0.901250	0.013656	65.99850	0.000000
gamma1	0.297212	0.029565	10.05280	0.000000
gamma2	-0.115841	0.041503	-2.79114	0.005252
gamma3	-0.070968	0.041422	-1.71330	0.086657
gamma4	0.227710	0.035633	6.39047	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t
mu	0.000709	0.000877	0.8
ar1	-0.121016	0.009508	-12.7
ar2	0.014107	0.022281	0.6
ar3	0.039614	0.021036	1.8
ar4	-0.006977	0.022503	-0.3
ma1	0.095360	0.009915	9.6
omega	-0.634276	0.345828	-1.8
alpha1	-0.038066	0.037632	-1.0
alpha2	-0.017477	0.047886	-0.3
alpha3	-0.001025	0.055053	-0.0
alpha4	0.009081	0.037656	0.2

Information Criteria

```
Akaike       -4.0037
Bayes        -3.9783
Shibata      -4.0038
Hannan-Quinn -3.9947
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	12.11	5.02e-04
Lag[2*(p+q)+(p+q)-1][14]	20.98	0.00e+00
Lag[4*(p+q)+(p+q)-1][24]	27.34	8.14e-06

d.o.f=5
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.02294	0.8796
Lag[2*(p+q)+(p+q)-1][14]	1.34062	0.9985
Lag[4*(p+q)+(p+q)-1][24]	2.89509	0.9997

d.o.f=5

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[6]	0.1936	0.500	2.000	0.6599
ARCH Lag[8]	0.4415	1.480	1.774	0.9158
ARCH Lag[10]	1.2635	2.424	1.650	0.8991

Sign Bias Test

	t-value	prob	sig
Sign Bias	1.2666	0.2054	
Negative Sign Bias	0.5879	0.5566	
Positive Sign Bias	1.1235	0.2613	
Joint Effect	1.9819	0.5762	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	568.5	2.192e-108
2	30	585.2	7.687e-105
3	40	613.1	2.832e-104
4	50	630.9	4.659e-102

Elapsed time : 0.7725019

Nyblom stability test

```
Joint Statistic:  4.0034
Individual Statistics:
mu      0.06519
ar1     0.21564
ar2     0.55549
ar3     0.09783
ar4     0.11932
ma1     0.19592
omega   0.60722
alpha1  0.15286
alpha2  0.09669
alpha3  0.12837
alpha4  0.04163
beta1   0.56435
gamma1  0.17083
gamma2  0.08391
gamma3  0.06141
gamma4  0.05182

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          3.46 3.75 4.3
Individual Statistic:     0.35 0.47 0.75
```

EGARCH(1,2) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                         statistic p-value
Lag[1]                     0.03882  0.8438
Lag[2*(p+q)+(p+q)-1][8]    2.12219  0.8397
Lag[4*(p+q)+(p+q)-1][14]   2.91445  0.9601
d.o.f=3

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[4]    0.8958 0.500 2.000  0.3439
ARCH Lag[6]    1.0004 1.461 1.711  0.7476
ARCH Lag[8]    1.1973 2.368 1.583  0.8933
```

```
Optimal Parameters
------------------------------------
        Estimate  Std. Error   t value Pr(>|t|)
mu      0.001295    0.000365  3.546024 0.000391
ar1     0.349124    0.100086  3.488243 0.000486
ma1    -0.386426    0.100278 -3.853567 0.000116
ma2     0.000519    0.022298  0.023258 0.981444
omega  -0.461387    0.053010 -8.703794 0.000000
alpha1 -0.044162    0.010835 -4.075740 0.000046
beta1   0.738797    0.021077 35.052605 0.000000
                    .020016  9.413418 0.000000
                    .021345 13.338025 0.000000

                    s:

Nyblom stability test
------------------------------------
Joint Statistic:  2.2643
Individual Statistics:
mu     0.13784
ar1    0.07782
ma1    0.07562
ma2    0.08792
omega  0.44259
alpha1 0.08322
beta1  0.43409
beta2  0.43318
gamma1 0.25286

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          2.1 2.32 2.82
Individual Statistic:     0.35 0.47 0.75
```

```
LogLikelihood : 7914.378

Information Criteria
------------------------------------

Akaike       -3.9946
Bayes        -3.9803
Shibata      -3.9946
Hannan-Quinn -3.9896

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      17.45 2.949e-05
Lag[2*(p+
Lag[4*(p+ Sign Bias Test
d.o.f=3   ------------------------------------
H0 : No s                  t-value   prob sig
          Sign Bias          0.03648 0.9709
          Negative Sign Bias 0.19768 0.8433
          Positive Sign Bias 0.36448 0.7155
          Joint Effect       0.20229 0.9772


          Adjusted Pearson Goodness-of-Fit Test:
          ------------------------------------
            group statistic p-value(g-1)
          1    20     581.2   4.724e-111
          2    30     602.7   1.752e-108
          3    40     625.5   8.352e-107
          4    50     662.7   1.804e-108


          Elapsed time : 0.491601
```

EGARCH(1,3) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(1,3)
Mean Model       : ARFIMA(1,0,3)
Distribution     : norm
Optimal Parameters
------------------------------------
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.000863    0.000323   2.6690 0.007608
ar1     0.343761    0.032132  10.6985 0.000000
ma1    -0.383219    0.032631 -11.7441 0.000000
ma2     0.013381    0.009891   1.3529 0.176101
ma3     0.040052    0.012920   3.1001 0.001935
omega  -0.593473    0.069261  -8.5687 0.000000
alpha1 -0.049841    0.016628  -2.9975 0.002722
beta1   0.330197    0.079204   4.1690 0.000031
beta2   0.050250    0.047045   1.0681 0.285461
beta3   0.525635    0.090446   5.8116 0.000000
gamma1  0.383655    0.025300  15.1640 0.000000
```

```
Robust Standard Errors:
        Estimate  Std. Error   t value Pr(>|t|)
mu      0.000863    0.000341   2.53335 0.011298
ar1     0.343761    0.011498  29.89744 0.000000
ma1    -0.383219    0.008494 -45.11560 0.000000
ma2     0.013381    0.007084   1.88882 0.058916
ma3     0.040052    0.020198   1.98293 0.047375
omega  -0.593473    0.159507  -3.72066 0.000199
alpha1 -0.049841    0.044808  -1.11233 0.265994
beta1   0.330197    0.615228   0.53671 0.591470
beta2   0.050250    0.124804   0.40263 0.687219
beta3   0.525635    0.708091   0.74233 0.457889
gamma1  0.383655    0.071807   5.34283 0.000000

LogLikelih
                Sign Bias Test
Informatic      ------------------------------------
----------                        t-value    prob sig
                Sign Bias          1.605435 0.10848
                Negative Sign Bias 0.006969 0.99444
                Positive Sign Bias 1.682976 0.09246   *
                Joint Effect       3.918686 0.27038
Akaike
Bayes
Shibata         Adjusted Pearson Goodness-of-Fit Test:
Hannan-Qui      ------------------------------------
                  group statistic p-value(g-1)
                1    20     578.0   2.204e-110
                2    30     597.6   1.995e-107
                3    40     624.2   1.521e-106
                4    50     651.8   2.867e-106


                Elapsed time : 0.5764251
```

```
Nyblom stability test
-------------------------------------
Joint Statistic:  2.8594
Individual Statistics:
mu      0.11617
ar1     0.03870
ma1     0.03492
ma2     0.40001
ma3     0.06403
omega   0.53063
alpha1  0.12081
beta1   0.50684
beta2   0.51346
beta3   0.52245
gamma1  0.21340

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          2.49 2.75 3.27
Individual Statistic:     0.35 0.47 0.75

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-----------------------------------
                        statistic p-value
Lag[1]                     0.2873  0.5920
Lag[2*(p+q)+(p+q)-1][11]   2.0579  0.9567
Lag[4*(p+q)+(p+q)-1][19]   3.3108  0.9917
d.o.f=4

Weighted ARCH LM Tests
-----------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[5]     1.332 0.500 2.000  0.2484
ARCH Lag[7]     1.884 1.473 1.746  0.5301
ARCH Lag[9]     2.077 2.402 1.619  0.7435
```

EGARCH(1,4) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(1,4)
Mean Model       : ARFIMA(1,0,4)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000706	0.000178	3.9700e+00	0.000072
ar1	0.988584	0.004319	2.2891e+02	0.000000
ma1	-1.020619	0.000001	-1.1797e+06	0.000000
ma2	0.033299	0.010555	3.1549e+00	0.001606
ma3	0.028164	0.005768	4.8828e+00	0.000001
ma4	-0.024488	0.007431	-3.2953e+00	0.000983
omega	-0.545460	0.058976	-9.2488e+00	0.000000
alpha1	-0.044196	0.010828	-4.0817e+00	0.000045
beta1	0.566489	0.034222	1.6553e+01	0.000000
beta2	0.164847	0.047528	3.4684e+00	0.000523
beta3	0.570282	0.039211	1.4544e+01	0.000000
beta4	-0.386981	0.077121	-5.0178e+00	0.000001
gamma1	0.309676	0.011973	2.5865e+01	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000706	0.000150	4.6992e+00	0.000003
ar1	0.988584	0.003900	2.5348e+02	0.000000
ma1	-1.020619	0.000001	-1.0372e+06	0.000000
ma2	0.033299	0.008349	3.9882e+00	0.000067
ma3	0.028164	0.002151	1.3093e+01	0.000000
ma4	-0.024488	0.007536	-3.2496e+00	0.001156
omega	-0.545460	0.155873	-3.4994e+00	0.000466
alpha1	-0.044196	0.041543	-1.0639e+00	0.287393
beta1	0.566489	0.139051	4.0740e+00	0.000046
beta2	0.164847	0.067751	2.4331e+00	0.014969
beta3	0.570282	0.185077	3.0813e+00	0.002061
beta4	-0.386981	0.329292	-1.1752e+00	0.239919
			2.5405e+00	0.011068

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic: 2.89 3.15 3.69
Individual Statistic: 0.35 0.47 0.75

Information Criteria

Akaike	-4.0037
Bayes	-3.9830
Shibata	-4.0037
Hannan-Quinn	-3.9963

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	14.24	1.613e-04
Lag[2*(p+q)+(p+q)-1][14]	23.06	0.000e+00
Lag[4*(p+q)+(p+q)-1][24]	28.11	3.640e-06

d.o.f=5
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.01923	0.8897
Lag[2*(p+q)+(p+q)-1][14]	1.98929	0.9914
Lag[4*(p+q)+(p+q)-1][24]	3.57007	0.9988

d.o.f=5

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[6]	0.01654	0.500	2.000	0.8977
ARCH Lag[8]	0.14729	1.480	1.774	0.9817
ARCH Lag[10]	0.53764	2.424	1.650	0.9828

Nyblom stability test

Joint Statistic: 3.1398
Individual Statistics:

mu	0.08477
ar1	0.18789
ma1	0.14965
ma2	0.13292
ma3	0.19402
ma4	0.17652
omega	0.58754
alpha1	0.08886
beta1	0.54891
beta2	0.55202
beta3	0.55459
beta4	0.55113
gamma1	0.15938

Sign Bias Test

	t-value	prob	sig
Sign Bias	1.2755	0.2022	
Negative Sign Bias	0.4264	0.6698	
Positive Sign Bias	1.1622	0.2452	
Joint Effect	2.0126	0.5698	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	574.2	1.400e-109
2	30	595.5	5.575e-107
3	40	615.7	8.159e-105
4	50	628.7	1.244e-101

Elapsed time : 0.9295678

EGARCH(2,2) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(2,2)
Mean Model       : ARFIMA(2,0,2)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001293	0.000278	4.64328	0.000003
ar1	-0.321973	0.023811	-13.52215	0.000000
ar2	0.231416	0.016612	13.93082	0.000000
ma1	0.284581	0.023513	12.10327	0.000000
ma2	-0.253033	0.016378	-15.44934	0.000000
omega	-0.454754	0.058046	-7.83441	0.000000
alpha1	-0.020435	0.022138	-0.92309	0.355960
alpha2	-0.028116	0.023475	-1.19768	0.231042
beta1	0.749865	0.019664	38.13470	0.000000
beta2	0.178386	0.017674	10.09290	0.000000
gamma1	0.294991	0.028987	10.17659	0.000000
gamma2	-0.012586	0.030762	-0.40916	0.682423

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001293	0.000202	6.41495	0.000000
ar1	-0.321973	0.010261	-31.37812	0.000000
ar2	0.231416	0.010686	21.65654	0.000000
ma1	0.284581	0.012117	23.48515	0.000000
ma2	-0.253033	0.010316	-24.52902	0.000000
omega	-0.454754	0.122113	-3.72404	0.000196
alpha1	-0.020435	0.039948	-0.51154	0.608975
alpha2	-0.028116	0.058243	-0.48273	0.629290
beta1	0.749865	0.006429	116.63391	0.000000
beta2	0.178386	0.013100	13.61759	0.000000
gamma1	0.294991	0.052421	5.62734	0.000000
gamma2	-0.012586	0.076709	-0.16408	0.869668

LogLikelihood : 7915.361

Information Criteria

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	17.25	3.269e-05
Lag[2*(p+q)+(p+q)-1][11]	31.54	0.000e+00
Lag[4*(p+q)+(p+q)-1][19]	38.21	1.605e-13

d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.02167	0.883
Lag[2*(p+q)+(p+q)-1][11]	2.52635	0.919
Lag[4*(p+q)+(p+q)-1][19]	3.49113	0.989

d.o.f=4

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[5]	0.1178	0.500	2.000	0.7314
ARCH Lag[7]	0.2329	1.473	1.746	0.9635
ARCH Lag[9]	0.5848	2.402	1.619	0.9775

Nyblom stability test

Joint Statistic: 3.379
Individual Statistics:

mu	0.14115
ar1	0.16940
ar2	0.15912
ma1	0.16212
ma2	0.15525
omega	0.45406
alpha1	0.08028
alpha2	0.05979
beta1	0.44484
beta2	0.44378
gamma1	0.25592
gamma2	0.16290

Asymptotic Critical Values (10% 5% 1%)

Joint Statistic:	2.69	2.96	3.51
Individual Statistic:	0.35	0.47	0.75

Sign Bias Test

	t-value	prob	sig
Sign Bias	0.1646334	0.8692	
Negative Sign Bias	0.0001102	0.9999	
Positive Sign Bias	0.6935759	0.4880	
Joint Effect	0.5358831	0.9109	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	583.8	1.326e-111
2	30	615.1	4.778e-111
3	40	631.4	5.095e-108
4	50	662.5	2.005e-108

Elapsed time : 0.6352179

EGARCH(2,3) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(2,3)
Mean Model       : ARFIMA(2,0,3)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001096	0.000231	4.74519	0.000002
ar1	-0.863687	0.008570	-100.77444	0.000000
ar2	-0.937906	0.005891	-159.21374	0.000000
ma1	0.824393	0.010196	80.85300	0.000000
ma2	0.892406	0.003308	269.79713	0.000000
ma3	-0.034120	0.004861	-7.01968	0.000000
omega	-0.895819	0.111926	-8.00370	0.000000
alpha1	-0.039564	0.016697	-2.36943	0.017815
alpha2	-0.038044	0.019350	-1.96611	0.049286
beta1	-0.065216	0.075414	-0.86477	0.387166
beta2	0.312186	0.057375	5.44118	0.000000
beta3	0.611584	0.041240	14.82983	0.000000
gamma1	0.330451	0.021222	15.57134	0.000000
gamma2	0.246141	0.043729	5.62882	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001096	0.000137	7.97079	0.000000
ar1	-0.863687	0.003627	-238.11439	0.000000
ar2	-0.937906	0.009845	-95.26497	0.000000
ma1	0.824393	0.005288	155.89299	0.000000
ma2	0.892406	0.010714	83.29422	0.000000
ma3	-0.034120	0.002256	-15.12520	0.000000
omega	-0.895819	0.243880	-3.67319	0.000240
alpha1	-0.039564	0.035876	-1.10280	0.270116
alpha2	-0.038044	0.036094	-1.05400	0.291883

Information Criteria

```
Akaike       -4.0023
Bayes        -3.9800
Shibata      -4.0023
Hannan-Quinn -3.9944
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	17.60	2.722e-05
Lag[2*(p+q)+(p+q)-1][14]	31.79	0.000e+00
Lag[4*(p+q)+(p+q)-1][24]	38.10	3.728e-11

d.o.f=5
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.008601	0.9261
Lag[2*(p+q)+(p+q)-1][14]	2.568350	0.9754
Lag[4*(p+q)+(p+q)-1][24]	4.138806	0.9967

d.o.f=5

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[6]	0.2326	0.500	2.000	0.6296
ARCH Lag[8]	0.2391	1.480	1.774	0.9637
ARCH Lag[10]	0.5861	2.424	1.650	0.9793

Nyblom stability test

```
Joint Statistic:  2.692
Individual Statistics:
mu     0.08130
ar1    0.28813
ar2    0.10358
ma1    0.2
ma2    0.1
ma3    0.0
omega  0.5
alpha1 0.1
alpha2 0.0
beta1  0.5
beta2  0.5
beta3  0.5
gamma1 0.2
gamma2 0.0
```

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          3.08 3.34 3.9
Individual Statistic:     0.35 0.47 0.75
```

Sign Bias Test

	t-value	prob	sig
Sign Bias	0.6344	0.5258	
Negative Sign Bias	0.1090	0.9132	
Positive Sign Bias	1.0761	0.2819	
Joint Effect	1.2103	0.7505	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	567.3	4.006e-108
2	30	604.4	7.898e-109
3	40	616.4	6.135e-105
4	50	637.8	1.888e-103

```
Elapsed time : 1.063746
```

EGARCH(2,4) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*
```

Conditional Variance Dynamics

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000962	0.000037	2.6180e+01	0.000000
ar1	1.713547	0.000382	4.4835e+03	0.000000
ar2	-0.716228	0.001592	-4.4980e+02	0.000000
ma1	-1.748504	0.000000	-4.2519e+09	0.000000
ma2	0.776389	0.000000	9.2444e+07	0.000000
ma3	-0.005599	0.000999	-5.6073e+00	0.000000
ma4	-0.018677	0.002870	-6.5089e+00	0.000000
omega	-0.544595	0.063743	-8.5435e+00	0.000000
alpha1	-0.038728	0.011036	-3.5093e+00	0.000449
alpha2	-0.005187	0.001380	-3.7599e+00	0.000170
beta1	0.561987	0.033040	1.7010e+01	0.000000
beta2	0.166600	0.048799	3.4140e+00	0.000640
beta3	0.575273	0.034924	1.6472e+01	0.000000
beta4	-0.389099	0.101860	-3.8199e+00	0.000133
gamma1	0.312030	0.003913	7.9752e+01	0.000000
gamma2	-0.000453	0.023497	-1.9260e-02	0.984634

Robust Standard Errors:

	Estimate	Std. Er
mu	0.000962	0.000
ar1	1.713547	0.001
ar2	-0.716228	0.002
ma1	-1.748504	0.000
ma2	0.776389	0.000
ma3	-0.005599	0.014
ma4	-0.018677	0.018
omega	-0.544595	0.196
alpha1	-0.038728	0.042536 -9.1048e-01 0.362567
alpha2	-0.005187	0.007164 -7.2[illegible]
beta1	0.561987	0.131208 4.2
beta2	0.166600	0.251703 6.6
beta3	0.575273	0.165871 3.4

Information Criteria

Akaike	-4.0028
Bayes	-3.9774
Shibata	-4.0029
Hannan-Quinn	-3.9938

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	15.19	9.698e-05
Lag[2*(p+q)+(p+q)-1][17]	24.82	0.000e+00
Lag[4*(p+q)+(p+q)-1][29]	31.14	6.002e-06

d.o.f=6
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residu

	statistic	p-value
Lag[1]	0.01462	0.9038
Lag[2*(p+q)+(p+q)-1][17]	2.33211	0.9964
Lag[4*(p+q)+(p+q)-1][29]	4.28679	0.9996

d.o.f=6

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[7]	0.1136	0.500	2.000	0.7361
ARCH Lag[9]	0.6000	1.485	1.796	0.8772
ARCH Lag[11]	0.6526	2.440	1.677	0.9755

Nyblom stability test

Joint Statistic: 3.1843
tistics:

Sign Bias Test

	t-value	prob	sig
Sign Bias	1.5129	0.1304	
Negative Sign Bias	0.5487	0.5832	
Positive Sign Bias	1.3514	0.1766	
Joint Effect	2.7930	0.4246	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	560.1	1.294e-106
2	30	585.2	7.577e-105
3	40	600.2	1.220e-101
4	50	623.8	1.215e-100

tical Values (10% 5% 1%)
c: 3.46 3.75 4.3
tistic: 0.35 0.47 0.75

Elapsed time : 1.744574

EGARCH(3,2) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(3,2)
Mean Model       : ARFIMA(3,0,2)
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001291	0.000298	4.3323e+00	0.000015
ar1	-0.053302	0.002282	-2.3359e+01	0.000000
ar2	0.985904	0.002138	4.6109e+02	0.000000
ar3	0.048353	0.002159	2.2401e+01	0.000000
ma1	0.006678	0.001159	5.7624e+00	0.000000
ma2	-0.985993	0.000000	-2.1419e+07	0.000000
omega	-0.763859	0.037383	-2.0433e+01	0.000000
alpha1	-0.018630	0.021959	-8.4841e-01	0.396212
alpha2	-0.068998	0.005442	-1.2678e+01	0.000000
alpha3	0.001063	0.023063	4.6095e-02	0.963234
beta1	-0.023677	0.001453	-1.6297e+01	0.000000
beta2	0.902931	0.001033	8.7407e+02	0.000000
gamma1	0.302105	0.029541	1.0227e+01	0.000000
gamma2	0.267300	0.019613	1.3629e+01	0.000000
gamma3	-0.072487	0.026862	-2.6985e+00	0.006965

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001291	0.000211	6.1097e+00	0.000000
ar1	-0.053302	0.001060	-5.0270e+01	0.000000
ar2	0.985904	0.002817	3.4995e+02	0.000000
ar3	0.048353	0.000865	5.5910e+01	0.000000
ma1	0.006678	0.000337	1.9831e+01	0.000000
ma2	-0.985993	0.000000	-1.1689e+07	0.000000
omega	-0.763859	0.038674	-1.9751e+01	0.000000
alpha1	-0.018630	0.044879	-4.1512e-01	0.678054
alpha2	-0.068998	0.017043	-4.0484e+00	0.000052
alpha3	0.001063	0.058462	1.8184e-02	0.985492
beta1	-0.023677	0.005253	-4.5076e+00	0.000007
beta2	0.902931	0.002381	3.7929e+02	0.000000

Information Criteria

```
Akaike       -3.9995
Bayes        -3.9757
Shibata      -3.9996
Hannan-Quinn -3.9911
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	21.51	3.518e-06
Lag[2*(p+q)+(p+q)-1][14]	33.50	0.000e+00
Lag[4*(p+q)+(p+q)-1][24]	39.83	4.282e-12

```
d.o.f=5
H0 : No serial correlation
```

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.03617	0.8492
Lag[2*(p+q)+(p+q)-1][14]	4.51011	0.8274
Lag[4*(p+q)+(p+q)-1][24]	6.28011	0.9605

```
d.o.f=5
```

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[6]	0.02641	0.500	2.000	0.8709
ARCH Lag[8]	1.53331	1.480	1.774	0.6231
ARCH Lag[10]	1.82483	2.424	1.650	0.8035

Sign Bias Test

	t-value	prob	sig
Sign Bias	0.2064	0.8365	
Negative Sign Bias	0.1638	0.8699	
Positive Sign Bias	0.5527	0.5805	
Joint Effect	0.6232	0.8911	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	588.4	1.394e-112
2	30	596.8	3.014e-107
3	40	621.8	4.759e-106
4	50	625.2	6.243e-101

```
Elapsed time : 1.190702
```

Nyblom stability test

```
Joint Statistic:  3.0749
Individual Statistics:
mu     0.17766
ar1    0.01895
ar2    0.07363
ar3    0.01935
ma1    0.02128
ma2    0.06062
omega  0.48813
alpha1 0.42539
alpha2 0.05522
alpha3 0.35227
beta1  0.47132
beta2  0.47747
gamma1 0.20257
gamma2 0.05120
gamma3 0.16655

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         3.26 3.54 4.07
Individual Statistic:    0.35 0.47 0.75
```

EGARCH(3,3) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(3,3)
Mean Model       : ARFIMA(3,0,3)
Distribution     : norm

Optimal Parameters
------------------------------------
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.000227    0.000358  6.3433e-01 0.525867
ar1     0.466165    0.006093  7.6507e+01 0.000000
ar2    -0.359572    0.008313 -4.3254e+01 0.000000
ar3     0.873429    0.006290  1.3886e+02 0.000000
ma1    -0.489979    0.000007 -7.0224e+04 0.000000
ma2     0.383132    0.006941  5.5195e+01 0.000000
ma3    -0.860415    0.000002 -4.6058e+05 0.000000
omega  -1.072036    0.127323 -8.4198e+00 0.000000
alpha1 -0.051911    0.013920 -3.7292e+00 0.000192
alpha2 -0.048402    0.020296 -2.3848e+00 0.017087
alpha3 -0.004450    0.016574 -2.6850e-01 0.788318
beta1  -0.175711    0.067577 -2.6001e+00 0.009318
beta2   0.323840    0.058670  5.5197e+00 0.000000
beta3   0.683195    0.044782  1.5256e+01 0.000000
gamma1  0.316849    0.025720  1.2319e+01 0.000000
gamma2  0.287095    0.040223  7.1376e+00 0.000000
gamma3  0.051849    0.031395  1.6515e+00 0.098629

Robust Standard Errors:
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.000227    0.000325  6.9884e-01 0.484651
ar1     0.466165    0.005975  7.8024e+01 0.000000
ar2    -0.359572    0.010762 -3.3412e+01 0.000000
ar3     0.873429    0.007055  1.2380e+02 0.000000
ma1    -0.489979    0.000020 -2.4734e+04 0.000000
ma2     0.383132    0.013352  2.8695e+01 0.000000
ma3    -0.860415    0.000007 -1.1913e+05 0.000000
omega  -1.072036    0.264552 -4.0523e+00 0.000051
alpha1 -0.051911    0.025338 -2.0487e+00 0.040487
alpha2 -0.048402    0.038715 -1.2502e+00 0.211220
alpha3 -0.004450    0.029047 -1.5320e-01 0.878240
beta1  -0.175711    0.127301 -1.3803e+00 0.167500
                                         01479
Information Criteria                     00000
------------------------------------     00000

Akaike       -4.0020                     00246
Bayes        -3.9750                     76816
Shibata      -4.0021
Hannan-Quinn -3.9925

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      11.70 0.0006253
Lag[2*(p+q)+(p+q)-1][17]    18.90 0.0000000
Lag[4*(p+q)+(p+q)-1][29]    24.26 0.0027504
d.o.f=6
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                   0.007344  0.9317
Lag[2*(p+q)+(p+q)-1][17] 3.510192  0.9742
Lag[4*(p+q)+(p+q)-1][29] 5.573068  0.9967
d.o.f=6

Weighted ARCH LM Tests
------------------------------------
                Statistic Shape Scale P-Value
ARCH Lag[7]       0.01504 0.500 2.000  0.9024
ARCH Lag[9]       0.41600 1.485 1.796  0.9244
ARCH Lag[11]      0.44317 2.440 1.677  0.9896

Sign Bias Test
------------------------------------
                   t-value   prob sig
Sign Bias           0.9888 0.3228
Negative Sign Bias  0.1279 0.8982
Positive Sign Bias  1.0716 0.2840
Joint Effect        1.4672 0.6899

Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     547.6   5.595e-104
2    30     571.5   5.062e-102
3    40     590.1    1.409e-99
4    50     607.7    2.107e-97

Elapsed time : 1.336442

Nyblom stability test
------------------------------------
Joint Statistic:  3.9328
Individual Statistics:
mu     0.07620
ar1    0.11305
ar2    0.21352
ar3    0.41766
ma1    0.06850
ma2    0.16786
ma3    0.36132
omega  0.55144
alpha1 0.17790
alpha2 0.05307
alpha3 0.23365
beta1  0.51583
beta2  0.51873
beta3  0.53378
gamma1 0.19523
gamma2 0.06064
gamma3 0.18145

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          3.64 3.95 4.51
Individual Statistic:     0.35 0.47 0.75
```

EGARCH(3,4) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(3,4)
Mean Model       : ARFIMA(3,0,4)
Distribution     : norm

Optimal Parameters
------------------------------------
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.000785    0.000252     3.11638 0.001831
ar1     0.091625    0.009603     9.54174 0.000000
ar2    -0.064383    0.010237    -6.28951 0.000000
ar3     0.936794    0.005186   180.64264 0.000000
ma1    -0.121400    0.002247   -54.03793 0.000000
ma2     0.061059    0.010592     5.76441 0.000000
ma3    -0.920441    0.000454 -2029.24233 0.000000
ma4     0.030509    0.013349     2.28558 0.022279
omega  -0.419769    0.062475    -6.71898 0.000000
alpha1 -0.056959    0.018104    -3.14628 0.001654
alpha2 -0.009122    0.020402    -0.44713 0.654780
alpha3  0.039180    0.019461     2.01324 0.044089
beta1   0.567216    0.003462   163.84975 0.000000
beta2   0.371170    0.005968    62.19474 0.000000
beta3   0.473727    0.012562    37.71240 0.000000
beta4  -0.477779    0.012305   -38.82749 0.000000
gamma1  0.296751    0.023920    12.40610 0.000000
gamma2  0.036572    0.026912     1.35893 0.174169
gamma3 -0.093501    0.030553    -3.06028 0.002211

Robust Standard Errors:
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.000785    0.000246     3.19671 0.001390
ar1     0.091625    0.011137     8.22676 0.000000
ar2    -0.064383    0.006433   -10.00855 0.000000
ar3     0.936794    0.006636   141.17403 0.000000
ma1    -0.121400    0.001651   -73.54574 0.000000
ma2     0.061059    0.006669     9.15546 0.000000
ma3    -0.920441    0.000280 -3283.69212 0.000000
ma4     0.030509    0.013044     2.33895 0.019338
omega  -0.419769    0.151093    -2.77822 0.005466
alpha1 -0.056959    0.032304    -1.76323 0.077861
alpha2 -0.009122    0.034654    -0.26324 0.792364
alpha3  0.039180    0.031387     1.24827 0.211932
beta1   0.567216    0.002337   242.69059 0.000000
beta2   0.371170    0.017386    21.34878 0.000000
beta3   0.473727    0.011344    41.75939 0.000000
beta4  -0.477779    0.011074   -43.14256 0.000000
gamma1  0.296751    0.062094     4.77905 0.000002
gamma2  0.036572    0.056995     0.64166 0.521095
gamma3 -0.093501    0.070372    -1.32867 0.183958

Information Criteria
------------------------------------

Akaike       -4.0028
Bayes        -3.9727
Shibata      -4.0029
Hannan-Quinn -3.9921

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      14.55 1.364e-04
Lag[2*(p+q)+(p+q)-1][20]    27.04 0.000e+00
Lag[4*(p+q)+(p+q)-1][34]    34.45 6.829e-06
d.o.f=7
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                    0.05435  0.8157
Lag[2*(p+q)+(p+q)-1][20]  3.19763  0.9957
Lag[4*(p+q)+(p+q)-1][34]  5.28815  0.9998
d.o.f=7

Weighted ARCH LM Tests
------------------------------------
             Statistic Shape Scale P-Value
ARCH Lag[8]    0.05225 0.500 2.000  0.8192
ARCH Lag[10]   0.76350 1.488 1.815  0.8363
ARCH Lag[12]   0.96049 2.451 1.700  0.9472

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         4.03 4.33 4.92
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value   prob sig
Sign Bias           1.6339 0.1024
Negative Sign Bias  0.5127 0.6082
Positive Sign Bias  1.1958 0.2318
Joint Effect        2.8937 0.4083


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     564.0   1.971e-107
2    30     583.4   1.791e-104
3    40     601.1   8.034e-102
4    50     626.6   3.360e-101


Elapsed time : 1.866613

Nyblom stability test
------------------------------------
Joint Statistic:  3.7847
Individual Statistics:
mu     0.05884
ar1    0.29473
ar2    0.12447
ar3    0.06117
ma1    0.20424
ma2    0.09741
ma3    0.06317
ma4    0.20990
omega  0.56760
alpha1 0.08862
alpha2 0.03584
alpha3 0.10842
beta1  0.53009
beta2  0.53292
beta3  0.53521
beta4  0.53224
gamma1 0.17988
gamma2 0.13168
gamma3 0.23437
```

EGARCH(4,2) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(4,2)
Mean Model       : ARFIMA(4,0,2)
Distribution     : norm

Optimal Parameters
------------------------------------
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.000656    0.000317  2.0677e+00 0.038669
ar1     0.556677    0.013314  4.1810e+01 0.000000
ar2     0.400706    0.010114  3.9619e+01 0.000000
ar3     0.041665    0.006532  6.3785e+00 0.000000
ar4    -0.027885    0.005750 -4.8496e+00 0.000001
ma1    -0.585145    0.000012 -5.0098e+04 0.000000
ma2    -0.376754    0.016644 -2.2636e+01 0.000000
omega  -0.734760    0.168390 -4.3634e+00 0.000013
alpha1 -0.037288    0.022099 -1.6873e+00 0.091544
alpha2 -0.030467    0.031770 -9.5898e-01 0.337569
alpha3  0.009993    0.023260  4.2964e-01 0.667460
alpha4 -0.001587    0.000943 -1.6837e+00 0.092231
beta1   0.622251    0.362508  1.7165e+00 0.086068
beta2   0.263223    0.340771  7.7243e-01 0.439857
gamma1  0.298400    0.029487  1.0120e+01 0.000000
gamma2 -0.030859    0.123044 -2.5080e-01 0.801970
gamma3 -0.109169    0.043645 -2.5013e+00 0.012374
gamma4  0.240619    0.032461  7.4125e+00 0.000000
```

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000656	0.000277	2.3666e+00	0.017952
ar1	0.556677	0.014401	3.8657e+01	0.000000
ar2	0.400706	0.015327	2.6144e+01	0.000000
ar3	0.041665	0.003450	1.2076e+01	0.000000
ar4	-0.027885	0.007737	-3.6040e+00	0.000313
ma1	-0.585145	0.000028	-2.0561e+04	0.000000
ma2	-0.376754	0.030649	-1.2293e+01	0.000000
omega	-0.734760	1.016700	-7.2269e-01	0.469870
alpha1	-0.037288	0.042397	-8.7950e-01	0.379129
alpha2	-0.030467	0.123478	-2.4674e-01	0.805110
alpha3	0.009993	0.043258	2.3102e-01	0.817303
alpha4	-0.001587	0.001767	-8.9815e-01	0.369104
beta1	0.622251	2.319861	2.6823e-01	0.788524
beta2	0.263223	2.166194	1.2151e-01	0.903284
gamma1	0.298400	0.069457	4.2962e+00	0.000017
gamma2	-0.030859	0.762530	-4.0470e-02	0.967719
gamma3	-0.109169	0.176930	-6.1702e-01	0.537221
gamma4	0.240619	0.110605	2.1755e+00	0.029594

LogLikelihood : 7942.212

Information Criteria

Akaike	-4.0041
Bayes	-3.9756
Shibata	-4.0042
Hannan-Quinn	-3.9940

Weighted Ljung-Box Test on Standardized Residuals
---

	statistic	p-value
Lag[1]	12.90	3.281e-04
Lag[2*(p+q)+(p+q)-1][17]	21.97	0.000e+00
Lag[4*(p+q)+(p+q)-1][29]	28.23	9.585e-05

d.o.f=6
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
---

	statistic	p-value
Lag[1]	0.05063	0.8220
Lag[2*(p+q)+(p+q)-1][17]	1.44812	0.9997
Lag[4*(p+q)+(p+q)-1][29]	3.41748	0.9999

d.o.f=6

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[7]	0.004182	0.500	2.000	0.9484
ARCH Lag[9]	0.991106	1.485	1.796	0.7714
ARCH Lag[11]	1.258367	2.440	1.677	0.9050

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic: 3.83 4.14 4.73
Individual Statistic: 0.35 0.47 0.75

Sign Bias Test
---

	t-value	prob	sig
Sign Bias	0.9632	0.3355	
Negative Sign Bias	0.4926	0.6223	
Positive Sign Bias	0.9637	0.3353	
Joint Effect	1.3080	0.7272	

Adjusted Pearson Goodness-of-Fit Test:
---

	group	statistic	p-value(g-1)
1	20	564.2	1.822e-107
2	30	578.5	1.824e-103
3	40	600.1	1.243e-101
4	50	622.4	2.311e-100

Elapsed time : 1.373199

Nyblom stability test

Joint Statistic: 4.332
Individual Statistics:

mu	0.06162
ar1	0.12552
ar2	0.07364
ar3	0.17415
ar4	0.14703
ma1	0.09634
ma2	0.05502
omega	0.61814
alpha1	0.16427
alpha2	0.07353
alpha3	0.15097
alpha4	0.04234
beta1	0.57343
beta2	0.57472
gamma1	0.19153
gamma2	0.09391
gamma3	0.10860
gamma4	0.05408

EGARCH(4,3) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(4,3)
Mean Model       : ARFIMA(4,0,3)
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000700	0.000261	2.6800e+00	0.007361
ar1	0.038243	0.003046	1.2555e+01	0.000000
ar2	-0.085884	0.004408	-1.9485e+01	0.000000
ar3	0.972186	0.003607	2.6955e+02	0.000000
ar4	0.032676	0.003160	1.0341e+01	0.000000
ma1	-0.065765	0.003842	-1.7117e+01	0.000000
ma2	0.090064	0.005149	1.7492e+01	0.000000
ma3	-0.962217	0.000000	-9.1208e+06	0.000000
omega	-0.706640	0.063994	-1.1042e+01	0.000000
alpha1	-0.037140	0.021821	-1.7021e+00	0.088742
alpha2	-0.030001	0.025704	-1.1672e+00	0.243135
alpha3	0.005745	0.027061	2.1230e-01	0.831874
alpha4	0.005966	0.022416	2.6613e-01	0.790141
beta1	0.621604	0.178446	3.4834e+00	0.000495
beta2	0.279119	0.167705	1.6643e+00	0.096044
beta3	-0.011005	0.001174	-9.3783e+00	0.000000
gamma1	0.294907	0.028593	1.0314e+01	0.000000
gamma2	-0.022360	0.042812	-5.2228e-01	0.601478
gamma3	-0.116962	0.034798	-3.3611e+00	0.000776
gamma4	0.238673	0.032172	7.4186e+00	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.000700	0.000153	4.5730e+00	0.000005
ar1	0.038243	0.001298	2.9468e+01	0.000000
ar2	-0.085884	0.003297	-2.6046e+01	0.000000
ar3	0.972186	0.004959	1.9606e+02	0.000000
ar4	0.032676	0.001531	2.1343e+01	0.000000
ma1	-0.065765	0.002649	-2.4829e+01	0.000000
ma2	0.090064	0.003224	2.7933e+01	0.000000
ma3	-0.962217	0.000000	-2.5154e+06	0.000000
omega	-0.706640	0.297293	-2.3769e+00	0.017458
alpha1	-0.037140	0.036373	-1.0211e+00	0.307218
alpha2	-0.030001	0.047667	-6.2940e-01	0.529086
alpha3	0.005745	0.046003	1.2488e-01	0.900615
alpha4	0.005966	0.036204	1.6478e-01	0.869121
beta1	0.621604	0.547476	1.1354e+00	0.256208
beta2	0.279119	0.530727	5.2592e-01	0.598945
beta3	-0.011005	0.001891	-5.8213e+00	0.000000

Information Criteria

```
Akaike       -4.0052
Bayes        -3.9735
Shibata      -4.0053
Hannan-Quinn -3.9939
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	13.15	0.0002870
Lag[2*(p+q)+(p+q)-1][20]	23.83	0.0000000
Lag[4*(p+q)+(p+q)-1][34]	31.21	0.0001272

d.o.f=7
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.06721	0.7954
Lag[2*(p+q)+(p+q)-1][20]	2.02956	0.9997
Lag[4*(p+q)+(p+q)-1][34]	4.20318	1.0000

d.o.f=7

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[8]	0.3492	0.500	2.000	0.5545
ARCH Lag[10]	1.4586	1.488	1.815	0.6533
ARCH Lag[12]	1.6147	2.451	1.700	0.8543

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic: 4.22 4.52 5.13
Individual Statistic: 0.35 0.47 0.75

Sign Bias Test

	t-value	prob	sig
Sign Bias	1.2593	0.2080	
Negative Sign Bias	0.5855	0.5582	
Positive Sign Bias	1.1449	0.2523	
Joint Effect	2.0024	0.5719	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	557.1	5.518e-106
2	30	572.7	2.987e-102
3	40	591.4	7.397e-100
4	50	609.0	1.136e-97

Elapsed time : 1.797881

Nyblom stability test

Joint Statistic: 3.8619
Individual Statistics:

mu	0.04050
ar1	0.51049
ar2	0.02610
ar3	0.13104
ar4	0.41688
ma1	0.34706
ma2	0.03126
ma3	0.15008
omega	0.59757
alpha1	0.14582
alpha2	0.06658
alpha3	0.13646
alpha4	0.03881
beta1	0.55708
beta2	0.55820
beta3	0.56149
gamma1	0.20241
gamma2	0.08704
gamma3	0.10496
gamma4	0.05612

EGARCH(4,4) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : eGARCH(4,4)
Mean Model       : ARFIMA(4,0,4)
Distribution     : norm

Optimal Parameters
------------------------------------
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.000745    0.000141  5.2998e+00 0.000000
ar1     0.742072    0.004652  1.5952e+02 0.000000
ar2    -1.218711    0.003591 -3.3936e+02 0.000000
ar3     0.618728    0.004122  1.5009e+02 0.000000
ar4    -0.887186    0.004519 -1.9634e+02 0.000000
ma1    -0.775239    0.000001 -8.0076e+05 0.000000
ma2     1.252187    0.001740  7.1964e+02 0.000000
ma3    -0.641626    0.000001 -1.1662e+06 0.000000
ma4     0.886606    0.000000  6.3790e+07 0.000000
omega  -0.633705    0.093585 -6.7714e+00 0.000000
alpha1 -0.056282    0.014101 -3.9914e+00 0.000066
alpha2 -0.012394    0.014817 -8.3644e-01 0.402906
alpha3  0.013318    0.004293  3.1023e+00 0.001920
alpha4  0.016546    0.019346  8.5524e-01 0.392416
beta1   0.687034    0.020212  3.3992e+01 0.000000
beta2   0.134571    0.078632  1.7114e+00 0.087005
beta3   0.290212    0.043115  6.7311e+00 0.000000
beta4  -0.210502    0.066677 -3.1570e+00 0.001594
gamma1  0.303333    0.027531  1.1018e+01 0.000000
gamma2 -0.037316    0.020681 -1.8043e+00 0.071179
gamma3 -0.086115    0.022128 -3.8917e+00 0.000100
gamma4  0.170147    0.037674  4.5163e+00 0.000006

LogLikelihood : 7950.592

Information Criteria
------------------------------------

Akaike       -4.0064
Bayes        -3.9714
Shibata      -4.0064
Hannan-Quinn -3.9940

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      15.13 1.003e-04
Lag[2*(p+q)+(p+q)-1][23]    27.03 0.000e+00
Lag[4*(p+q)+(p+q)-1][39]    34.96 8.393e-05
d.o.f=8
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                            statistic p-value
Lag[1]                        0.04386  0.8341
Lag[2*(p+q)+(p+q)-1][23]      2.60840  0.9998
Lag[4*(p+q)+(p+q)-1][39]      5.09176  1.0000
d.o.f=8

Weighted ARCH LM Tests
------------------------------------
             Statistic Shape Scale P-Value
ARCH Lag[9]     0.8689 0.500  2.00  0.3513
ARCH Lag[11]    0.9764 1.490  1.83  0.7820
ARCH Lag[13]    1.3562 2.459  1.72  0.8984

Asymptotic Critical Values (10% 5% 1%)
Individual Statistic:     0.35 0.47 0.75

Sign Bias Test
------------------------------------
                    t-value   prob sig
Sign Bias            1.0351 0.3007
Negative Sign Bias  0.3385 0.7350
Positive Sign Bias  0.9165 0.3595
Joint Effect         1.2947 0.7304


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     549.1   2.737e-104
2    30     568.3   2.426e-101
3    40     596.1   8.127e-101
4    50     620.2   6.383e-100


Elapsed time : 2.880297

Nyblom stability test
------------------------------------
Joint Statistic:  no.parameters>20 (not available)
Individual Statistics:
mu     0.09090
ar1    0.08856
ar2    0.24960
ar3    0.29686
ar4    0.14816
ma1    0.12484
ma2    0.21781
ma3    0.14351
ma4    0.06806
omega  0.61384
alpha1 0.14880
alpha2 0.07238
alpha3 0.17190
alpha4 0.04970
beta1  0.56339
beta2  0.56640
beta3  0.56930
beta4  0.56588
gamma1 0.14810
gamma2 0.09398
gamma3 0.11169
gamma4 0.04664
```

GJR-GARCH(0,1) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(0,1)
Mean Model       : ARFIMA(0,0,1)
Distribution     : norm
```

```
Optimal Parameters
------------------------------------
       Estimate  Std. Error    t value Pr(>|t|)
mu     0.001397    0.000408     3.4214 0.000623
ma1   -0.022517    0.010705    -2.1034 0.035430
omega  0.000008    0.000000 20144.5401 0.000000
beta1  0.994100    0.000072 13787.1423 0.000000

Robust Standard Errors:
       Estimate  Std. Error    t value Pr(>|t|)
mu     0.001397    0.000416     3.3627 0.000772
ma1   -0.022517    0.010850    -2.0754 0.037952
omega  0.000008    0.000000 14522.6279 0.000000
beta1  0.994100    0.000140  7125.2300 0.000000

LogLikelihood : 7550.942
```

```
Information Criteria
------------------------------------

Akaike       -3.8135
Bayes        -3.8072
Shibata      -3.8135
Hannan-Quinn -3.8113
```

```
Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statisti p-value
Lag[1]                  0.000158
Lag[2*(p+q)+(p+q)-1][2] 0.272894
Lag[4*(p+q)+(p+q)-1][5] 1.050514
d.o.f=1
H0 : No serial correlation
```

```
Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      60.33 7.994e-15
Lag[2*(p+q)+(p+q)-1][2]     64.98 0.000e+00
Lag[4*(p+q)+(p+q)-1][5]     82.51 0.000e+00
d.o.f=1

Weighted ARCH LM Tests
------------------------------------
           Statistic Shape Scale   P-Value
ARCH Lag[2]     9.297 0.500 2.000 2.296e-03
ARCH Lag[4]    25.284 1.397 1.611 5.259e-07
ARCH Lag[6]    37.378 2.222 1.500 7.215e-10
```

```
Nyblom stability test
------------------------------------
Joint Statistic:  377.4626
Individual Statistics:
mu      0.09559
ma1     0.09951
omega 375.68039
beta1   1.88115

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         1.07 1.24 1.6
Individual Statistic:    0.35 0.47 0.75
```

```
Sign Bias Test
------------------------------------
                   t-value      prob sig
Sign Bias            0.849 3.959e-01
Negative Sign Bias   7.606 3.521e-14 ***
Positive Sign Bias   3.109 1.893e-03 ***
Joint Effect        68.970 7.092e-15 ***


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     871.5   1.337e-172
2    30     911.2   4.668e-173
3    40     937.0   3.283e-171
4    50     945.9   2.440e-166


Elapsed time : 1.155902
```

GJR-GARCH(2,1) Model Fit - 1

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(2,1)
Mean Model       : ARFIMA(2,0,1)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001374	0.000511	2.68948	0.007156
ar1	0.607749	0.360127	1.68760	0.091488
ar2	0.034377	0.016805	2.04559	0.040797
ma1	-0.618105	0.359113	-1.72120	0.085215
omega	0.000059	0.000009	6.50493	0.000000
alpha1	0.093858	0.020837	4.50434	0.000007
alpha2	0.000000	0.024005	0.00000	1.000000
beta1	0.843291	0.016755	50.33035	0.000000
gamma1	0.088680	0.037055	2.39320	0.016702
gamma2	-0.035597	0.036562	-0.97362	0.330245

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001374	0.000509	2.69976	0.006939
ar1	0.607749	0.398445	1.52530	0.127184
ar2	0.034377	0.023721	1.44923	0.147273
ma1	-0.618105	0.402529	-1.53555	0.124648
omega	0.000059	0.000024	2.43702	0.014809
alpha1	0.093858	0.034651	2.70868	0.006755
alpha2	0.000000	0.035842	0.00000	1.000000
beta1	0.843291	0.043545	19.36617	0.000000
gamma1	0.088680	0.062836	1.41128	0.158161
gamma2	-0.035597	0.083533	-0.42615	0.670001

```
LogLikelihood : 7905.729

Information Criteria
------------------------------------

Akaike       -3.9898
Bayes        -3.9739
Shibata      -3.9898
Hannan-Quinn -3.9841
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	7.785	5.268e-03
Lag[2*(p+q)+(p+q)-1][8]	10.280	7.344e-13
Lag[4*(p+q)+(p+q)-1][14]	15.241	1.466e-03

d.o.f=3
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.07204	0.7884
Lag[2*(p+q)+(p+q)-1][8]	3.02023	0.6818
Lag[4*(p+q)+(p+q)-1][14]	4.02884	0.8775

d.o.f=3

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[4]	1.371	0.500	2.000	0.2417
ARCH Lag[6]	1.383	1.461	1.711	0.6412
ARCH Lag[8]	1.455	2.368	1.583	0.8480

GJR-GARCH(2,1) Model Fit - 2

```
Nyblom stability test
------------------------------------
Joint Statistic:  2.0588
Individual Statistics:
mu      0.07980
ar1     0.12085
ar2     0.07848
ma1     0.11703
omega   0.35876
alpha1 0.33585
alpha2 0.23165
beta1   0.34731
gamma1 0.11399
gamma2 0.10437
```

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          2.29 2.54 3.05
Individual Statistic:     0.35 0.47 0.75

Sign Bias Test
------------------------------------
                    t-value   prob sig
Sign Bias           0.82743 0.4080
Negative Sign Bias 0.06968 0.9445
Positive Sign
Joint Effect
```

```
Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     569.4   1.424e-108
2    30     605.7   4.329e-109
3    40     618.0   2.869e-105
4    50     642.5   2.164e-104
```

```
Elapsed time : 1.03023
```

GJR-GARCH(3,1) Model Fit - 1

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(3,1)
Mean Model       : ARFIMA(3,0,1)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001437	0.000494	2.90894	0.003627
ar1	0.446125	0.238910	1.86733	0.061855
ar2	0.034246	0.016150	2.12047	0.033966
ar3	0.010326	0.016693	0.61861	0.536175
ma1	-0.475852	0.233914	-2.03430	0.041921
omega	0.000073	0.000000	194.23955	0.000000
alpha1	0.058178	0.001801	32.30369	0.000000
alpha2	0.000000	0.000497	0.00000	1.000000
alpha3	0.046944	0.009176	5.11597	0.000000
beta1	0.824567	0.000348	2372.12865	0.000000
gamma1	0.149030	0.000096	1552.51884	0.000000
gamma2	-0.197066	0.000027	-7331.00534	0.000000
gamma3	0.097106	0.014996	6.47535	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001437	0.000513	2.80139	0.005088
ar1	0.446125	0.474511	0.94018	0.347126
ar2	0.034246	0.020149	1.69958	0.089210
ar3	0.010326	0.030487	0.33872	0.734818
ma1	-0.475852	0.468214	-1.01632	0.309480
omega	0.000073	0.000003	27.78601	0.000000
alpha1	0.058178	0.002769	21.00719	0.000000
alpha2	0.000000	0.000948	0.00000	1.000000
alpha3	0.046944	0.016193	2.89911	0.003742
beta1	0.824567	0.000753	1095.73277	0.000000
gamma1	0.149030	0.000288	516.93277	0.000000
gamma2	-0.197066	0.000067	-2952.50197	0.000000
gamma3	0.097106	0.040377	2.40501	0.016172

```
ogLikelihood : 7887.194

nformation Criteria
-----------------------------------

kaike       -3.9789
ayes        -3.9582
hibata      -3.9789
annan-Quinn -3.9716

eighted Ljung-Box Test on Standardized Residuals
-----------------------------------
                         statistic   p-value
ag[1]                        12.72 3.608e-04
ag[2*(p+q)+(p+q)-1][11]      18.45 0.000e+00
ag[4*(p+q)+(p+q)-1][19]      24.03 6.053e-06
.o.f=4
0 : No serial correlation
```

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.2746	0.6003
Lag[2*(p+q)+(p+q)-1][11]	4.6628	0.6321
Lag[4*(p+q)+(p+q)-1][19]	6.2735	0.8512

d.o.f=4

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[5]	0.1222	0.500	2.000	0.7266
ARCH Lag[7]	0.2051	1.473	1.746	0.9695
ARCH Lag[9]	0.7906	2.402	1.619	0.9574

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	585.8	4.897e-112
2	30	615.0	4.883e-111
3	40	648.5	1.645e-111
4	50	668.2	1.395e-109

```
  Elapsed time : 1.447099
```

	t-value	prob	sig
Sign Bias	0.66031	0.5091	
Negative Sign Bias	0.05702	0.9545	
Positive Sign Bias	0.12980	0.8967	
Joint Effect	0.87506	0.8314	

```
Nyblom stability test
-----------------------------------
Joint Statistic:  4.1243
Individual Statistics:
mu     0.07895
ar1    0.03858
ar2    0.16301
ar3    0.04733
ma1    0.03952
omega  0.42080
alpha1 0.22300
alpha2 0.22129
alpha3 0.34766
beta1  0.39190
gamma1 0.24603
gamma2 0.24671
gamma3 0.12626
```

GJR-GARCH(3,1) Model Fit - 2

Nyblom stability test

Joint Statistic:	4.1243
Individual Statistics:	
mu	0.07895
ar1	0.03858
ar2	0.16301
ar3	0.04733
ma1	0.03952
omega	0.42080
alpha1	0.22300
alpha2	0.22129
alpha3	0.34766
beta1	0.39190
gamma1	0.24603
gamma2	0.24671
gamma3	0.12626

Asymptotic Critical Values (10% 5% 1%)	
Joint Statistic:	2.89 3.15 3.69
Individual Statistic:	0.35 0.47 0.75

Sign Bias Test

	t-value	prob	sig
Sign Bias	0.66031	0.5091	
Negative Sign Bias	0.05702	0.9545	
Positive Sign Bias	0.12980	0.8967	
Joint Effect	0.87506	0.8314	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	585.8	4.897e-112
2	30	615.0	4.883e-111
3	40	648.5	1.645e-111
4	50	668.2	1.395e-109

Elapsed time : 1.508597

GJR-GARCH(4,1) Model Fit - 1

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(4,1)
Mean Model       : ARFIMA(4,0,1)
LogLikelihood : 7927.026

Information Criteria
-----------------------------------

Akaike       -3.9975
Bayes        -3.9721
Shibata      -3.9975
Hannan-Quinn -3.9885
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t
mu	0.001369	0.000508	2.695433	0.0070
ar1	-0.907574	0.148406	-6.115490	0.0000
ar2	0.028967	0.024539	1.180457	0.2378
ar3	0.061482	0.023123	2.658865	0.0078
ar4	0.032012	0.016667	1.920651	0.0547
ma1	0.902631	0.147061	6.137794	0.0000
omega	0.000103	0.000018	5.743894	0.0000
alpha1	0.072475	0.017400	4.165227	0.0000
alpha2	0.000000	0.034827	0.000000	1.0000
alpha3	0.000000	0.035509	0.000001	0.9999
alpha4	0.086246	0.026727	3.226864	0.0012
beta1	0.743594	0.032009	23.231098	0.0000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001369	0.000507	2.701676	0.006899
ar1	-0.907574	0.090912	-9.983007	0.000000
ar2	0.028967	0.025064	1.155716	0.247797
ar3	0.061482	0.026198	2.346860	0.018932
ar4	0.032012	0.018026	1.775942	0.075742
ma1	0.902631	0.087555	10.309330	0.000000
omega	0.000103	0.000063	1.624978	0.104167
alpha1	0.072475	0.029626	2.446318	0.014432
alpha2	0.000000	0.058315	0.000000	1.000000
alpha3	0.000000	0.059722	0.000001	0.999999
alpha4	0.086246	0.054735	1.575692	0.115097
beta1	0.743594	0.118317	6.284740	0.000000
gamma1	0.119559	0.060649	1.971334	0.048686
gamma2	-0.057501	0.079934	-0.719364	0.471917
gamma3	-0.048662	0.077418	-0.628563	0.529635
gamma4	0.051825	0.077392	0.669642	0.503086

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	6.975	0.008267
Lag[2*(p+q)+(p+q)-1][14]	14.685	0.000000
Lag[4*(p+q)+(p+q)-1][24]	20.904	0.003250

d.o.f=5
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Resi

	statistic	p-value
Lag[1]	0.2485	0.6181
Lag[2*(p+q)+(p+q)-1][14]	2.4776	0.9787
Lag[4*(p+q)+(p+q)-1][24]	4.3419	0.9955

d.o.f=5

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[6]	0.2651	0.500	2.000	0.6067
ARCH Lag[8]	0.4620	1.480	1.774	0.9106
ARCH Lag[10]	1.3460	2.424	1.650	0.8862

GJR-GARCH(4,1) Model Fit - 2

```
Nyblom stability test
------------------------------------
Joint Statistic:  5.7233
Individual Statistics:
mu      0.06502
ar1     0.22195
ar2     0.46702
ar3     0.60017
ar4     0.85968
ma1     0.21557
omega   0.37202
alpha1 0.47880
alpha2 0.13563
alpha3 0.19732
alpha4 0.30044
beta1   0.42593
gamma1 0.09828
gamma2 0.07822
gamma3 0.05744
gamma4 0.10079
```

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          3.46 3.75 4.3
Individual Statistic:     0.35 0.47 0.75
```

```
Sign Bias Test
---------------------------
                    t-value
Sign Bias           0.33340
Negative Sign Bias 0.05857
Positive Sign Bias 0.09935
Joint Effect        0.18188
```

```
Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     547.6   5.433e-104
2    30     578.6   1.784e-103
3    40     591.0   8.941e-100
4    50     618.9    1.158e-99
```

```
Elapsed time : 1.950433
```

GJR-GARCH(0,3) Model Fit

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          1.89 2.11 2.59
Individual Statistic:     0.35 0.47 0.75

Sign Bias Test
-----------------------------------
                   t-value      prob sig
Sign Bias            2.634 8.476e-03 ***
Negative Sign Bias   8.416 5.370e-17 ***

Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     873.6   4.769e-173
2    30     908.0   2.220e-172
3    40     939.4   1.024e-171
4    50     957.5   9.614e-169


Elapsed time : 0.5756979
```

```
Robust Standard Errors:
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.001406    0.000136    10.37362  0.00000
ma1    -0.022002    0.022222    -0.99011  0.32212
ma2     0.010899    0.019597     0.55618  0.57809
ma3     0.010833    0.017611     0.61514  0.53846
omega   0.000008    0.000000  7478.74027  0.00000
beta1   0.647179    0.000123  5263.33140  0.00000
beta2   0.280869    0.000005 56671.49248  0.00000
beta3   0.065655    0.000080   817.43308  0.00000

Optimal Parameters
------------------------------------
        Estimate  Std. Error     t value Pr
mu      0.001406    0.000109    12.89868
ma1    -0.022002    0.015763    -1.39578
ma2     0.010899    0.015648     0.69653

Ljung-Box Test on Standardized Squared Residuals
------------------------------
                 statistic   p-value
                     61.18 5.218e-15
+q)+(p+q)-1][8]     106.65 0.000e+00
+q)+(p+q)-1][14]    137.47 0.000e+00

ARCH LM Tests
------------------------------
    Statistic Shape Scale   P-Value
[4]     23.53 0.500 2.000 1.231e-06
[6]     37.52 1.461 1.711 1.436e-09
[8]     64.71 2.368 1.583 2.220e-16
```

```
LogLikelihood : 7551.499

Information Criteria
------------------------------
Akaike       -3.8118
Bayes        -3.7991
Shibata      -3.8118
Hannan-Quinn -3.8073

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                          statistic p-value
Lag[1]                     0.003701  0.9515
Lag[2*(p+q)+(p+q)-1][8]    3.311138  0.9821
Lag[4*(p+q)+(p+q)-1][14]   8.938670  0.2293
d.o.f=3
H0 : No serial correlation
```

```
Nyblom stability test
------------------------------------
Joint Statistic:  -50.1745
Individual Statistics:
mu      0.09036
ma1     0.09365
ma2     0.36249
ma3     0.09557
omega 319.86609
beta1   1.93391
beta2   1.93390
beta3   1.93389
```

GJR-GARCH(0,4) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(0,4)
Mean Model       : ARFIMA(0,0,4)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001401	0.000574	2.4395e+00	0.014707
ma1	-0.024319	0.015900	-1.5294e+00	0.126155
ma2	0.012432	0.015929	7.8046e-01	0.435119
ma3	0.011224	0.015405	7.2860e-01	0.466244
ma4	0.011484	0.016379	7.0115e-01	0.483208
omega	0.000001	0.000000	9.4699e+01	0.000000
beta1	0.000000	0.027487	0.0000e+00	1.000000
beta2	0.000000	0.017661	0.0000e+00	1.000000
beta3	0.000000	0.031925	2.0000e-06	0.999998
beta4	0.999000	0.000002	4.6149e+05	0.000000

Robust Standard Er

	Estimate	S
mu	0.001401	
ma1	-0.024319	
ma2	0.012432	
ma3	0.011224	
ma4	0.011484	
omega	0.000001	
beta1	0.000000	
beta2	0.000000	
beta3	0.000000	
beta4	0.999000	

LogLikelihood : 75

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	773.1	1.124e-151
2	30	817.7	2.267e-153
3	40	851.7	1.893e-153
4	50	879.8	9.926e-153

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	55.24	1.069e-13
Lag[2*(p+q)+(p+q)-1][11]	114.76	0.000e+00
Lag[4*(p+q)+(p+q)-1][19]	143.87	0.000e+00

d.o.f=4

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[5]	8.791	0.500	2.000	3.026e-03
ARCH Lag[7]	43.960	1.473	1.746	6.147e-11
ARCH Lag[9]	49.333	2.402	1.619	5.918e-12

Nyblom stability test

Joint Statistic: 665.74

Individual Statistics:

mu	0.08731
ma1	0.10059
ma2	0.34864
ma3	0.07757
ma4	0.18959
omega	5.03288
beta1	2.59857
beta2	2.59860
beta3	2.59832
beta4	2.59832

Elapsed time : 1.269272

Asymptotic Critical Values (10% 5% 1%)

Joint Statistic:	2.29	2.54	3.05
Individual Statistic:	0.35	0.47	0.75

Sign Bias Test

	t-value	prob	sig
Sign Bias	2.514	1.198e-02	**
Negative Sign Bias	8.194	3.380e-16	***
Positive Sign Bias	2.247	2.467e-02	**
Joint Effect	72.199	1.443e-15	***

GJR-GARCH(4,1) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(4,1)
Mean Model       : ARFIMA(4,0,1)
Distribution     : norm

LogLikelihood : 7927.026

Information Criteria
------------------------------------

Akaike       -3.9975
Bayes        -3.9721
Shibata      -3.9975
Hannan-Quinn -3.9885

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic  p-value
Lag[1]                      6.975 0.008267
Lag[2*(p+q)+(p+q)-1][14]   14.685 0.000000
Lag[4*(p+q)+(p+q)-1][24]   20.904 0.003250
d.o.f=5
H0 : No serial correlation
```

```
Optimal Parameters
------------------------------------
        Estimate  Std. Error   t value Pr(>|t|)
mu      0.001369    0.000508  2.695433 0.007030
ar1    -0.907574    0.148406 -6.115490 0.000000
ar2     0.028967    0.024539  1.180457 0.237819
ar3     0.061482    0.023123  2.658865 0.007840
ar4     0.032012    0.016667  1.920651 0.054776
ma1     0.902631    0.147061  6.137794 0.000000
omega   0.000103    0.000018  5.743894 0.000000
alpha1  0.072475    0.017400  4.165227 0.000031
alpha2  0.000000    0.034827  0.000000 1.000000
alpha3  0.000000    0.035509  0.000001 0.999999
alpha4  0.086246    0.026727  3.226864 0.001252
beta1   0.743594    0.032009 23.231098 0.000000
gamma1  0.119559    0.036869  3.242802 0.001184
gamma2 -0.057501    0.047797 -1.203022 0.228968
gamma3 -0.048662    0.043248 -1.125194 0.260507
gamma4  0.051825    0.031077  1.667648 0.095386
```

```
Weighted Ljung-Box Test on Standardized Squared R
------------------------------------
                        statistic p-value
Lag[1]                     0.2485  0.6181
Lag[2*(p+q)+(p+q)-1][14]   2.4776  0.9787
Lag[4*(p+q)+(p+q)-1][24]   4.3419  0.9955
d.o.f=5

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[6]    0.2651 0.500 2.000  0.6067
ARCH Lag[8]    0.4620 1.480 1.774  0.9106
ARCH Lag[10]   1.3460 2.424 1.650  0.8862
```

```
Robust Standard Errors:
        Estimate  Std. Error
mu      0.001369    0.000507  2
ar1    -0.907574    0.090912 -9
ar2     0.028967    0.025064  1
ar3     0.061482    0.026198  2
ar4     0.032012    0.018026  1
ma1     0.902631    0.087555 10
omega   0.000103    0.000063  1
alpha1  0.072475    0.029626  2
alpha2  0.000000    0.058315  0
alpha3  0.000000    0.059722  0
alpha4  0.086246    0.054735  1.575692 0.115097
beta1   0.743594    0.118317  6.284740 0.000000
gamma1  0.119559    0.060649  1.971334 0.048686
gamma2 -0.057501    0.079934 -0.719364 0.471917
                                                35
                                                86
```

```
Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     547.6   5.433e-104
2    30     578.6   1.784e-103
3    40     591.0   8.941e-100
4    50     618.9    1.158e-99

Elapsed time : 1.877523
```

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          3.46 3.75 4.3
Individual Statistic:     0.35 0.47 0.75

Sign Bias Test
------------------------------------
                    t-value   prob sig
Sign Bias           0.33340 0.7389
Negative Sign Bias  0.05857 0.9533
Positive Sign Bias  0.09935 0.9209
Joint Effect        0.18188 0.9805
```

```
Nyblom stability test
------------------------------------
Joint Statistic:  5.7233
Individual Statistics:
mu     0.06502
ar1    0.22195
ar2    0.46702
ar3    0.60017
ar4    0.85968
ma1    0.21557
omega  0.37202
alpha1 0.47880
alpha2 0.13563
alpha3 0.19732
alpha4 0.30044
beta1  0.42593
gamma1 0.09828
gamma2 0.07822
gamma3 0.05744
gamma4 0.10079
```

GJR-GARCH(1,2) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(1,2)
Mean Model       : ARFIMA(1,0,2)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001375	0.000511	2.6885	0.007177
ar1	0.654898	0.329995	1.9846	0.047192
ma1	-0.665595	0.329891	-2.0176	0.043630
ma2	0.034879	0.017818	1.9575	0.050290
omega	0.000074	0.000010	7.0454	0.000000
alpha1	0.117041	0.015819	7.3986	0.000000
beta1	0.551915	0.091970	6.0010	0.000000
beta2	0.250124	0.081036	3.0866	0.002025
gamma1	0.071148	0.020227		

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001375	0.000509	2.7036	0.006859
ar1	0.654898	0.369812	1.7709	0.076578
ma1	-0.665595	0.372532	-1.7867	0.073989
ma2	0.034879	0.026621	1.3102	0.190126
omega	0.000074	0.000022	3.2836	0.001025
alpha1	0.117041	0.028826	4.0602	0.000049
beta1	0.551915	0.119704	4.6107	0.000004

```
LogLikelihood : 7907.645

Information Criteria
-----------------------------------

Akaike       -3.9912
Bayes        -3.9769
Shibata      -3.9912
Hannan-Quinn -3.9862

Weighted Ljung-Box Test on Standardized Residuals
-----------------------------------
```

```
Weighted Ljung-Box Test on Standardized Squared Residuals
-----------------------------------
                         statistic p-value
Lag[1]                    0.003003  0.9563
Lag[2*(p+q)+(p+q)-1][8]   2.753546  0.7305
Lag[4*(p+q)+(p+q)-1][14]  3.754402  0.9025
d.o.f=3

Weighted ARCH LM Tests
-----------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[4]     1.410 0.500 2.000  0.2351
ARCH Lag[6]     1.425 1.461 1.711  0.6301
ARCH Lag[8]     1.492 2.368 1.583  0.8411
```

```
Nyblom stability test
-----------------------------------
Joint Statistic:  1.7926
Individual Statistics:
mu      0.08668
ar1     0.12602
ma1     0.12151
ma2     0.08310
omega   0.35766
alpha1 0.34370
beta1   0.34293
beta2   0.34826
gamma1 0.11928
```

```
Adjusted Pearson Goodness-of-Fit Test:
-----------------------------------
  group statistic p-value(g-1)
1    20     565.2   1.073e-107
2    30     596.9   2.803e-107
3    40     620.0   1.120e-105
4    50     638.9   1.128e-103


Elapsed time : 0.6851459
```

```
-----------------------------------
                    t-value   prob sig
Sign Bias          0.845012 0.3982
Negative Sign Bias 0.003182 0.9975
Positive Sign Bias 1.033750 0.3013
Joint Effect       1.265489 0.7373
```

GJR-GARCH(1,3) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(1,3)
Mean Model       : ARFIMA(1,0,3)
Distribution     : norm

LogLikelihood : 7925.436

Information Criteria
------------------------------------

Akaike       -3.9992
Bayes        -3.9817
Shibata      -3.9992
Hannan-Quinn -3.9930

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic   p-value
Lag[1]                      7.678 0.0055898
Lag[2*(p+q)+(p+q)-1][11]   13.804 0.0000000
Lag[4*(p+q)+(p+q)-1][19]   19.726 0.0005067
d.o.f=4
H0 : No serial correlation
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001354	0.000491	2.756481	0.005843
ar1	-0.022390	0.659454	-0.033952	0.972916
ma1	0.009056	0.658744	0.013747	0.989032
ma2	0.020274	0.019005	1.066770	0.286076
ma3	0.030737	0.020812	1.476877	0.139709
omega	0.000095	0.000013	7.534321	0.000000
alpha1	0.168208	0.020235	8.312717	0.000000
beta1	0.241708	0.048643	4.969044	0.000001
beta2	0.016553	0.055778	0.296760	0.766650
beta3	0.469736	0.044338	10.59430	
gamma1	0.094162	0.028073	3.35413	

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001354	0.000506	2.678581	0.007393
ar1	-0.022390	0.761152	-0.029416	0.976533
ma1	0.009056	0.762161	0.011882	0.990520
ma2	0.020274	0.019476	1.040952	0.297898
ma3	0.030737	0.020883	1.471879	0.141054
omega	0.000095	0.000033	2.915032	0.003557
alpha1	0.168208	0.039929	4.212666	0.000025
beta1	0.241708	0.131046	1.844451	0.065117

```
Weighted Ljung-Box Test on Standardized S
------------------------------------
                          statistic p
Lag[1]                       0.4028 (
Lag[2*(p+q)+(p+q)-1][11]     1.5705 (
Lag[4*(p+q)+(p+q)-1][19]     2.8628 (
d.o.f=4

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-
ARCH Lag[5]    0.1011 0.500 2.000 0
ARCH Lag[7]    0.2956 1.473 1.746 0
ARCH Lag[9]    0.6339 2.402 1.619 0
```

```
Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     557.1   5.600e-106
2    30     584.1   1.257e-104
3    40     600.2   1.185e-101
4    50     611.0    4.574e-98

Elapsed time : 0.8873119
```

GJR-GARCH(1,4) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(1,4)
Mean Model       : ARFIMA(1,0,4)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001435	0.000677	2.1178e+00	0.034190
ar1	0.990069	0.002612	3.7906e+02	0.000000
ma1	-1.002597	0.000001	-1.0155e+06	0.000000
ma2	0.032961	0.017013	1.9374e+00	0.052699
ma3	0.010593	0.023551	4.4977e-01	0.652877
ma4	-0.026741	0.015853	-1.6868e+00	0.091646
omega	0.000097	0.000014	7.0773e+00	0.000000
alpha1	0.163890	0.022089	7.4195e+00	0.000000
beta1	0.237129	0.078113	3.0357e+00	0.002399
beta2	0.018396	0.055062	3.3409e-01	0.738312
beta3	0.471760	0.051483	9.1634e+00	0.000000
beta4	0.000001	0.069490	1.0000e-05	0.999992
gamma1	0.100412	0.029737	3.3767e+00	0.000734

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001435	0.000695	2.0635e+00	0.039062
ar1	0.990069	0.003510	2.8211e+02	0.000000
ma1	-1.002597	0.000001	-1.1737e+06	0.000000
ma2	0.032961	0.016654	1.9791e+00	0.047802
ma3	0.010593	0.023412	4.5244e-01	0.650948
ma4	-0.026741	0.015529	-1.7220e+00	0.085071
omega	0.000097	0.000033	2.9523e+00	0.003154
alpha1	0.163890	0.044588	3.6756e+00	0.000237
beta1	0.237129	0.162330	1.4608e+00	0.144075
beta2	0.018396	0.100968	1.8219e-01	0.855432
beta3	0.471760	0.108632	4.3427e+00	0.000014
beta4	0.000001	0.138965	5.0000e-06	0.999996
gamma1	0.100412	0.079794	1.2584e+00	0.208252

```
LogLikelihood : 7926.488

Information Criteria
------------------------------------
Akaike       -3.9987
Bayes        -3.9781
Shibata      -3.9988
Hannan-Quinn -3.9914

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic  p-value
Lag[1]                      7.247 0.007103
Lag[2*(p+q)+(p+q)-1][14]   14.314 0.000000
Lag[4*(p+q)+(p+q)-1][24]   19.497 0.009806
d.o.f=5
H0 : No serial correlation
```

```
Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.3593  0.5489
Lag[2*(p+q)+(p+q)-1][14]   1.9192  0.9926
Lag[4*(p+q)+(p+q)-1][24]   3.8137  0.9981
d.o.f=5

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[6]    0.2165 0.500 2.000  0.6418
ARCH Lag[8]    0.2819 1.480 1.774  0.9543
ARCH Lag[10]   0.7983 2.424 1.650  0.9599
```

```
Nyblom stability test
------------------------------------
Joint Statistic:  2.3825
Individual Statistics:
mu      0.07329
ar1     0.23802
ma1     0.21874
ma2     0.19406
ma3     0.21056
ma4     0.18682
omega   0.34842
alpha1 0.36140
beta1   0.41417
beta2   0.36728
beta3   0.35112
beta4   0.38634
gamma1 0.14299
```

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          2.89 3.15 3.69
Individual Statistic:     0.35 0.47 0.75
```

```
Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     561.4   7.082e-107
2    30     580.5   7.230e-104
3    40     603.5   2.623e-102
4    50     606.8    3.132e-97


Elapsed time : 1.257403
```

```
:
----------------------
      t-value   prob sig
       1.1053 0.2691
Bias   0.4611 0.6448
Bias   1.6444 0.1002
       3.3544 0.3401
```

GJR-GARCH(2,2) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(2,2)
Mean Model       : ARFIMA(2,0,2)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001399	0.000492	2.84201	0.004483
ar1	1.745918	0.004022	434.04398	0.000000
ar2	-0.969951	0.003595	-269.79244	0.000000
ma1	-1.738006	0.003818	-455.22995	0.000000
ma2	0.969240	0.000966	1003.75659	0.000000
omega	0.000078	0.000033	2.39249	0.016735
alpha1	0.126139	0.026366	4.78418	0.000002
alpha2	0.000000	0.051548	0.00000	1.000000
beta1	0.472664	0.394760	1.19734	0.231173
beta2	0.315335	0.315339	0.99999	0.317317
gamma1	0.046443	0.065556	0.70845	0.478663
gamma2	0.027849	0.087595	0.31793	0.750540

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001399	0.000502	2.789251	0.005283
ar1	1.745918	0.007143	244.411856	0.000000
ar2	-0.969951	0.004735	-204.867412	0.000000
ma1	-1.738006	0.003345	-519.528595	0.000000
ma2	0.969240	0.001595	607.610881	0.000000
omega	0.000078	0.000163	0.480138	0.631130
alpha1	0.126139	0.049955	2.525023	0.011569
alpha2	0.000000	0.215837	0.000000	1.000000
beta1	0.472664	1.913239	0.247049	0.804870
beta2	0.315335	1.501489	0.210015	0.833656
gamma1	0.046443	0.292112	0.158991	0.873676
gamma2	0.027849	0.454771	0.061237	0.951171

LogLikelihood : 7913.511

Information Criteria

Akaike	-3.9927
Bayes	-3.9736
Shibata	-3.9927
Hannan-Quinn	-3.9859

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	3.578	5.854e-02
Lag[2*(p+q)+(p+q)-1][11]	10.249	2.055e-09
Lag[4*(p+q)+(p+q)-1][19]	17.157	5.182e-03

d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standa

	statisti
Lag[1]	0.0156
Lag[2*(p+q)+(p+q)-1][11]	3.4671
Lag[4*(p+q)+(p+q)-1][19]	4.8001

d.o.f=4

Weighted ARCH LM Tests

	Statistic	Shape	Scale
ARCH Lag[5]	0.00126	0.500	2.000
ARCH Lag[7]	0.02547	1.473	1.746
ARCH Lag[9]	0.50654	2.402	1.619

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	516.8	1.666e-97
2	30	549.2	2.061e-97
3	40	570.7	1.232e-95
4	50	591.1	4.380e-94

Elapsed time : 1.936076

cal Values (10% 5% 1%)
2.69 2.96 3.51
stic: 0.35 0.47 0.75

	t-value	prob sig
	0.7062	0.4801
is	0.0639	0.9491
is	1.0708	0.2843
	1.1929	0.7547

GJR-GARCH(2,3) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(2,3)
Mean Model       : ARFIMA(2,0,3)
```

LogLikelihood : 7930.166

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	546.4	1.002e-103
2	30	554.3	1.854e-98
3	40	590.0	1.477e-99
4	50	604.8	7.867e-97

Elapsed time : 1.529593

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001394	0.000480	2.90164	0.003712
ar1	-0.359835	0.120954	-2.97498	0.002930
ar2	-0.802769	0.087985	-9.12392	0.000000
ma1	0.348102	0.120292	2.89380	0.003806
ma2	0.819118	0.090150	9.08616	0.000000
ma3	0.025771	0.021557	1.19546	0.231907
omega	0.000099	0.000016	6.29898	0.000000
alpha1	0.166033	0.026883	6.17603	0.000000
alpha2	0.009808	0.029046	0.33768	0.735604

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001394	0.000496	2.806819	0.005003
ar1	-0.359835	0.191468	-1.879351	0.060197
ar2	-0.802769	0.127770	-6.282942	0.000000
ma1	0.348102	0.186852	1.862985	0.062464
ma2	0.819118	0.139575	5.868674	0.000000
ma3	0.025771	0.025365	1.016012	0.309624
omega	0.000099	0.000040	2.509392	0.012094
alpha1	0.166033	0.048592	3.416910	0.000633
alpha2	0.009808	0.048962	0.200327	0.841225
beta1	0.220533	0.195346	1.128935	0.258925
beta2	0.014549	0.145781	0.099798	0.920504

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic: 3.08 3.34 3.9
Individual Statistic: 0.35 0.47 0.75

Weighted Ljung-B
Lag[1]
Lag[2*(p+q)+(p+q
Lag[4*(p+q)+(p+q
d.o.f=5

Weighted ARCH LM
ARCH Lag[6]
ARCH Lag[8]
ARCH Lag[10]

Sign Bias Test

	t-value	prob sig
Sign Bias	0.3155	0.7524
Negative Sign Bias	0.8155	0.4148
Positive Sign Bias	1.3660	0.1720
Joint Effect	2.5313	0.4697

stability test
tatistic: 2.7268
ual Statistics:
0.09155
0.10312
0.21275
0.13054
0.17570
0.59480
0.34203
0.34471
0.56778
0.41676
0.36550
0.34933
0.13249
0.22721

GJR-GARCH(2,4) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(2,4)
Mean Model       : ARFIMA(2,0,4)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001335	0.000074	1.8062e+01	0.000000
ar1	-0.923008	0.000022	-4.1281e+04	0.000000
ar2	-0.998410	0.000028	-3.5900e+04	0.000000
ma1	0.856159	0.000018	4.7796e+04	0.000000
ma2	0.953074	0.000026	3.7079e+04	0.000000
ma3	-0.056554	0.000008	-6.9153e+03	0.000000
ma4	0.012922	0.000014	9.0263e+02	0.000000
omega	0.000100	0.000011	9.2719e+00	0.000000
alpha1	0.137692	0.001597	8.6242e+01	0.000000
alpha2	0.029346	0.009161	3.2034e+00	0.001358
beta1	0.004911	0.032833	1.4958e-01	0.881098
be				
be				
be				
ga				
ga				

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001335	0.000396	3.37018	0.000751
ar1	-0.923008	0.001116	-826.77380	0.000000
ar2	-0.998410	0.001443	-692.00457	0.000000
ma1	0.856159	0.001380	620.25947	0.000000
ma2	0.953074	0.000483	1972.29044	0.000000
ma3	-0.056554	0.000229	-247.16308	0.000000
ma4	0.012922	0.000133	97.52441	0.000000
omega	0.000100	0.000073	1.38167	0.167074
alpha1	0.137692	0.070550	1.95168	0.050976
alpha2	0.029346	0.117619	0.24950	0.802976
beta1	0.004911	0.394771	0.01244	0.990074
ta2	0.124534	0.167957	0.74146	0.458413
ta3	0.454159	0.099156	4.58024	0.000005
ta4	0.144321	0.424304	0.34014	0.733754
mma1	0.152582	0.167783	0.90940	0.363138
mma2	-0.061238	0.001366	-44.83727	0.000000

```
LogLikelihood : 7917.026

Information Criteria
-------------------------------------

Akaike       -3.9924
Bayes        -3.9670
Shibata      -3.9925
Hannan-Quinn -3.9834
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	27.86	1.301e-07
Lag[2*(p+q)+(p+q)-1][17]	42.76	0.000e+00
Lag[4*(p+q)+(p+q)-1][29]	51.03	5.551e-16

d.o.f=6
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.1605	0.6887
Lag[2*(p+q)+(p+q)-1][17]	5.5111	0.8475
Lag[4*(p+q)+(p+q)-1][29]	8.5400	0.9461

d.o.f=6

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[7]	0.004975	0.500	2.000	0.9438
ARCH Lag[9]	0.493608	1.485	1.796	0.9049
ARCH Lag[11]	1.395822	2.440	1.677	0.8842

Nyblom stability test

Joint Statistic: 7.5296

Individual Statistics:

mu	0.02432
ar1	0.07776
ar2	0.01334
ma1	0.03127
ma2	0.10868
ma3	0.03564
ma4	0.04154
omega	0.27085
alpha1	0.19881
alpha2	0.27924
beta1	0.27576
beta2	0.23622
beta3	0.23709
beta4	0.27736
gamma1	0.12213
gamma2	0.32489

Asymptotic Critical Values (10% 5% 1%)

Joint Statistic:	3.46	3.75	4.3
Individual Statistic:	0.35	0.47	

Sign Bias Test

	t-value	prob
Sign Bias	1.0075	0.3138
Negative Sign Bias	1.1500	0.2502
Positive Sign Bias	0.0644	0.9487
Joint Effect	1.7262	0.6311

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	548.1	4.316e-104
2	30	586.6	3.952e-105
3	40	595.5	1.101e-100
4	50	610.0	7.293e-98

Elapsed time : 3.289039

GJR-GARCH(3,2) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(3,2)
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001277	0.000473	2.6969e+00	0.006999
ar1	-0.447125	0.001315	-3.4005e+02	0.000000
ar2	-0.812561	0.000578	-1.4054e+03	0.000000
ar3	0.032449	0.011857	2.7366e+00	0.006207
ma1	0.433474	0.004579	9.4657e+01	0.000000
ma2	0.817134	0.000278	2.9374e+03	0.000000
omega	0.000070	0.000003	2.7265e+01	0.000000
alpha1	0.126324	0.002307	5.4750e+01	0.000000
alpha2	0.003743	0.008930	4.1911e-01	0.675137
alpha3	0.000000	0.003069	2.5000e-05	0.999980
	[illegible]	0.003575	5.5843e+01	0.000000
	[illegible]4	0.000665	8.9848e+02	0.000000
	[illegible]4	0.000228	3.1485e+02	0.000000
	[illegible]	[illegible]	[illegible]	[illegible]

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001277	0.000486	2.625645	0.008648
ar1	-0.447125	0.002401	-186.206396	0.000000
ar2	-0.812561	0.001056	-769.705142	0.000000
ar3	0.032449	0.018453	1.758444	0.078672
ma1	0.433474	0.009028	48.012203	0.000000
ma2	0.817134	0.000553	1476.894623	0.000000
omega	0.000070	0.000005	14.816761	0.000000
alpha1	0.126324	0.004262	29.636859	0.000000
alpha2	0.003743	0.018733	0.199782	0.841651
alpha3	0.000000	0.006355	0.000012	0.999990
beta1	0.199654	0.005861	34.066285	0.000000
beta2	[illegible]	[illegible]	[illegible]	[illegible]
gamma1	[illegible]	[illegible]	[illegible]	[illegible]

Adjusted Pearson Goodness-of-Fit Test:

group	statistic	p-value(g-1)
20	553.8	2.806e-105
30	564.2	1.692e-100
40	597.7	3.953e-101
50	628.0	1.767e-101

Elapsed time : 2.653827

nted Ljung-Box Test on Standardized

	statistic	p-va
1]	0.005515	0.9
2*(p+q)+(p+q)-1][14]	5.567268	0.6
4*(p+q)+(p+q)-1][24]	8.119766	0.8

f=5

nted ARCH LM Tests

	Statistic	Shape	Scale	P-Va
Lag[6]	0.07641	0.500	2.000	0.7
Lag[8]	0.67697	1.480	1.774	0.8
Lag[10]	1.09662	2.424	1.650	0.9

Asymptotic Critical Values (10% 5% 1%)

Joint Statistic:	3.26	3.54	4.07
Individual Statistic:	0.35	0.47	0.75

Sign Bias Test

	t-value	prob	sig
Sign Bias	0.7132	0.4758	
Negative Sign Bias	0.1102	0.9123	
Positive Sign Bias	1.0469	0.2952	
Joint Effect	1.2132	0.7498	

GJR-GARCH(3,3) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(3,3)
Mean Model       : ARFIMA(3,0,3)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001442	0.000627	2.30021	0.021436
ar1	0.668000	0.009254	72.18220	0.000000
ar2	-0.546121	0.010992	-49.68456	0.000000
ar3	0.845616	0.009261	91.30909	0.000000
ma1	-0.690476	0.000647	-1067.17049	0.000000
ma2	0.586757	0.000670	875.23507	0.000000
ma3	-0.853351	0.000568	-1502.29610	0.000000
omega	0.000110	0.000023	4.89780	0.000001
alpha1	0.152575	0.026575	5.74129	0.000000
alpha2	0.025275	0.035602	0.70993	0.477750
alpha3	0.019296	0.025816	0.74742	0.454810
beta1	0.000000	0.192028	0.00000	1.000000
beta2	0.254559	0.138504	1.83792	0.066075
beta3	0.431440	0.066832	6.45562	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001442	0.000644	2.2378e+00	0.025234
ar1	0.668000	0.009558	6.9888e+01	0.000000
ar2	-0.546121	0.013961	-3.9118e+01	0.000000
ar3	0.845616	0.013245	6.3845e+01	0.000000
ma1	-0.690476	0.000015	-4.5826e+04	0.000000
ma2	0.586757	0.000015	3.9125e+04	0.000000
ma3	-0.853351	0.000015	-5.7158e+04	0.000000
omega	0.000110	0.000077	1.4345e+00	0.151441
alpha1	0.152575	0.053155	2.8704e+00	0.004100
alpha2	0.025275	0.122285	2.0669e-01	0.836251
alpha3	0.019296	0.042632	4.5261e-01	0.650831
beta1	0.000000	0.754614	0.0000e+00	1.000000
beta2	0.254559	0.452642	5.6239e-01	0.573854
beta3	0.431440	0.195563	2.2061e+00	0.027374
gamma1	0.100336	0.090330	1.1108e+00	0.266665
			5676e-01	0.647843
			1093e+00	0.267305

```
LogLikelihood : 7933.372

Information Criteria
-------------------------------------

Akaike       -4.0002
Bayes        -3.9732
Shibata      -4.0002
Hannan-Quinn -3.9906
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	10.93	0.0009448
Lag[2*(p+q)+(p+q)-1][17]	17.01	0.0000000
Lag[4*(p+q)+(p+q)-1][29]	22.46	0.0104325

d.o.f=6
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residu

	statistic	p-value
Lag[1]	0.2042	0.6514
Lag[2*(p+q)+(p+q)-1][17]	2.8419	0.9903
Lag[4*(p+q)+(p+q)-1][29]	5.3103	0.9977

d.o.f=6

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	557.9	3.803e-106
2	30	582.0	3.459e-104
3	40	600.7	9.621e-102
4	50	615.5	5.665e-99

Elapsed time : 2.547912

```
otic Critical Values (10% 5% 1%)
 Statistic:         3.64 3.95 4.51
dual Statistic:     0.35 0.47 0.75

ias Test
--------------------------------

             t-value   prob sig
ias           0.7321 0.4641
ve Sign Bias  1.1888 0.2346
ve Sign Bias  0.7123 0.4763
 Effect       2.4881 0.4774
```

```
y test
--------------------
:  3.0907
istics:
```

GJR-GARCH(3,4) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(3,4)
Mean Model       : ARFIMA(3,0,4)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001422	0.000612	2.3227e+00	0.020195
ar1	0.605028	0.017802	3.3986e+01	0.000000
ar2	-0.441062	0.014876	-2.9648e+01	0.000000
ar3	0.804064	0.013410	5.9961e+01	0.000000
ma1	-0.617141	0.000727	-8.4928e+02	0.000000
ma2	0.473720	0.000787	6.0171e+02	0.000000
ma3	-0.791944	0.000504	-1.5724e+03	0.000000
ma4	-0.022791	0.016993	-1.3412e+00	0.179866
omega	0.000104	0.000014	7.6368e+00	0.000000
alpha1	0.152972	0.002448	6.2493e+01	0.000000
alpha2	0.021904	0.015075	1.4530e+00	0.146222
alpha3	0.013522	0.004788	2.8244e+00	0.004737
beta1	0.000001	0.006545	1.3200e-04	0.999895
beta2	0.285349	0.000024	1.2066e+04	0.000000
beta3	0.417910	0.041904	9.9730e+00	0.000000
beta4	0.000000	0.044709	0.0000e+00	1.000000
gamma1	0.100956	0.021681	4.6563e+00	0.000003
gamma2	0.093377	0.022977	4.0640e+00	0.000048
gamma3	-0.099631	0.000456	-2.1849e+02	0.000000

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001422	0.000607	2.3426e+00	0.019149
ar1	0.605028	0.017148	3.5283e+01	0.000000
ar2	-0.441062	0.017468	-2.5250e+01	0.000000
ar3	0.804064	0.015931	5.0473e+01	0.000000
ma1	-0.617141	0.000069	-8.9058e+03	0.000000
ma2	0.473720	0.000024	1.9883e+04	0.000000
ma3	-0.791944	0.000132	-5.9918e+03	0.000000
ma4	-0.022791	0.016144	-1.4117e+00	0.158030
omega	0.000104	0.000033	3.1027e+00	0.001918
alpha1	0.152972	0.003137	4.8769e+01	0.000000
alpha2	0.021904	0.025051	8.7440e-01	0.381898
alpha3	0.013522	0.007348	1.8403e+00	0.065726
beta1	0.000001	0.022909	3.8000e-05	0.999970
beta2	0.285349	0.000044	6.5070e+03	0.000000
beta3	0.417910	0.163014	2.5637e+00	0.010358
beta4	0.000000	0.176085	0.0000e+00	1.000000
gamma1	0.100956	0.026689	3.7826e+00	0.000155
gamma2	0.093377	0.054114	1.7256e+00	0.084422
gamma3	-0.099631	0.000606	-1.6429e+02	0.000000

LogLikelihood : 7931.843

Information Criteria

```
Akaike       -3.9984
Bayes        -3.9682
Shibata      -3.9985
Hannan-Quinn -3.9877
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	7.734	5.418e-03
Lag[2*(p+q)+(p+q)-1][20]	15.795	3.220e-15
Lag[4*(p+q)+(p+q)-1][34]	22.794	5.334e-02

d.o.f=7
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.2428	0.6222
Lag[2*(p+q)+(p+q)-1][20]	3.4387	0.9935
Lag[4*(p+q)+(p+q)-1][34]	6.1439	0.9991

d.o.f=7

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[8]	0.05463	0.500	2.000	0.8152
ARCH Lag[10]	0.88844	1.488	1.815	0.8027
ARCH Lag[12]	1.41155	2.451	1.700	0.8865

Nyblom stability test

```
Joint Statistic:  4.0878
Individual Statistics:
mu     0.06643
ar1    0.13227
ar2    0.19008
ar3    0.34713
ma1    0.10396
ma2    0.16552
ma3    0.32796
ma4    0.08016
omega  0.29001
alpha1 0.16737
alpha2 0.36326
alpha3 0.08898
beta1  0.28232
beta2  0.22327
beta3  0.28191
beta4  0.31095
gamma1 0.05141
gamma2 0.18531
gamma3 0.08161
```

```
Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:          4.03 4.33 4.92
Individual Statistic:     0.35 0.47 0.75
```

Sign Bias Test

	t-value	prob	sig
Sign Bias	0.6686	0.5038	
Negative Sign Bias	0.5622	0.5740	
Positive Sign Bias	1.3859	0.1659	
Joint Effect	2.3281	0.5072	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	553.1	3.878e-105
2	30	573.9	1.664e-102
3	40	598.3	2.891e-101
4	50	611.4	3.751e-98

```
Elapsed time : 3.437108
```

GJR-GARCH(4,2) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(4,2)
Mean Model       : ARFIMA(4,0,2)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001311	0.000481	2.72845	0.006363
ar1	-0.495752	0.128871	-3.84690	0.000120
ar2	-0.827226	0.120866	-6.84415	0.000000
ar3	0.029626	0.021617	1.37047	0.170541
ar4	0.016162	0.021751	0.74304	0.457456
ma1	0.488756	0.127392	3.83664	0.000125
ma2	0.847416	0.118383	7.15823	0.000000
omega	0.000106	0.000015	6.96282	0.000000
alpha1	0.122096	0.023733	5.14452	0.000000
alpha2	0.009749	0.013823	0.70527	0.480642
alpha3	0.000000	0.025839	0.00000	1.000000
alpha4	0.071811	0.018809	3.81796	0.000135
beta1	0.000000	0.036645	0.00000	1.000000
beta2	0.688808	0.033310	20.67867	0.000000
gamma1	0.07			
gamma2	0.07			
gamma3	-0.05			
gamma4	0.04			

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001311	0.000494	2.65564	0.007916
ar1	-0.495752	0.218311	-2.27086	0.023156
ar2	-0.827226	0.215524	-3.83821	0.000124
ar3	0.029626	0.024177	1.22537	0.220437
ar4	0.016162	0.029113	0.55515	0.578794
ma1	0.488756	0.216018	2.26257	0.023662
ma2	0.847416	0.212899	3.98037	0.000069
omega	0.000106	0.000038	2.80205	0.005078
alpha1	0.122096	0.040704	2.99963	0.002703
alpha2	0.009749	0.023526	0.41440	0.678582
alpha3	0.000000	0.040113	0.00000	1.000000
alpha4	0.071811	0.027588	2.60295	0.009243
beta1	0.000000	0.117334	0.00000	1.000000
beta2	0.688808	0.093603	7.35884	0.000000
	0.077590	0.066599	1.16503	0.244006
	0.077143	0.048471	1.59153	0.111491
	0.099502	0.06200		
	0.041640	0.08305		

```
LogLikelihood : 7945.641

Information Criteria
-------------------------------------

Akaike       -4.0059
Bayes        -3.9773
Shibata      -4.0059
Hannan-Quinn -3.9957
```

Weighted Ljung-Box Test on Standardized Residuals

	statistic	p-value
Lag[1]	7.787	0.005263
Lag[2*(p+q)+(p+q)-1][17]	17.060	0.000000
Lag[4*(p+q)+(p+q)-1][29]	24.164	0.002972

Weighted Ljung-Box Test on Standardized Squared Residuals

	statistic	p-value
Lag[1]	0.0001255	0.9911
Lag[2*(p+q)+(p+q)-1][17]	4.5640473	0.9230
Lag[4*(p+q)+(p+q)-1][29]	7.7237580	0.9701

d.o.f=6

Nyblom stability test

```
Joint Statistic:  5.0942
Individual Statistics:
mu     0.09028
ar1    0.05956
ar2    0.15638
ar3    0.10280
ar4    0.04803
ma1    0.06962
ma2    0.17937
omega  0.36428
```

Asymptotic Critical Values (10% 5% 1%)

```
Joint Statistic:          3.83 4.14 4.73
Individual Statistic:     0.35 0.47 0.75
```

Sign Bias Test

	t-value	prob	sig
Sign Bias	0.6040	0.5459	
Negative Sign Bias	0.1787	0.8582	
Positive Sign Bias	0.9659	0.3341	
Joint Effect	1.0368	0.7924	

Weighted ARCH LM Tests

	Statistic	Shape	Sc	
ARCH Lag[7]	0.503	0.500	2	
ARCH Lag[9]	1.692	1.485	1.796	0.5910
ARCH Lag[11]	1.819	2.440	1.677	0.8131

Elapsed time : 2.810474

```
gamma2 0.11228
gamma3 0.11476
gamma4 0.07729
```

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	544.6	2.419e-103
2	30	567.4	3.661e-101
3	40	611.7	5.558e-104
4	50	616.7	3.312e-99

GJR-GARCH(4,3) Model Fit

```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(4,3)
Mean Model       : ARFIMA(4,0,3)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001291	0.000502	2.573933	0.010055
ar1	0.526269	0.286089	1.839530	0.065837
ar2	-0.547182	0.109100	-5.015437	0.000001
ar3	0.697698	0.207103	3.368851	0.000755
ar4	-0.011713	0.013956	-0.839294	0.401305
ma1	-0.534728	0.285748	-1.871326	0.061300
ma2	0.586705	0.110275	5.320375	0.000000
ma3	-0.693829	0.224854	-3.085686	0.002031
omega	0.000112	0.000016	7.202026	0.000000
alpha1	0.107692	0.004324	24.903752	0.000000
alpha2	0.000000	0.019299	0.000000	1.000000
alpha3	0.000000	0.007079	0.000002	0.999999
alpha4	0.090679	0.025342	3.578191	0.000346
beta1	0.237526	0.037532	6.328587	0.000000
beta2	0.460127	0.000357	1288.211630	0.000000
beta3	0.000000	0.040307	0.000001	0.999999
gamma1	0.089735	0.003313	27.085642	0.000000
gamma2	0.050124	0.016210	3.092232	0.001987
gamma3	-0.125493	0.000217	-577.252042	0.000000
gamma4	0.058297	0.074317	0.784442	0.027480

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>
mu	0.001291	0.000503	2.568910	0.01
ar1	0.526269	0.413381	1.273086	0.20
ar2	-0.547182	0.099242	-5.513626	0.00
ar3	0.697698	0.306709	2.274786	0.02
ar4	-0.011713	0.021993	-0.532582	0.59
ma1	-0.534728	0.412642	-1.295864	0.19
ma2	0.586705	0.099020	5.925095	0.00
ma3	-0.693829	0.322896	-2.148766	0.03
omega	0.000112	0.000045	2.479761	0.01
alpha1	0.107692	0.002818	38.218287	0.00
alpha2	0.000000	0.029836	0.000000	1.00
alpha3	0.000000	0.009052	0.000001	0.99
alpha4	0.090679	0.047837	1.895579	0.05
beta1	0.237526	0.047808	4.968305	0.00
beta2	0.460127	0.000685	671.888090	0.00
beta3	0.000000	0.082916	0.000001	1.00
gamma1	0.089735	0.004661	19.251748	0.00
gamma2	0.050124	0.019221	2.607764	0.00
gamma3	-0.125493	0.000178	-703.498685	0.00
gamma4	0.058297	0.074317	0.784442	0.43

LogLikelihood : 7940.344

Information Criteria

Akaike	-4.0022
Bayes	-3.9704
Shibata	-4.0022
Hannan-Quinn	-3.9909

Weighted Ljung-Box Test on Standardized Residuals
---
	statistic	p-value
Lag[1]	0.0119	0.9131
Lag[2*(p+q)+(p+q)-1][20]	3.1046	0.9963
Lag[4*(p+q)+(p+q)-1][34]	6.0800	0.9991

d.o.f=7

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[8]	0.398	0.500	2.000	0.5281
ARCH Lag[10]	1.369	1.488	1.815	0.6758
ARCH Lag[12]	1.874	2.451	1.700	0.8101

Nyblom stability test

Joint Statistic: 5.0137
Individual Statistics:

mu	0.09069
ar1	0.08345
ar2	0.07681
ar3	0.19452
ar4	0.10624
ma1	0.09864
ma2	0.06976
ma3	0.20957
omega	0.33900
alpha1	0.28714
alpha2	0.18860
alpha3	0.11171
alpha4	0.30769
beta1	0.37029
beta2	0.32534
beta3	0.36023
gamma1	0.05413
gamma2	0.08489

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic: 4.22 4.52 5.13
Individual Statistic: 0.35 0.47 0.75

Sign Bias Test
---
	t-value	prob sig
Sign Bias	0.09543	0.9240
Negative Sign Bias	0.29566	0.7675
Positive Sign Bias	0.59624	0.5510
Joint Effect	0.45240	0.9292

Adjusted Pearson Goodness-of-Fit Test:
---
	group	statistic	p-value(g-1)
1	20	545.9	1.243e-103
2	30	549.3	1.973e-97
3	40	602.1	4.953e-102
4	50	606.7	3.244e-97

Elapsed time : 3.239443

GJR-GARCH(4,4) Model Fit

```
*----------------------------------*
*          GARCH Model Fit         *
*----------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model      : gjrGARCH(4,4)
Mean Model       : ARFIMA(4,0,4)
Distribution     : norm
```

Optimal Parameters

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001289	0.000545	2.3678e+00	0.017897
ar1	0.992266	0.002276	4.3592e+02	0.000000
ar2	-1.000130	0.002993	-3.3415e+02	0.000000
ar3	0.734920	0.000904	8.1327e+02	0.000000
ar4	0.161815	0.003274	4.9430e+01	0.000000
ma1	-0.999603	0.000062	-1.6077e+04	0.000000
ma2	1.024379	0.000010	1.0514e+05	0.000000
ma3	-0.751375	0.000051	-1.4733e+04	0.000000
ma4	-0.143184	0.000066	-2.1553e+03	0.000000
omega	0.000123	0.000027	4.5476e+00	0.000005
alpha1	0.144410	0.023076	6.2579e+00	0.000000
alpha2	0.012139	0.039216	3.0954e-01	0.756908
alpha3	0.000000	0.012913	3.0000e-06	0.999998
alpha4	0.080976	0.017074	4.7428e+00	0.000002
beta1	0.000000	0.265316	0.0000e+00	1.000000
beta2	0.400685	0.193436	2.0714e+00	
beta3	0.000000	0.219921	0.0000e+00	
beta4	0.228326	0.137623	1.6591e+00	
gamma1	0.050845	0.031516	1.6133e+00	
gamma2	0.077306	0.024004	3.2206e+00	
gamma3	-0.075594	0.036534	-2.0691e+00	
gamma4	0.075966	0.030711	2.4735e+00	

Robust Standard Errors:

	Estimate	Std. Error	t value	Pr(>|t|)
mu	0.001289	0.000570	2.2606e+00	0.023785
ar1	0.992266	0.002286	4.3401e+02	0.000000
ar2	-1.000130	0.003223	-3.1031e+02	0.000000
ar3	0.734920	0.001039	7.0719e+02	0.000000
ar4	0.161815	0.002887	5.6047e+01	0.000000
ma1	-0.999603	0.000085	-1.1825e+04	0.000000
ma2	1.024379	0.000007	1.4984e+05	0.000000
ma3	-0.751375	0.000090	-8.3378e+03	0.000000
ma4	-0.143184	0.000084	-1.7127e+03	0.000000
omega	0.000123	0.000141	8.7325e-01	0.382528
alpha1	0.144410	0.052315	2.7604e+00	0.005774
alpha2	0.012139	0.160085	7.5830e-02	0.939554
alpha3	0.000000	0.096037	0.0000e+00	1.000000
alpha4	0.080976	0.056362	1.4367e+00	0.150797
beta1	0.000000	1.008772	0.0000e+00	1.000000

```
LogLikelihood : 7953.188

Information Criteria
------------------------------------

Akaike       -4.0077
Bayes        -3.9727
Shibata      -4.0077
Hannan-Quinn -3.9953
```

Nyblom stability test

```
------------------------------------
Joint Statistic:  no.parameters>20 (not available)
Individual Statistics:
mu      0.07911
ar1     0.25521
ar2     0.03515
ar3     0.22417
ar4     0.07081
ma1     0.11422
ma2     0.05096
ma3     0.11001
ma4     0.06598
omega   0.32596
alpha1 0.30541
alpha2 0.40625
alpha3 0.15352
alpha4 0.24253
beta1   0.25256
beta2   0.32305
beta3   0.23014
beta4   0.34583
gamma1 0.16439
gamma2 0.13100
gamma3 0.19182
gamma4 0.07679
```

Weighted Ljung-Box Test on Standardized Residuals

```
------------------------------------
                        statistic p-value
Lag[1]                      7.312 0.00685
Lag[2*(p+q)+(p+q)-1][23]   18.946 0.00000
Lag[4*(p+q)+(p+q)-1][39]   28.220 0.01206
d.o.f=8
H0 : No serial correlation
```

Weighted Ljung-Box Test on Standardized Squared Residuals

```
------------------------
                        Asymptotic Critical Values (10% 5% 1%)
                        Individual Statistic:    0.35 0.47 0.75
Lag[1]
Lag[2*(p+q)+(p+q)-1][23]
Lag[4*(p+q)+(p+q)-1][39]
d.o.f=8
```

Weighted ARCH LM Tests

```
------------------------------------
```

	Statistic	Shape	Scale	P-Value
ARCH Lag[9]	0.8773	0.500	2.00	0.3490
ARCH Lag[11]	1.1359	1.490	1.83	0.7398
ARCH Lag[13]	2.0813	2.459	1.72	0.7796

Sign Bias Test

```
------------------------------------
                     t-value   prob sig
Sign Bias           0.65032 0.5155
Negative Sign Bias 0.08053 0.9358
Positive Sign Bias 1.30026 0.1936
Joint Effect        1.70060 0.6368
```

Adjusted Pearson Goodness-of-Fit Test:

```
------------------------------------
  group statistic p-value(g-1)
1    20     534.6   3.031e-101
2    30     551.7    6.549e-98
3    40     585.8    1.020e-98
4    50     599.0    1.133e-95
```

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